Installation guide

OneSumX Regulatory Engine 4.7.0 Installation Guide

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1 Preface

1.1 About the Regulatory Engine Installation Guide

1.1.1 Audience

This guide is intended primarily for support personnel, administrators, and system administrators. The following sections provide information about using this guide.

1.1.2 Related documentation

- OneSumX Regulatory Engine Hardware and Software Requirements
- OneSumX Regulatory Engine User Manual
- OneSumX Regulatory Engine Administrator Manual

1.1.3 Online resources and support

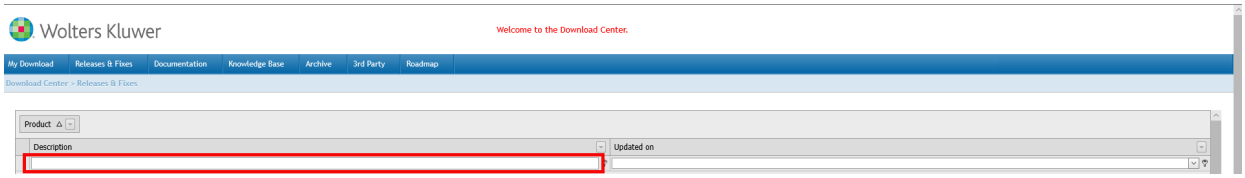
1.1.3.1 Downloading files from the Product Download Center

Context

This procedure is generic for everyone who wants to download files from the Product Download Center.

Procedure

1. Log in to the Product Download Center.
2. Click **Releases & Fixes**.
3. Search for the correct file via the **Description** field.



4. Save the file locally.

2 Installation requirements

The installation of the solution should not be delegated to limited users because the solution misconfiguration could lead to security issues.

The requirements apply to the installation of the standard edition OneSumX Regulatory Engine.

2.1 Hardware requirements

For detailed hardware requirements, refer to the *OneSumX Regulatory Engine Hardware and Software Requirements*.

2.2 Software requirements

For detailed software requirements, refer to the *OneSumX Regulatory Engine Hardware and Software Requirements*.

2.3 Installation prerequisites

2.3.1 Java installation

Context

OneSumX Regulatory Engine requires Java 64-bit runtime (JRE 11) to be installed on the application server where it will run.

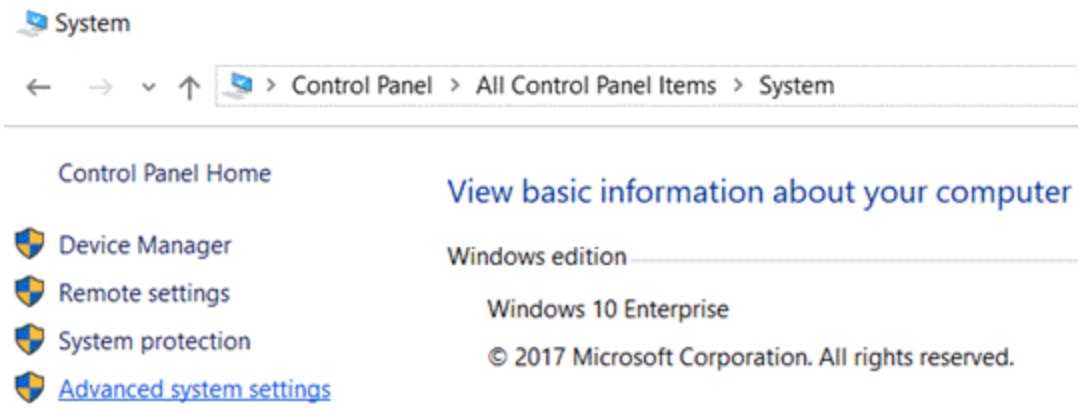
Warning:
In case the Java version was updated, the RE certificate needs to be imported into the Java *cacerts* file.

Procedure

1. Download Java (**Windows Offline (64-bit)**) from https://www.java.com/en/download/manual.jsp.
2. Right-click the downloaded EXE file and, in the context menu, click **Run as administrator**.
3. Click **Install**.

   Note:
   If you want to change the default *C\:Program Files* target destination, select the option to do so before clicking the installation button.

4. Recommended only for the distribution with embedded container: create the WK_RE_JAVA_HOME environment variable as follows:
   a. Via the **Windows Start** menu, go to **Control Panel** > **All Control Panel Items** > **System** > **Advanced System Settings**.



b. Near the bottom of the **System Properties** screen, click **Environment Variables**.
c. In the **Environment Variables** screen, click **New** under the system variables overview.
d. In the **New System Variable** screen, enter the following values:

- **Variable Name**: `WK_RE_JAVA_HOME`
- **Variable Value**: The path where you installed Java (for example, *C:\Program Files\Java\jdk11.0.13_8*)

e. Click **OK** to save the settings for the new system variable.

5. If WK_RE_JAVA_HOME isn't set, then the default JRE will be used (default JRE means that the path to it is set in PATH environment variable).
6. If there is no WK_RE_JAVA_HOME environment variable, nor a default JRE, then the JAVA_HOME environment variable will be used.

Proceed to the following step only if you are using Java IBM8 version:

7. Add a new environment variable with the following values:

- **Variable Name**: `IBM_JAVA_OPTIONS`
- **Variable Value**: *-Dcom.ibm.jsse2.overrideDefaultTLS=true -Dcom.ibm.jsse2.overrideDefaultProtocol=TLSv12*

2.3.2 Identity Providers

OneSumX Regulatory Engine supports single sign-on (SSO) authentication through SAML. For the SAML authentication to work, you need to configure an Identity Provider (IdP).

There are various Identity Providers available, two of which can be used for the OneSumX Regulatory Engine authentication, namely:

- Keycloak Identity Provider
- Active Directory Identity Provider

If OneSumX Regulatory Engine and the Identity Provider server are running on different computers, then the time settings of the computers must be synchronized. If the time settings are not synchronized, and they differ by more than 60 seconds, then the authentication process may fail. For details, refer to topic Authentication process fails if OneSumX Regulatory Engine and the Identity Provider server are not synced.

2.3.2.1 Keycloak Identity Provider

2.3.2.1.1 Install Keycloak Identity Provider

Procedure

1. Go to https://www.keycloak.org/archive/downloads-6.0.1.html and download Keycloak.
2. Install the Keycloak authentication server.

2.3.2.1.2 Change the Keycloak port

Context

To start Keycloak on the same server as the WildFly-based application server, change the port for Keycloak to avoid conflicts.

Procedure

1. Open the *[drive]:/keycloak-[version]\standalone\configuration\standalone.xml* file.
2. Add the following extra property between the `</extensions>` and `<management>` tags:

```
</extensions>
<system-properties>
<property name="jboss.socket.binding.port-offset" value="1" />
</system-properties>
<management>
```

2.3.2.1.3 Starting Keycloak

1. If Keycloak is located on a different machine than OneSumX Regulatory Engine, then follow the procedure described in topic Change Keycloak settings if on a different machine.

2. Start Keycloak choosing one of the following options, depending on your configuration:

   - Start Keycloak as a service

   - Start Keycloak via a script

2.3.2.1.3.1 Change Keycloak settings if on a different machine

Context

In case Keycloak is located on a different machine than OneSumX Regulatory Engine, then follow this procedure before starting Keycloak.

Procedure

1. Open the *[drive]:/keycloak-[version]\standalone\configuration\standalone.xml* file.
2. Modify the interface tag:

   Change:

```
<interface name="public">
        <inet-address value="${jboss.bind.address.management:127.0.0.1}"/>
</interface>
```

To:

```
<interface name="public">
        <any-address/>
</interface>
```

2.3.2.1.3.2 Start Keycloak as a service

Prerequisites

If Keycloak is located on a different machine than OneSumX Regulatory Engine, then follow the procedure described in topic Change Keycloak settings if on a different machine.

Context

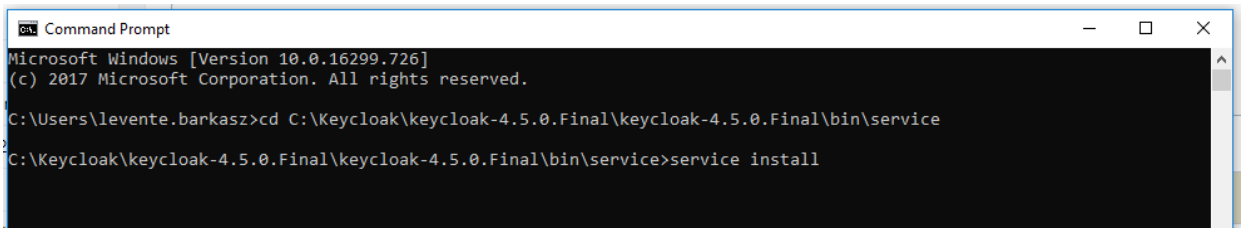
If you registered Keycloak as a Windows service, then start it using this procedure.

Procedure

1. Go to the directory where Keycloak is extracted and browse for */docs/contrib/scripts*.
2. Copy the *service* folder to the *bin* folder of the Keycloak package (same place where *standalone.bat* can be found).
3. Open the *service* folder, and then open the *service.bat* file with a notepad editor.
4. Change the `SHORTNAME`, `DISPLAYNAME`, and `DESCRIPTION` values to be more descriptive.

```
set SHORTNAME=Keycloak
rem NO quotes around the display name here !
set DISPLAYNAME=Keycloak
rem NO quotes around the description here !
set DESCRIPTION=Wildfly Application Server
```

5. Open the Windows **Command Prompt** as an Administrator, navigate to where the service directory is now located, and run the `service install` command to install the service.



Results

You should receive a confirmation message that the service has been created and the Keycloak service should now be present in Windows services (**Control Panel** / **Administrative Tools** / **Services**).

2.3.2.1.3.3 Start Keycloak via a script

Prerequisites

If Keycloak is located on a different machine than OneSumX Regulatory Engine, then follow the procedure described in topic Change Keycloak settings if on a different machine.

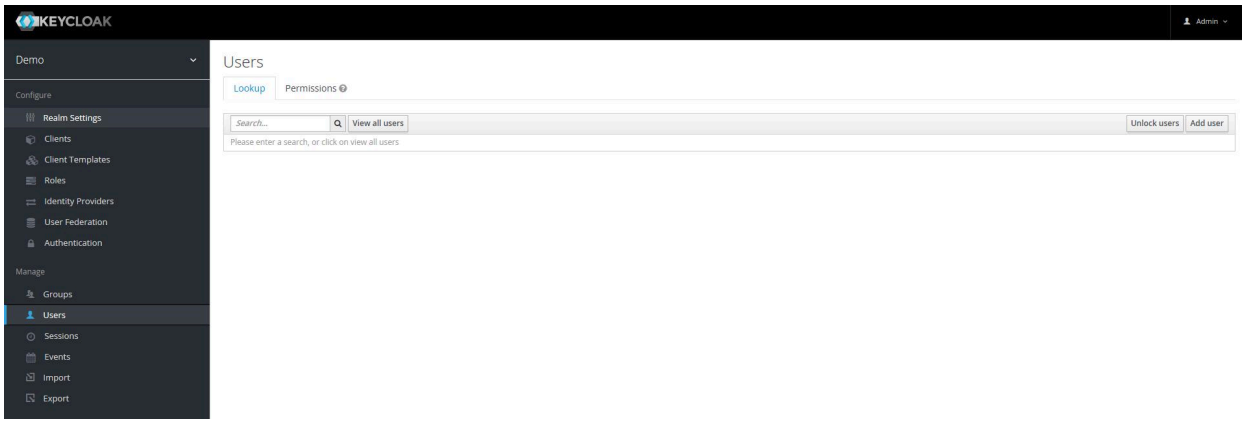
Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Type `[drive]:/keycloak-[version]/bin`.
3. Type standalone.bat.

2.3.2.1.4 Create a realm and a default user in Keycloak

Procedure

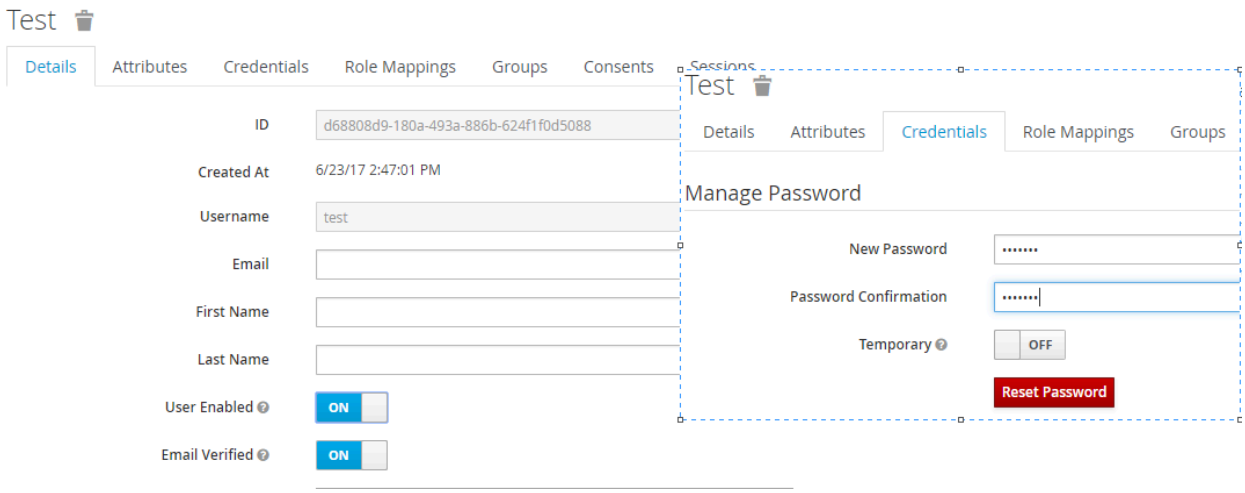
1. Log on to Keycloak with your admin account: http://localhost:8081/auth/admin/.
2. Click **Realm Settings**, then click **Create new realm**.
3. In the realm's main menu, navigate to **Users** and add a new user.



4. Ensure that you have credentials to access the realm corresponding to the value of the property admin.defaultuser from *install.properties*.

   If you don't have a user belonging to the appropriate realm, then you need to create it. For example, you have `admin.defaultuser=admin`, you need to create user "admin" in step 3.

   Note:
   - To change the password, go to the **Credentials** tab of the user's page.
   - Ensure not to select **Temporary**.
   - The user name must have the same value as the admin.defaultuser property configured in *install.properties*.



Tip:

If you need to have signatures present in all SAML responses, change the configuration as follows:

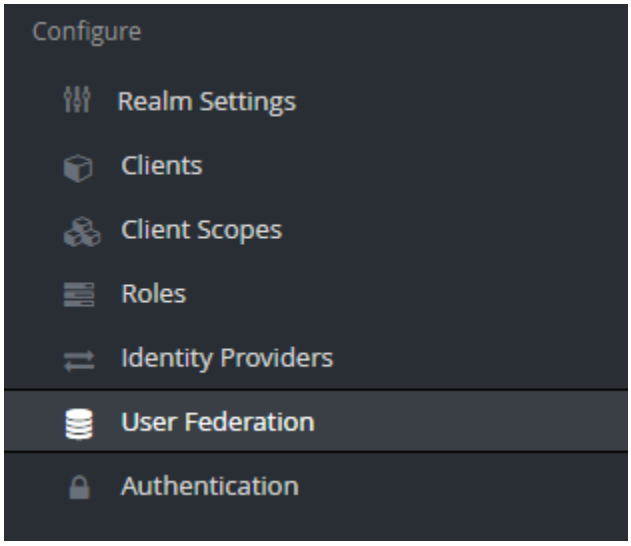
- In all Service Provider metadata XML files (AdminConsole, ReportingHub, ReportGenerator, Lineage), set WantAssertionsSigned to "true".

2.3.2.1.5 Create an LDAP authentication

Context

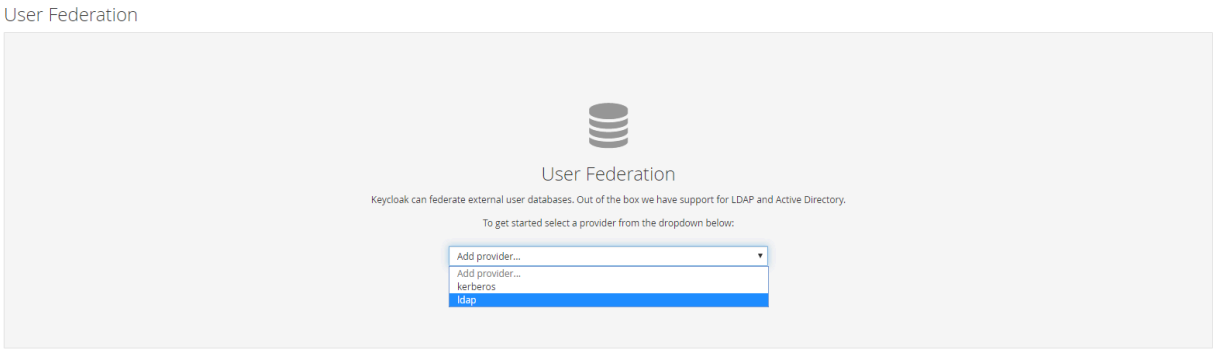
Creating LDAP authentication is an optional step.

Domain users can be authorized in Keycloak. To that purpose, LDAP authentication must be configured via option **User Federation** of the realm. As a prerequisite for this unexpired service an account should be created. This will be used for LDAP requests in the domain.

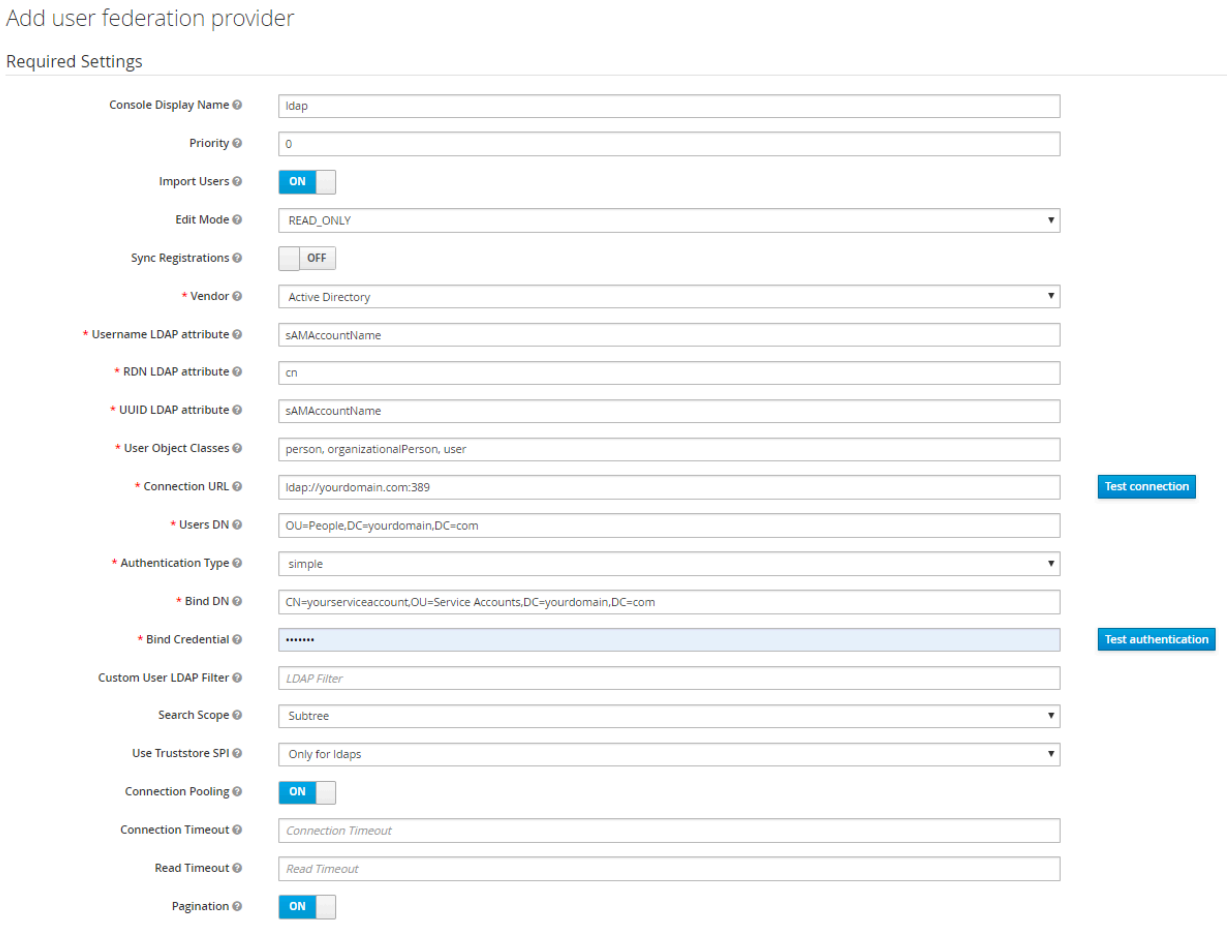


Procedure

1. Choose **ldap** as provider.



2. To set up LDAP for your domain, provide the information in the form as illustrated here and select **Save** to store the settings.



2.3.2.1.6 Create Kerberos authentication

Prerequisites

Creating a Kerberos authentication is an optional step.

Procedure

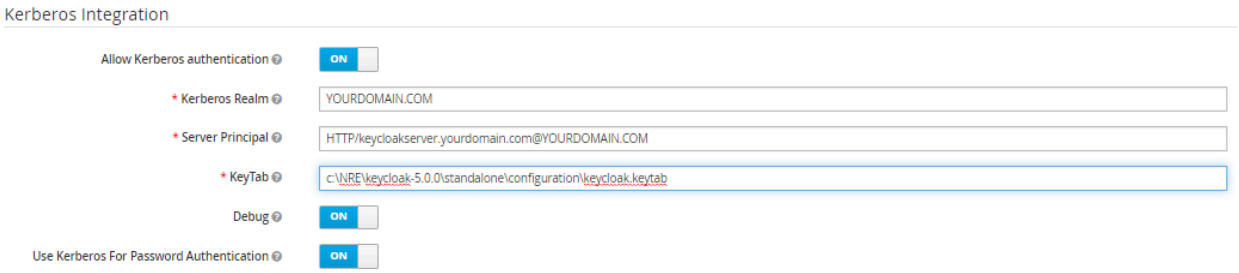
1. Create a Keytab file on one of the domain controller servers using the command:

```
ktpass -out keycloak.keytab -princ HTTP/<full FQDN for host or DNS
alias>@<domain name in upper case> -mapUser <service user account>@<domain
name in upper case> -pass <service user password> -kvno 0 -ptype
KRB5_NT_PRINCIPAL -crypto All
```

Example

```
ktpass -out keycloak.keytab -princ
HTTP/keycloakserver.yourdomain.com@YOURDOMAIN.COM -mapUser
yourserviceaccount@YOURDOMAIN.COM -pass PASSWORD -kvno 0 -ptype
KRB5_NT_PRINCIPAL -crypto All
```

2. Copy file *keycloak.keytab* to the *Keycloak* folder.
3. Activate Kerberos authentication as illustrated here.



4. Verify that Kerberos authentication is indeed turned on.

2.3.2.2 Active Directory Identity Provider

Having AD FS installed is a prerequisite, but the installation steps are not in the scope of this documentation.

To implement AD FS as Identity Provider, follow the steps in this section.

3 Installation process

There are two distribution types:

- An embedded app server package - A self hosted distribution, where OneSumX Regulatory Engine is managed by the WildFly application server.
- An external app server package- An external application server based distribution that provides WAR files, configuration files and security configuration for OneSumX Regulatory Engine. An application server is required to manage OneSumX Regulatory Engine. The supported servers are JBoss EAP and WebSphere Liberty.

3.1 Installing the embedded app server package

The installation process of the embedded app server package consists of the following actions:

1. Downloading and extracting the distribution file
2. Configuring the installation properties: setting the communication properties, generating the HTTPS keystore, creating the database schema, setting the security configuration
3. Installing OneSumX Regulatory Engine
4. Configuring the Identity Provider for the SAML authentication

3.1.1 Download and extract the distribution file

Prerequisites

- You have an Oracle or MSSQL instance running and accessible on the computer where you perform the installation.
- You have sufficient privileges to create other users in the database.

Procedure

1. Download the latest version of OneSumX Regulatory Engine Standard Edition from the Product Download Center. For details, refer to topic Downloading files from the Product Download Center.
2. Extract the downloaded file to *[installation drive:]\[RE installation folder]*.

   The expected content structure is as follows:

   - The Installer web application installation scripts are stored in the */bin* folder. In this folder you will also find the services to start and stop the WildFly application server.
   - The */conf* directory contains the *install.properties* file that will be used during the installation.
   - During the installation, the local file repository will be created in directory */etc/contentstore/repository* and then used at runtime.

   Warning:
   If you unpack and trigger the OneSumX Regulatory Engine installation from a very long path (more than 200 characters) you may face issues with certain functionalities of the legacy engine.

3.1.2 Download the Semansys package

Prerequisites

- You downloaded and extracted the distribution package for OneSumX Regulatory Engine.

Context

Starting from OneSumX Report Generator Web (TNG) version 2.24.0, the Semansys libraries are not distributed along with the engine.

These libraries are required if you need to perform a XBRL delivery.

Procedure

1. Go to the Product Download Center and download the following files: *RG_Semansys_artifacts* and *RG_Semansys_license*.
2. Extract them in the following location: *[installation drive:]\[RE installation folder]\deploy\legacy\WK ReportGenerator\bin*

3.1.3 Configuring the installation properties

You can edit the OneSumX Regulatory Engine installation properties in the *install.properties* file.

The following aspects are important:

- Avoid using spaces in any path definition. This may cause errors in specific situations.
- Avoid using backslashes in any path properties in the *install.properties* file, because this may cause errors. If it can't be avoided, you should use a double backslash (\\), otherwise the backslash will be lost in result value.
- To avoid losing any of the created artifacts and to be able to use the application after future upgrades, ensure that the wolterskluwer.ssl.keystore.path, workspaces.legacy.root, and workspaces.workdir.root properties are set to a path outside the application installation folder.
- For OneSumX Regulatory Engine, external app server package, for the WebSphere application server, property wolterskluwer.datasource.jndi.prefix is mandatory and must be set to the following value: `jdbc/wolterskluwer/datasources`.

3.1.3.1 Setting the communication properties

Context

Define the addresses that will be used to access the application in the installation properties file.

Procedure

1. Go to *[installation drive:]\[RE installation folder]\Conf* and open the *install.properties* file.
2. Modify the following properties:

- wolterskluwer.application.host - define the host that will be used to access the application. Make sure that underscore "_" is NOT used within the machine/alias name. The java.net.URI doesn't support underscores in host names.
- wolterskluwer.locally.accessible.host -
- wolterskluwer.application.port - the port used by OneSumX Regulatory Engine can have values between 1 and 65535.
- For HTTPS, provide additional properties and a certificate, as described in topic Configuring the HTTPS protocol.

3.1.3.1.1 Configuring the HTTPS protocol

OneSumX Regulatory Engine only supports the HTTPS protocol. You need a JAVA keystore with the certificate that will be used to encrypt HTTPS. If you don't have one, then it must be generated.

3.1.3.1.1.1 Generate the HTTPS certificate

Context

You can generate the keystore using the keytool utility, which is part of JRE.

Procedure

1. Open Command Prompt or PowerShell.
2. To run the keytool, add *%JAVA_HOME%/bin* to the PATH system variable of Windows.
3. Open *install.properties* and add the appropriate values to the following properties:
   - wolterskluwer.ssl.keystore.alias = certificate alias
   - wolterskluwer.ssl.keystore.path = path to keystore
   - wolterskluwer.ssl.keystore.password = keystore password
4. Replace the placeholders with actual values used in *install.properties* for the following properties *${wolterskluwer.ssl.keystore.alias}*, *${wolterskluwer.ssl.keystore.path}*, and *${wolterskluwer.ssl.keystore.password}* and then run the script:

   ```
   keytool -genkey -alias ${wolterskluwer.ssl.keystore.alias} -keystore
   ${wolterskluwer.ssl.keystore.path}
   -storepass ${wolterskluwer.ssl.keystore.password} -keyalg RSA -validity 360
   -keysize 2048
   ```

5. Answer the suite of questions that follow.

   It's important to have the value of the first and last name the same as the value of the host of your application (wolterskluwer.application.host) from the installer properties. For the other settings, provide any values as appropriate.

   To reply to the question “Is ... correct?”, type `yes` and press **ENTER**.

```
C:\certificates>keytool -genkey -alias nre-https -keystore nre.keystore -storepass changeme -keyalg RSA -validity 360 -keysize 2048
What is your first and last name?
  [Unknown]:  localhost
What is the name of your organizational unit?
  [Unknown]:  bank
What is the name of your organization?
  [Unknown]:  bank
What is the name of your City or Locality?
  [Unknown]:  City
What is the name of your State or Province?
  [Unknown]:  State
What is the two-letter country code for this unit?
  [Unknown]:  EU
Is CN=localhost, OU=bank, O=bank, L=City, ST=State, C=EU correct?
  [no]:  yes

Enter key password for <nre-https>
        (RETURN if same as keystore password):
```

3.1.3.1.1.2 Generate a certificate signing request

Context

This is an optional step. Run these commands only when signed certificates are needed.

Procedure

1. Run the following command:

```
keytool -keystore ${wolterskluwer.ssl.keystore.path} -certreq -alias
${wolterskluwer.ssl.keystore.alias} -keyalg rsa -file signing_request.csr
```

Results

The signed certificate is generated.

2. The generated certificate can be imported in the keystore file and in *cacerts* using the import command:

```
keytool -import -alias ${wolterskluwer.ssl.keystore.alias} -file
{certificate.file.name} -keystore "${server.java.home}/jre/lib/security/
cacerts" -
storepass ${java.trustore.password} -noprompt
```

3.1.3.1.1.3 Importing HTTPS certificate into JRE cacerts manually

Context

This is an optional step. The installer will perform these operations, but in case you want to do it manually, follow this procedure.

Procedure

1. Extract the certificate from the created keystore, where *${certificate.file.name}* (for example, *secure_https.cer*) represents the path to the file that will be created and that will contain the certificate:

```
keytool -export -alias ${wolterskluwer.ssl.keystore.alias} -keystore
```

```
${wolterskluwer.ssl.keystore.path} -rfc -file ${certificate.file.name} -
storepass ${wolterskluwer.ssl.keystore.password}
```

Note:
If you already have a certificate with *${server.alias}* and you want to modify it, then you first have to delete it. See topic Delete the certificate from the keystore and cacerts file.

2. Import the certificate to the JAVA_HOME trust store created certificate.

```
keytool -import -alias ${wolterskluwer.ssl.keystore.alias} -file
{certificate.file.name} -keystore "%JAVA_HOME%/jre/lib/security/cacerts" -
storepass ${java.trustore.password} -noprompt
```

The default value of *${java.trust.store.password}* is `changeit`.

3.1.3.1.1.4 Delete the certificate from the keystore and cacerts file

Context

When you are importing a certificate manually into the keystore and/or cacerts file, you may see an error, such as "Certificate not imported, alias already exists". In that case, you may need to delete the certificate from the keystore and/or the cacerts file.

Procedure

1. Set the values for the following parameters according to the values used in topic Generate the HTTPS certificate:
   - ${wolterskluwer.ssl.keystore.alias}
   - ${java.trustore.password}
   - ${wolterskluwer.ssl.keystore.path}
   - ${wolterskluwer.ssl.keystore.password}
   - ${server.java.home}
2. Run the following commands:

```
keytool -delete -alias ${wolterskluwer.ssl.keystore.alias} -keystore
"${server.java.home}/jre/lib/security/cacerts" -storepass
${java.trustore.password}
```

```
keytool -delete -alias ${wolterskluwer.ssl.keystore.alias} -keystore
${wolterskluwer.ssl.keystore.path} -storepass
${wolterskluwer.ssl.keystore.password}
```

3.1.3.2 Creating the database schema

Before starting the installation process of OneSumX Regulatory Engine, it is necessary to prepare the database environment. The application supports the following types of databases:

- Oracle Server
- SQL Server

Note:
To avoid ambiguity with time zones, the machine where the application is installed and the database server must have the same time zone settings.

Starting with version 4.1.0, there are two supported authentication configurations for Microsoft SQL Server (further referred to as "SQL Server"):

- Setting the database properties for the username/password authentication on SQL Server
- Setting the database properties for the integrated authentication on SQL Server

There are three options to start using a database with OneSumX Regulatory Engine:

- Using an existing database
- Automatically creating a database
- Manually creating a database

3.1.3.2.1 Setting the database properties for the username/password authentication on SQL Server

Procedure

1. Go to *install.properties*.
2. Provide the appropriate values for the following properties:
   - wolterskluwer.system.db.username (schema manipulation user)
   - wolterskluwer.system.db.password
   - wolterskluwer.application.db.username (data manipulation user)
   - wolterskluwer.application.db.password
   - wolterskluwer.db.connection.url
   - wolterskluwer.db.sqlserver.name - this

     property should concatenate the host name and the SQL Server instance name in case you want to use a named instance (for example, *localhost\SQLSERVER_CI*). For the default instance, it should contain the host name and the port (such as *localhost,1433*). This configuration should be provided from the UI during the creation of a workspace.
   - wolterskluwer.sqlserver.default.driver
   - wolterskluwer.datasource.jndi.prefix - for WebSphere Liberty, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*.
3. Create a privileged user (schema manipulation user) login: this must be declared at wolterskluwer.system.db.username, with password wolterskluwer.system.db.password. This is a user who has the rights needed to create a new schema, users, grant permissions to users.

Warning:
If the password contains symbol "\", issues may occur during an engine upgrade process. It is recomended to not use the "\" symbol in the wolterskluwer.system.db.password definition.

The privileged user login can be created in several ways; one method is using the “sqlcmd” utility.

The following script is an example of creating a privileged user login via sqlcmd, using the name "RE_ADMIN" and the password "P@ssw0rd", and granting all privileges:

```
CREATE LOGIN [RE_ADMIN] WITH PASSWORD='P@ssw0rd', DEFAULT_DATABASE=[master],
CHECK_EXPIRATION=OFF, CHECK_POLICY=ON
ALTER SERVER ROLE [bulkadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [dbcreator] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [diskadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [processadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [securityadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [serveradmin] ADD MEMBER [RE_ADMIN
ALTER SERVER ROLE [setupadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [sysadmin] ADD MEMBER [RE_ADMIN]
```

4. Create a less privileged user (data manipulation user) login – this must be declared at wolterskluwer.application.db.username, with password wolterskluwer.application.db.password. This is a user who has rights to read and write data from tables. This user will be further used by the application to access the database.

   The less privileged user login can be created in several ways; one method is using the “sqlcmd” utility.

   The following script is an example of creating a less privileged user login, as an example with the name "RE_USER" and password "P@ssw0rd":

```
CREATE LOGIN [RE_USER] WITH PASSWORD='P@ssw0rd', DEFAULT_DATABASE=[master],
CHECK_EXPIRATION=OFF, CHECK_POLICY=ON
```

What to do next

Choose to proceed with one of the following options:

- The database will be created automatically for you - refer to topic Automatic database creation using user/password authentication.
- Create a database manually - refer to topic Manually create a database using username/password authentication.

3.1.3.2.1.1 Automatic database creation using user/password authentication

The database is created automatically during the OneSumX Regulatory Engine installation. In case you choose to manually create a database, refer to topic Manually create a database using username/password authentication.

The OneSumX Regulatory Engine installer uses system user credentials to automatically create the database and the user. The database name is obtained from wolterskluwer.db.connection.URL.

3.1.3.2.1.2 Manually create a database using username/password authentication

Prerequisites

Set the database properties for the username/password authentication. For details, refer to topic Setting the database properties for the username/password authentication on SQL Server.

Context

File *db_scripts/mssql/CREATE_SCHEMA.sql* contains a SQL script to create the database.

File *db_scripts/mssql/CREATE_USER_FOR_LOGIN.sql* contains a SQL script to create the user for the login and to grant permissions to read and write to the database.

Procedure

1. Copy the *CREATE_SCHEMA.sql* script to the database client executor.
2. Provide the actual values for the placeholders.
3. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

   ```
   CREATE DATABASE [${databaseName}];
   ```

   databaseName

   Name of the database where the tables are created by the reporting installer.

4. Copy the *CREATE_USER_FOR_LOGIN.sql* script to the database client executor.
5. Provide the actual values for the placeholders.
6. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

   ```
   use ${databaseName};
   go
   CREATE USER [${userName}] FOR LOGIN [${userName}];
   go
   -- granting required privileges
   EXEC sp_addrolemember 'db_datareader', [${userName}];
   EXEC sp_addrolemember 'db_datawriter', [${userName}];
   ```

   databaseName

   Name of the database where the tables are created by the reporting installer.

   userName

   Name of the less privileged database user (that will be used by the application to access the database).

3.1.3.2.2 Setting the database properties for the integrated authentication on SQL Server

Procedure

1. Open *install.properties* and add values to the following properties:

   - wolterskluwer.sqlserver.integrated.security (by default it is set to false. To use integrated security, the property value needs to be set to true.)
   - wolterskluwer.db.connection.url
   - wolterskluwer.sqlserver.default.driver
   - wolterskluwer.db.sqlserver.name - this

     property should concatenate the host name and the SQL Server instance name in case you want to use a named instance (for example, *localhost\SQLSERVER_CI*). For the default instance, it should contain the host name and the port (such as *localhost,1433*). This configuration should be provided from the UI during the creation of a workspace.
   - wolterskluwer.datasource.jndi.prefix - for WebSphere Liberty, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*.

2. Remove the following properties, because they are not needed anymore:

   - wolterskluwer.system.db.username
   - wolterskluwer.system.db.password
   - wolterskluwer.application.db.username
   - wolterskluwer.application.db.password

What to do next

Choose to proceed with one of the following options:

- The database will be created automatically for you - refer to topic Automatic database creation with integrated security on SQL Server.
- Create a database manually - refer to topic Manually create a database with integrated security on SQL Server.

3.1.3.2.2.1 Automatic database creation with integrated security on SQL Server

The database is created automatically during the OneSumX Regulatory Engine installation. In case you choose to manually create a database, refer to topic Manually create a database with integrated security on SQL Server.

The OneSumX Regulatory Engine installer uses system user credentials to automatically create the database and the user. The database name is obtained from wolterskluwer.db.connection.URL.

3.1.3.2.2.2 Manually create a database with integrated security on SQL Server

Prerequisites

Set the database properties for the integrated security. For details, refer to topic Setting the database properties for the integrated authentication on SQL Server.

Context

File *db_scripts/mssql/CREATE_SCHEMA.sql* contains a SQL script to create the database.

Procedure

1. Copy this script into the database client executor.
2. Provide the actual values for the placeholders.
3. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

   ```
   CREATE DATABASE [${databaseName}];
   ```

   databaseName

   Name of the database where the tables are created by the reporting installer.

3.1.3.2.3 Setting the properties for the database creation on Oracle Server

Context

The installer uses system user credentials to create the user.

If a database already exists, then that database will be used during installation.

Procedure

1. Open *install.properties* and add values to the following properties:
   - wolterskluwer.db.connection.url
   - wolterskluwer.oracle.default.driver
   - wolterskluwer.oracle.tnsNames.folder -*tnsnames.ora* file location
   - wolterskluwer.system.db.username
   - wolterskluwer.system.db.password
   - wolterskluwer.datasource.jndi.prefix - for OneSumX Regulatory Engine, external app server package, for WebSphere Liberty, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*.
2. Change the value of property wolterskluwer.db.connection.url according to the database connection

(remote or local).

3. Ensure that property wolterskluwer.oracle.tnsNames.folder points to the folder of the Oracle client/server that contains *tnsnames.ora*.

Note:

If the Oracle server is on another machine, this path should point to the Oracle client location on the machine where OneSumX Regulatory Engine is being installed.

What to do next

Choose to proceed with one of the following options:

- The database will be created automatically for you - refer to topic Automatic database creation on Oracle Server.
- Create a database manually - refer to topic Manually create a database on Oracle Server.

3.1.3.2.3.1 Automatic database creation on Oracle Server

The database is created automatically during the OneSumX Regulatory Engine installation. In case you choose to manually create a database, refer to topic Manually create a database on Oracle Server.

The OneSumX Regulatory Engine installer uses system user credentials to automatically create the database and the user. The database name is obtained from wolterskluwer.db.connection.URL.

3.1.3.2.3.2 Manually create a database on Oracle Server

Prerequisites

Set the properties for the database creation. For details, refer to topic Setting the properties for the database creation on Oracle Server.

Procedure

1. Open file *db_scripts/oracle/CREATE_SCHEMA.sql*. This file contains a SQL script to create a database.
2. Copy this script into the database client executor.
3. Provide the actual values for the placeholders.
4. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

```
CREATE USER ${userName} IDENTIFIED BY "${userPwd}" DEFAULT TABLESPACE users
QUOTA UNLIMITED ON users;
```

userName

name of the database user that will be used by the application to access the database.

userPwd

password of the database user.

3.1.3.3 Security configuration

To make the application secure, update the installation properties in the security section for the applications listed below.

- Admin application
- Reporting hub application
- ReportGenerator application
- Lineage application
- Application database password encryption

Note:
If you didn't generate the service provider files, they will be generated by the installer. The service provider files will be used to configure the SAML clients in the Identity Provider. Refer to topic Configuring the Identity Provider.

3.1.3.3.1 Admin application

admin.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided.

admin.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

admin.keystore.password

Password for the keystore.

admin.certificate.password

Password for the certificate that will be sent to the Identity Provider.

admin.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided.

admin.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider after installation and moved to that path.

admin.auth.entityId

Unique name of the application in the Identity Provider.

admin.defaultuser

The identifier of a default admin user with access to the Admin application. This user will be created automatically during installation.

3.1.3.3.2 Reporting hub application

hub.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided.

hub.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

hub.keystore.password

Password for the keystore.

hub.certificate.password

Password for the certificate that will be sent to the Identity Provider.

hub.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided.

hub.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider after installation and moved to that path.

hub.auth.entityId

Unique name of application in the Identity Provider.

3.1.3.3.3 ReportGenerator application

reportgenerator.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided by the IT Admin.

reportgenerator.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

reportgenerator.keystore.password

Password for the keystore.

reportgenerator.certificate.password

Password for the certificate that will be sent to the Identity Provider.

reportgenerator.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided by IT Admin.

reportgenerator.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider

after installation and moved to that path.

reportgenerator.auth.entityId

Unique name of application in the Identity Provider.

3.1.3.3.4 Lineage application

lineage.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided.

lineage.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

lineage.keystore.password

Password for the keystore.

lineage.certificate.password

Password for the certificate that will be sent to the Identity Provider.

lineage.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided.

lineage.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider after installation and moved to that path.

lineage.auth.entityId

Unique name of the application in the Identity Provider.

3.1.3.3.5 Application database password encryption

Context

Starting with OneSumX Report Generator Web (TNG) version 2.24.0, the value of property wolterskluwer.application.db.password from the installation properties file needs to be encrypted in the connection.password property, in the *conf/init.xml* file.

The password will be encrypted during the installation or upgrade process, using the following two properties from the *install.properties* file:

- wolterskluwer.db.passphrase.encryption.cipher.algorithm, the default value is AES
- wolterskluwer.db.passphrase.encryption.cipher.securityKey, empty

Follow the procedure to generate the security key.

Procedure

1. Choose a 16 characters long string as a security key.

   ### Example

   16CharSecureKey!

2. Encode it to the Base64 format using any online tool or software that provides this encoding capability.

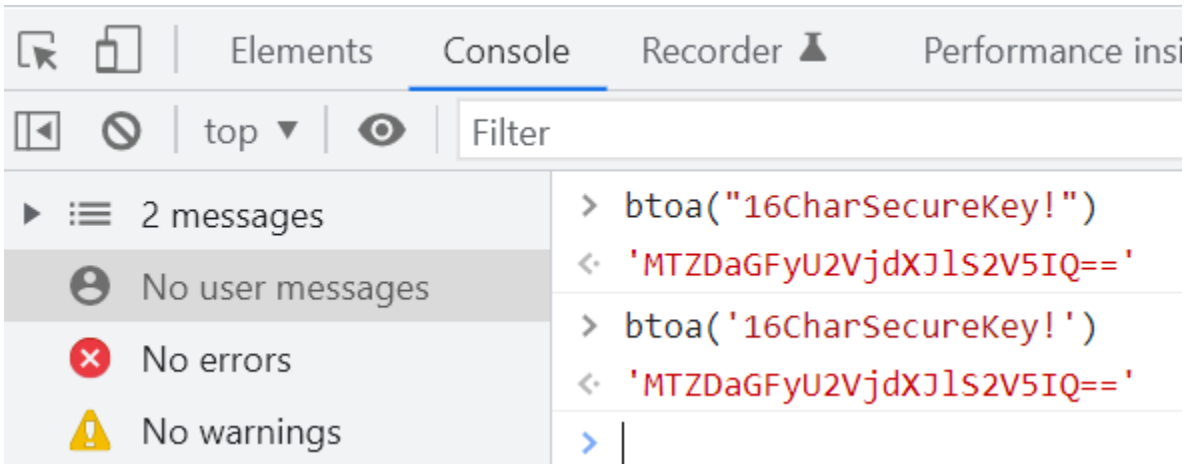
   ### Example

   a. Open the developer tools (press key F12) in the browser and click the **Console** tab.

   b. Use the `btoa` JavaScript function with your chosen string as parameter

   `btoa('16CharSecureKey!')`

   c. Copy the output `MTZDaGFyU2VjdXJlS2V5IQ==`



3. Save the Base64 encoded value into installation property woltersskluwer.db.passphrase.encryption.cipher.securityKey or provide it in the command line during the upgrade or installation (for example `-Dwoltersskluwer.db.passphrase.encryption.cipher.securityKey=MTZDaGFyU2VjdXJlS2V5IQ==`).

3.1.3.4 Adding users to OneSumX Regulatory Engine from the Identity Provider

Prerequisites

- Ensure that the Identity Provider mappers are configured correctly as described in topic [Configure SAML mappers](#).

Context

You can enable the automatic user creation in OneSumX Regulatory Engine using the Identity Provider.

Procedure

1. Go to the installation folder of OneSumX Regulatory Engine and open the *install.properties* file.
2. Set property wolterskluwer.automatic.user.assignment to *true*.

   Note:
   After setting this property to "true", the user roles and resource groups will be in sync with the Identity Provider. This means that each time you add/remove a user from a role/resource group in OneSumX Regulatory Engine, this needs to be reflected in the Identity Provider.

3.1.3.5 Registering upload .cmd client

Context

This keystore will be generated automatically. You can follow the steps in this section to create the client keystore that will be able to connect to the application API.

Note:
If the client keystores are already configured, skip this step.

The generated file will be used to call the following script: *upload client.cmd*.

Procedure

1. Open Command Prompt or PowerShell.
2. Open *install.properties* and add the appropriate values to the following properties:
   - upload.keystore.file = path to keystore
   - upload.keystore.password = keystore password
   - upload.certificate.alias = certificate alias
3. Replace the placeholders with actual values used in *install.properties* for the following properties upload.keystore.file, upload.keystore.password, upload.certificate.alias, and then run the script:

   ```
   keytool -genkey -alias ${upload.certificate.alias} -keyalg RSA -keystore
   ${upload.keystore.file} -keysize 2048 -validity 360 -storepass
   ${upload.keystore.password}
   ```

4. Answer the suite of questions that follow.

   It's important to have the value of the first and last name the same as the value of the host of your application (wolterskluwer.application.host) from the installer properties. For the other settings, provide any values as appropriate.

   To reply to the question “Is ... correct?”, type `yes` and press **ENTER**.

```
C:\certificates>keytool -genkey -alias nre-https -keystore nre.keystore -storepass changeme -keyalg RSA -validity 360 -keysize 2048
What is your first and last name?
  [Unknown]:  localhost
What is the name of your organizational unit?
  [Unknown]:  bank
What is the name of your organization?
  [Unknown]:  bank
What is the name of your City or Locality?
  [Unknown]:  City
What is the name of your State or Province?
  [Unknown]:  State
What is the two-letter country code for this unit?
  [Unknown]:  EU
Is CN=localhost, OU=bank, O=bank, L=City, ST=State, C=EU correct?
  [no]:  yes

Enter key password for <nre-https>
        (RETURN if same as keystore password):
```

3.1.3.6 Installer properties

All the properties that are required for the installation are provided in *install.properties*.

Property	Example	Description
wolterskluwer.application.host	• ***Localhost*** • **server-01**	**URL address of the application. Make sure that underscore "_" is NOT used within the machine/alias name. The java.net.URI doesn't support underscores in host names.**
wolterskluwer.locally.accessible.host	• ***Localhost*** • **server-01** • **server-01.local**	**URL used for communication between the application's components. For example, when using load balancers, a different value than the one set in wolterskluwer.application.host is needed. The URL needs to be generated using the locally accessible domain name.**
wolterskluwer.application.port	**8080**	**Port of the application**
wolterskluwer.ssl.keystore.path	***path/to/keystore/application.keystore***	**Path to the JKS file that will be used for the HTTPS configuration**
wolterskluwer.ssl.keystore.alias	**server**	**Alias in the JKS file for the HTTPS configuration**
wolterskluwer.ssl.keystore.password	**password**	**Password from JKS for HTTPS**
wolterskluwer.java.cacerts.password	**changeit**	**Password for the Java *cacerts* file**
wolterskluwer.jre32.cacerts.password	**changeit**	**Password for JRE 32 bit*cacerts*. The path is configured through this property: rgtng.jre32.path.**

Property	Example	Description
wolterskluwer.db.connection.url	• *jdbc:oracle:thin:@//servername:1521/ORACLE*, or • *jdbc:sqlserver://servername:1433;instanceName=Sql2017;databaseName=SYSTEM_DB* • *jdbc:sqlserver://servername:1437;instanceName=Sql2017;databaseName=SYSTEM_DB*	**Required database URL of your application.** **For the SQL server, thedatabaseName property is required at the end of the URL.** **If an instance is used for the SQL server, use the 2nd variant.** **If a custom port is used for the SQL server, use the 3rd variant.**
wolterskluwer.application.db.username	**SYSTEM_DB**	**Database application user name**
wolterskluwer.application.db.password	**P@sSW0rD!**	**Database application password.** **Optionally, you can leave it as parameters when running the install command in the following format:** `install-platform.cmd -Dwolterskluwer.application.db.password=system` ***Note:*** **To ensure that the password is not readable after the installation, refer to section 5.2. from the OneSumX ReportGenerator Administration Guide.**
wolterskluwer.db.sqlserver.name	***localhost\\InstanceName*** ***localhost,1437***	**MSSQL service address. For remote it could be:** `<<MSSQL-HOST>>@<<InstanceName>>` . **In case of default instance, use the 2nd option.**
wolterskluwer.system.db.username	**system**	**A user who has permission to create a schema, users, and grant permissions to users.**
wolterskluwer.system.db.password	**system**	**The password for the user who has permission to create a schema, users, and grant permissions to users.**
wolterskluwer.oracle.tnsNames.folder	***c:/app/oracle/product/19.3.0/client_1/***	**Path to the folder with TNS names of the Oracle database client. This parameter is needed for the**

Property	Example	Description
	NETWORK/ADMIN/	configuration of RG TNG.
wolterskluwer.oracle.default.driver	Oracle in OraClient19Home1_32bit	This property will contain the default driver name for the Oracle 19 database server. This property should be updated in case the default driver name was changed.
wolterskluwer.sqlserver.default.driver	ODBC Driver 17 forSQL Server	This property will contain the default driver name for the SQL Server 2017 database server. This property should be updated in case the default driver name was changed.
wolterskluwer.sqlserver.integrated.security	false	Integrated security authentication for SQL Server
Property	Example	Description
wolterskluwer.application.host	*Localhost*	URL address of the application. Make sure that underscore "_" is NOT used within the machine/alias name. The java.net.URI doesn't support underscores in host names.
wolterskluwer.application.port	8080	Port of the application
wolterskluwer.ssl.keystore.path	*path/to/keystore/application.keystore*	Path to the JKS file that will be used for the HTTPS configuration
wolterskluwer.ssl.keystore.alias	server	Alias in the JKS file for the HTTPS configuration
wolterskluwer.ssl.keystore.password	password	Password from JKS for HTTPS
wolterskluwer.java.cacerts.password	changeit	Password for the Java *cacerts* file
wolterskluwer.jre32.cacerts.password	changeit	Password for JRE 32 bit*cacerts*. The path is configured through this property: rgtng.jre32.path.
wolterskluwer.db.connection.url	• *jdbc:oracle:thin:@//servername:1521/ORACLE*, or	Required database URL of your application. For the SQL server, thedatabaseName property is required at the end of the URL.

Property	Example	Description
	• *jdbc:sqlserver://servername:1433;instanceName=Sql2017;databaseName=SYSTEM_DB* • *jdbc:sqlserver://servername:1437;instanceName=Sql2017;databaseName=SYSTEM_DB*	If an instance is used for the SQL server, use the 2nd variant. If a custom port is used for the SQL server, use the 3rd variant.
woltersskluwer.application.db.username	SYSTEM_DB	Database application user name
woltersskluwer.application.db.password	P@sSW0rD!	Database application password. Optionally you can leave it as parameters when running install command in the following format: `install-platform.cmd -Dwoltersskluwer.application.db.password=system` ***Note:*** To ensure the password will not be readable after installation please refer to chapter 5.2. from the OneSumX ReportGenerator Administration Guide.
woltersskluwer.db.sqlserver.name	*localhost\\InstanceName* *localhost,1437*	MSSQL service address. For remote it could be: `<<MSSQL-HOST>>@<<InstanceName>>` . In case of default instance, use the 2nd option.
woltersskluwer.system.db.username	system	A user who has permission to create a schema, users, and grant permissions to users.
woltersskluwer.system.db.password	system	The password for the user who has permission to create a schema, users, and grant permissions to users.
woltersskluwer.oracle.tnsNames.folder	*c:/app/oracle/product/12.1.0/client_1/NETWORK/ADMIN/*	Path to folder with TNS names of Oracle database client. This parameter is needed for configuration of RG TNG.
woltersskluwer.oracle.defa	Oracle in	This property will contain default driver name for

Property	Example	Description
ult.driver	OraClient19Home1_32bit	Oracle 19 database server. This property should be updated in case that default driver name was changed.
wolterskluwer.sqlserver.default.driver	ODBC Driver 17 for SQL Server	This property will contain default driver name for SQL Server 2017 database server. This property should be updated in case that default driver name was changed.
wolterskluwer.sqlserver.integrated.security	false	Integrated security authentication for SQL Server
maestro.application.drillback.url	*localhost*	URL address of the Maestro application. For use in Lineage
maestro.application.drillback.database.name	MAESTRO_DB	Maestro database name. For use in Lineage
wolterskluwer.db.passphrase.encryption.cipher.algorithm	• AES • AES256	The algorithm used for encryption
wolterskluwer.db.passphrase.encryption.cipher.securityKey		The key to be used for encryptng the application database password (wolterskluwer.application.db.password property) in the *conf/init.xml* file. It is a required, not an empty parameter, and the value must be a string with the length of 128-bit (16 bytes or characters) for AES or 256-bit (32 bytes) for AES 256, encoded to the Base64 format. Refer to topic Application database password encryption.

3.1.3.6.1 Security part for AdminConsole application

Property	Example	Description
admin.keystore.file	*conf/security/RegulatoryPlatform.jks*	Path to JKS file used for SAML authentication. ***Note:*** If it doesn't exist, it will be generated by installer.

Property	Example	Description
admin.keystore.key	wkfs-saml-key	Certificate path in keystore. ***Note:*** If it doesn't exist, it will be generated by installer.
admin.keystore.password	changeme	Keystore password
admin.certificate.password	changeme	Certificate password
admin.serviceprovider.metadata	*conf/security/regulatory-platform-localhost-sp.xml*	Path to Service Provider metadata file. ***Note:*** If it doesn't exist, it will be generated by installer.
admin.identityprovider.metadata	*conf/security/client-tailored-saml-idp-metadata.xml*	Path to Identity Provider metadata file. Must be provided by the user.
admin.auth.entityId	wkfs-regapp-sp	Specifies ID referenced in URI and tokens
admin.defaultuser	Admin	Specifies ID of the predefined user with a System Administrator role ***Note:*** Property admin.defaultuser support capital letters only from version 4.7.0. Upgrade of version 3.1.0 should be performed using 4.7.0 distribution. Upgrades from version 3.0.0 to version 4.7.0 are not possible. Changing the user from one version to another is also not possible.

3.1.3.6.2 Security part for Reporting Hub application

Property	Example	Description
hub.keystore.file	*conf/security/RegulatoryPlatform.jks*	Path to JKS file used for SAML authentication.

Property	Example	Description
		Note: **If it doesn't exist, it will be generated by installer.**
hub.keystore.key	**wkfs-saml-key**	**Certificate path inside keystore.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
hub.keystore.password	**changeme**	**Keystore password**
hub.certificate.password	**changeme**	**Certificate password**
hub.serviceprovider.metadata	***conf/security/regulatory-platform-localhost-sp.xml***	**Path to Service Provider metadata file.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
hub.identityprovider.metadata	***conf/security/client-tailored-saml-idp-metadata.xml***	**Path to Identity Provider metadata file.** **Must be provided by user.**
hub.auth.entityId	**rh-regapp-sp**	**Specifies ID referenced in URI and tokens**

3.1.3.6.3 Security part for ReportGenerator application

Property	Example	Description
reportgenerator.keystore.file	***conf/security/rgtng_client.jks***	**Path to JKS file used for SAML authentication.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
reportgenerator.keystore.key	**rgtng_cert_alias**	**Certificate path inside keystore.**

Property	Example	Description
		Note: **If it doesn't exist, it will be generated by installer.**
reportgenerator.keystore.password	**12345678**	**Keystore password**
reportgenerator.certificate.password	**12345678**	**Certificate password**
reportgenerator.serviceprovider.metadata	***Reportgenerator-localhost-sp.xml***	**Path to Service Provider metadata file.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
reportgenerator.identityprovider.metadata	***conf/security/client-tailored-saml-idp-metadata.xml***	**Path to Identity Provider metadata file.** **Must be provided by user.**
reportgenerator.auth.entityId	**rgtng-client**	**Specifies ID referenced in URI and tokens**
hmac.secret	**welcome**	**This parameter is one of the parameters used to create a new Security Token.**
truststore.path	***conf/security/trust_store.jks***	**Path to the trust store that stores all the components certificates**
truststore.password	**changeme**	**Password to trust store**
upload.client.id	**upload-client**	**Client ID for *install-repository-client.cmd* that uploads reporting modules**
upload.keystore.file	***conf/security/upload_client.jks***	**Path to keystore for *install-repository-client.cmd***
upload.keystore.password	**changeme**	**Password to keystore for *install-repository-client.cmd***
upload.certificate.alias	**upload**	**Alias certificate name that will be uploaded in the trust store**

Property	Example	Description
rgtng.file.browser.root	*C:/WK*	Path where files trusted to be used are stored, its default value is equal to the value of workspaces.legacy.root.
workspaces.legacy.root	*C:/WK/Artifacts*	Path where reporting modules files will be placed after installation
workspaces.workdir.root	*C:/WK/SHAREDRWDIRECTORY*	Path where all artifact files generated in runtime will be stored
reportgenerator.default.user.role	TemplateRole	Reportgenerator default user role. This property is used in case of automatic user provisioning.
remove.sec.roles.links	True/False	In case this property is set to "True", when a user is removed from the Resource Group, the link between the user and the role from reportgenerator will be removed. In case the property is set to "False", the link remains.
rgtng.jre32.path	*C:/Program Files/Java/jre_32-1.8.0_191*	Path to 32 bit java Runtime Environment
install.upgrade.from.folder.with.previous.installation	*C:/path/to/previous/installation*	Property to be defined when the upgrade workflow will be used and you have a folder with previous installation
install.upgrade.from.zip.with.config	*C:/path/to/file/configuration.zip*	Property to be defined when the upgrade workflow will be used, and you have a ZIP file with configuration. This file is generated by the uninstall workflow.
install.upgrade.from.zip.with.content.store	*C:/path/to/file/content_store.zip*	Property to be defined when the upgrade workflow will be used and you have a ZIP file with content store. This file is generated by the uninstall workflow.

3.1.3.6.4 Security part for Lineage application

Property	Example	Description
lineage.keystore.file	*conf/security/Lineage.jks*	Path to the JKS file used for SAML authentication. If it doesn't exist, it will be generated by installer.
lineage.keystore.key	lineage-alias	Certificate path inside keystore. If it doesn't exist, it

Property	Example	Description
		will be generated by installer.
lineage.keystore.password	changeme	Keystore password
lineage.certificate.password	changeme	Certificate password
lineage.serviceprovider.metadata	*lineage-localhost-sp.xml*	Path to the Service Provider metadata file. If it doesn't exist, it will be generated by installer.
lineage.identityprovider.metadata	*conf/security/client-tailored-saml-idp-metadata.xml*	Path to Identity Provider metadata file. Must be provided by the user.
lineage.auth.entityId	lineage	Specifies ID referenced in URI and tokens

3.1.3.6.5 Default properties that can be redefined

Default properties that can be redefined by inserting them in *install.properties:*

Property	Example	Description
webAppFolder	*deploy/webapps*	Path to folder that contains WAR files to be deployed onto application server
outputConfigurationPath	*conf/wkfs-external.properties*	Path where properties file needed for application launch will be generated
processingEngineConfigurationPath	*conf/wkfs-processing-engine.properties*	Path where processing engine properties file will be generated
standalone.path	*conf/standalone.xml*	Path where *standalone.xml* file for standalone installation will be generated
yaml.path	*conf/config.yml*	Path where *config.yml* file for standalone installation will be generated
contentStoreFolder	*etc/contentstore/repository*	Folder where content store files will be stored

Property	Example	Description
admin-webapp.context	admin-webapp	Web context for admin console
reporting-webapp.context	reporting-webapp	Web context for reporting hub
rgtng-webapp.context	reportgenerator	Web context for ReportGenerator
lineage-webapp.context	lineage-webapp	Web context for lineage
rgtng.max.number.of.instances	30	Maximum number of Report Generator instances that can be active at a given time ***Warning:*** This property needs to be adapted based on the available resources of the machine where the engine is installed.
rgtng.file.browser.root	C:/WK/SharedRWDirectory	Can be redefined to any path including the path defined by the workspaces.legacy.root property. This parameter is needed for the configuration of RG TNG.
auth.adminGroup	Admin	Security group name for administrators
token.validity	PT8H	This parameter is one of the parameters used to create a new Security Token.
token.algorithm	HMAC	This parameter is one of the parameters used to create a new Security Token. Only HMAC is supported.
hmac.size	512	This parameter is one of the parameters used to create a new Security Token. The recommended value is 512, higher values are not supported.
wolterskluwer.datasource.jndi.prefix	java:/wolterskluwer/datasources	Prefix of JNDI name ***Warning:*** In case of the WebSphere application server, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*
wolterskluwer.datasource	SYSTEM_DB	Data source name for System database

Property	Example	Description
.jndi.datasource.name		
rgtng.install.location	***deploy/legacy/WK ReportGenerator***	**Folder where ReportGenerator exe should be deployed.** **If left empty, the default location will be: *deploy/legacy/WK ReportGenerator***
apikey.validity	**P1Y**	**Time of validity of the token for external systems.** **By default, 1 Year**
access-control.cache.poll-interval	**300**	**Time interval between refreshes of access control information in seconds.**
container.tmpDir	***%LOCALAPPDATA%/WoltersKluwer/temp_regulatory_engine***	**Path to the temporary folder used by the solution.**
processing.engine.memory.size	**1G, 512M, 4G, etc**	**Off-heap memory size used by in-memory processing.** **It is not present by default in *install.properties*, it's default value 8G.**
wkfs.saml.max.authentication.age	**7200, 28800, etc.**	**Configures the client saml maximum authentication age that defaults to 7200 secs(2 hrs). The value should be the same as the value for the session expiration in the IdP settings. For example, the ADFS server token is valid by default for 8 hrs.**

Note:

In addition, the configuration properties can also be passed on to the installer as command line parameters.

Note:

In case you encounter performance issues during specific processes (enrichment, transmission etc.), verify that property `com.wolterskluwer.reg.processing.system.offheap.memory.size` is set to an appropriate value (for example, 30G or higher if needed) in startup-swarm, for the embedded container.

Please perform any needed changes for this property for the external app server setup, if required.

3.1.4 Installing OneSumX Regulatory Engine

Prerequisites

Uninstall all Wolters Kluwer ReportGenerator applications that are installed on the machine where you are performing the installation.

- Run *appwiz.cpl* (Programs and Features) and then uninstall all ReportGenerator applications from the list.

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Go to the installation directory by typing
   `cd [installation drive:]\[RE installation folder]\bin`.
3. Generate the security key as described in topic Application database password encryption.
4. Execute *install-platform.cmd*.

Results

The progress will be shown in a dialog box.

5. Ensure that the installation is complete by checking whether the following files are present in the related (sub)folders:

Folder	Filename
conf	***config.yml***
	standalone.xml
	wkfs-external.properties
conf/security	***AdminConsole.jks***
	AdminConsole-localhost-sp.xml

Note:
The log files are found in the *[installation drive:]\[RE installation folder]/logs* folder.

Warning:
Ensure that you don't change the value of the admin.defaultuser property from *install.properties* after the *install-platform.cmd* script was run.

Warning:
If the publicly accessible domain name of the solution differs from the locally accessible domain name of the application server, then take the following action:

- In the *wkfs-external.properties* file, update the domain contained in the URLs set up for the following properties from the publicly accessible domain name to the locally accessible domain name:
  - wkfs.content-store.uri
  - wkfs.component.lineage.provider.processing-engine.uri
  - com.wolterskluwer.reg.reporting.uri
  - com.wolterskluwer.reg.reportgenerator.uri
  - com.wolterskluwer.reg.admin.uri
  - com.wolterskluwer.common.processing-engine.uri
  - wkfs.component.lineage.application.context.uri

6. For security reasons, the TLSv1.2 protocol was enforced. This can be checked from the *standalone.xml* that was just created. Under the `server-ssl-context` tag, `protocols="TLSv1.2"` should be present.

3.2 Installing the external app server package

The installation process of the external app server package consists of the following actions:

1. Downloading and extracting the external application server.
2. Downloading and extracting the OneSumX Regulatory Engine distribution file.
3. Configuring the installation properties.
4. Generating HTTPS keystore
5. Installing OneSumX Regulatory Engine
6. Configuring an Identity Provider for the SAML authentication
7. Configuring the application server

3.2.1 Downloading and extracting the app server

When you install the external app server package you need to first install and configure WebSphere Liberty or JBoss EAP before you run the installation process for OneSumX Regulatory Engine. For details, refer to the following sections:

- Configuring JBoss
- Configuring WebSphere Liberty

3.2.1.1 Download and extract WebSphere Liberty

Context

WebSphere Liberty is the application server that can run OneSumX Regulatory Engine in case of an external distribution type. For details, see Installation process.

Note:
Make sure that you fill in the installation properties according to your own setup.

Procedure

1. Go to the IBM WebSphere Liberty official download page and download WebSphere Liberty.

   Note:
   Version 4.7.0 of OneSumX Regulatory Engine only supports WebSphere Liberty application server version 20 or higher with Java EE 7.

2. Extract the downloaded archive content into a folder, for example, *C:\server\wlp*.

bin	9/15/2020 2:50 PM	File folder	
clients	9/15/2020 2:50 PM	File folder	
dev	9/15/2020 2:49 PM	File folder	
lafiles	1/27/2021 4:36 PM	File folder	
lib	1/27/2021 4:36 PM	File folder	
templates	9/15/2020 2:50 PM	File folder	
usr	9/15/2020 2:50 PM	File folder	
CHANGES.TXT	9/15/2020 1:49 PM	Text Document	2 KB
Copyright.txt	9/15/2020 1:16 PM	Text Document	1 KB
README.TXT	9/15/2020 1:49 PM	Text Document	13 KB

What to do next

After installing WebSphere Liberty, start it as described in topic Start WebSphere Liberty.

3.2.1.2 Download and extract JBoss EAP

Context

JBoss EAP is the application server that can run OneSumX Regulatory Engine.

Note:
Make sure that you fill in the installation properties according to your own setup.

Procedure

1. Go to the Red Hat Developer official download page and download JBoss EAP.

   Note:
   Version 4.7.0 of OneSumX Regulatory Engine only supports version 7.3.

2. Extract the downloaded archive content into a folder, for example *C:\server\jboss-eap-7.3*.

.installation	2/18/2020 9:43 AM	File folder	
.well-known	2/18/2020 9:43 AM	File folder	
appclient	2/18/2020 9:43 AM	File folder	
bin	3/10/2021 5:14 PM	File folder	
docs	2/18/2020 9:44 AM	File folder	
domain	2/18/2020 9:44 AM	File folder	
migration	2/18/2020 9:43 AM	File folder	
modules	3/10/2021 3:16 PM	File folder	
standalone	3/10/2021 12:08 PM	File folder	
welcome-content	2/18/2020 9:43 AM	File folder	
JBossEULA.txt	2/18/2020 9:43 AM	Text Document	1 KB
jboss-modules.jar	2/18/2020 9:43 AM	Executable Jar File	478 KB
LICENSE.txt	2/18/2020 9:43 AM	Text Document	26 KB
version.txt	2/18/2020 9:43 AM	Text Document	1 KB

3.2.2 Download and extract the distribution file

Prerequisites

- You have an Oracle or MSSQL instance running and accessible on the computer where you perform the installation.
- You have enough privileges to create other users.

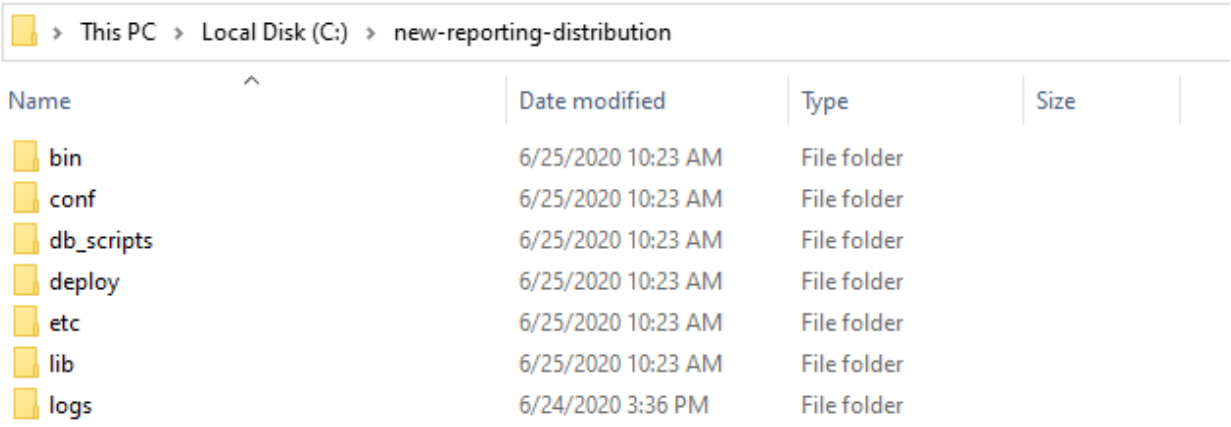
Procedure

1. Download the latest version of OneSumX Regulatory Engine for the external application server from the Product Download Center. For details, refer to topic Downloading files from the Product Download Center.
2. Extract the downloaded file to *[installation drive:]\[RE installation folder]*.

   The expected content structure is as follows:

   - The installation scripts are stored in the */bin* folder.
   - The */conf* directory contains the *install.properties* file that will be used during the installation.
   - During the installation, the local file repository will be created in directory */etc/contentstore/repository* and then used at runtime.

   Figure 1. Example of content for the unpacked distribution



Warning:
If you unpack and trigger the OneSumX Regulatory Engine installation from a very long path (more than 200 characters) you may face issues with certain functionalities of the legacy engine.

3.2.3 Download the Semansys package

Prerequisites

- You downloaded and extracted the distribution package for OneSumX Regulatory Engine.

Context

Starting from OneSumX Report Generator Web (TNG) version 2.24.0, the Semansys libraries are not distributed along with the engine.

These libraries are required if you need to perform a XBRL delivery.

Procedure

1. Go to the Product Download Center and download the following files: *RG_Semansys_artifacts* and *RG_Semansys_license*.
2. Extract them in the following location: *[installation drive:]\[RE installation folder]\deploy\legacy\WK ReportGenerator\bin*

3.2.4 Configuring the installation properties

You can edit the OneSumX Regulatory Engine installation properties in the *install.properties* file.

The following aspects are important:

- Avoid using spaces in any path definition. This may cause errors in specific situations.
- Avoid using backslashes in any path properties in the *install.properties* file, because this may cause errors. If it can't be avoided, you should use a double backslash (\\), otherwise the backslash will be lost in result value.

- To avoid losing any of the created artifacts and to be able to use the application after future upgrades, ensure that the wolterskluwer.ssl.keystore.path, workspaces.legacy.root, and workspaces.workdir.root properties are set to a path outside the application installation folder.
- For OneSumX Regulatory Engine, external app server package, for the WebSphere application server, property wolterskluwer.datasource.jndi.prefix is mandatory and must be set to the following value: `jdbc/wolterskluwer/datasources`.

3.2.4.1 Setting the communication properties

Context

Define the addresses that will be used to access the application in the installation properties file.

Procedure

1. Go to *[installation drive:]\[RE installation folder]\Conf* and open the *install.properties* file.
2. Modify the following properties:
   - wolterskluwer.application.host - define the host that will be used to access the application. Make sure that underscore "_" is NOT used within the machine/alias name. The java.net.URI doesn't support underscores in host names.
   - wolterskluwer.locally.accessible.host -
   - wolterskluwer.application.port - the port used by OneSumX Regulatory Engine can have values between 1 and 65535.
   - For HTTPS, provide additional properties and a certificate, as described in topic Configuring the HTTPS protocol.

3.2.4.2 Configuring the HTTPS protocol

OneSumX Regulatory Engine only supports the HTTPS protocol. You need a JAVA keystore with the certificate that will be used to encrypt HTTPS. If you don't have one, then it must be generated.

3.2.4.2.1 Generate the HTTPS certificate

Context

You can generate the keystore using the keytool utility, which is part of JRE.

Procedure

1. Open Command Prompt or PowerShell.
2. To run the keytool, add *%JAVA_HOME%/bin* to the PATH system variable of Windows.
3. Open *install.properties* and add the appropriate values to the following properties:
   - wolterskluwer.ssl.keystore.alias = certificate alias
   - wolterskluwer.ssl.keystore.path = path to keystore
   - wolterskluwer.ssl.keystore.password = keystore password

4. Replace the placeholders with actual values used in *install.properties* for the following properties *${wolterskluwer.ssl.keystore.alias}*, *${wolterskluwer.ssl.keystore.path}*, and *${wolterskluwer.ssl.keystore.password}* and then run the script:

```
keytool -genkey -alias ${wolterskluwer.ssl.keystore.alias} -keystore
${wolterskluwer.ssl.keystore.path}
-storepass ${wolterskluwer.ssl.keystore.password} -keyalg RSA -validity 360
-keysize 2048
```

5. Answer the suite of questions that follow.

   It's important to have the value of the first and last name the same as the value of the host of your application (wolterskluwer.application.host) from the installer properties. For the other settings, provide any values as appropriate.

   To reply to the question “Is ... correct?”, type `yes` and press **ENTER**.

```
C:\certificates>keytool -genkey -alias nre-https -keystore nre.keystore -storepass changeme -keyalg RSA -validity 360 -keysize 2048
What is your first and last name?
  [Unknown]:  localhost
What is the name of your organizational unit?
  [Unknown]:  bank
What is the name of your organization?
  [Unknown]:  bank
What is the name of your City or Locality?
  [Unknown]:  City
What is the name of your State or Province?
  [Unknown]:  State
What is the two-letter country code for this unit?
  [Unknown]:  EU
Is CN=localhost, OU=bank, O=bank, L=City, ST=State, C=EU correct?
  [no]:  yes

Enter key password for <nre-https>
        (RETURN if same as keystore password):
```

3.2.4.2.2 Generate a certificate signing request

Context

This is an optional step. Run these commands only when signed certificates are needed.

Procedure

1. Run the following command:

```
keytool -keystore ${wolterskluwer.ssl.keystore.path} -certreq -alias
${wolterskluwer.ssl.keystore.alias} -keyalg rsa -file signing_request.csr
```

Results

The signed certificate is generated.

2. The generated certificate can be imported in the keystore file and in *cacerts* using the import command:

```
keytool -import -alias ${wolterskluwer.ssl.keystore.alias} -file
{certificate.file.name} -keystore "${server.java.home}/jre/lib/security/
```

```
cacerts" -
storepass ${java.trustore.password} -noprompt
```

3.2.4.2.3 Importing HTTPS certificate into JRE cacerts manually

Context

This is an optional step. The installer will perform these operations, but in case you want to do it manually, follow this procedure.

Procedure

1. Extract the certificate from the created keystore, where *${certificate.file.name}* (for example, *secure_https.cer*) represents the path to the file that will be created and that will contain the certificate:

   ```
   keytool -export -alias ${wolterskluwer.ssl.keystore.alias} -keystore
   ${wolterskluwer.ssl.keystore.path} -rfc -file ${certificate.file.name} -
   storepass ${wolterskluwer.ssl.keystore.password}
   ```

   Note:
   If you already have a certificate with *${server.alias}* and you want to modify it, then you first have to delete it. See topic Delete the certificate from the keystore and cacerts file.

2. Import the certificate to the JAVA_HOME trust store created certificate.

   ```
   keytool -import -alias ${wolterskluwer.ssl.keystore.alias} -file
   {certificate.file.name} -keystore "%JAVA_HOME%/jre/lib/security/cacerts" -
   storepass ${java.trustore.password} -noprompt
   ```

   The default value of *${java.trust.store.password}* is `changeit`.

3.2.4.2.4 Delete the certificate from the keystore and cacerts file

Context

When you are importing a certificate manually into the keystore and/or cacerts file, you may see an error, such as "Certificate not imported, alias already exists". In that case, you may need to delete the certificate from the keystore and/or the cacerts file.

Procedure

1. Set the values for the following parameters according to the values used in topic Generate the HTTPS certificate:

   - ${wolterskluwer.ssl.keystore.alias}
   - ${java.trustore.password}
   - ${wolterskluwer.ssl.keystore.path}

- ${wolterskluwer.ssl.keystore.password}
- ${server.java.home}

2. Run the following commands:

```
keytool -delete -alias ${wolterskluwer.ssl.keystore.alias} -keystore
"${server.java.home}/jre/lib/security/cacerts" -storepass
${java.trustore.password}
```

```
keytool -delete -alias ${wolterskluwer.ssl.keystore.alias} -keystore
${wolterskluwer.ssl.keystore.path} -storepass
${wolterskluwer.ssl.keystore.password}
```

3.2.4.3 Creating the database schema

Before starting the installation process of OneSumX Regulatory Engine, it is necessary to prepare the database environment. The application supports the following types of databases:

- Oracle Server
- SQL Server

Note:
To avoid ambiguity with time zones, the machine where the application is installed and the database server must have the same time zone settings.

Starting with version 4.1.0, there are two supported authentication configurations for Microsoft SQL Server (further referred to as "SQL Server"):

- Setting the database properties for the username/password authentication on SQL Server
- Setting the database properties for the integrated authentication on SQL Server

There are three options to start using a database with OneSumX Regulatory Engine:

- Using an existing database
- Automatically creating a database
- Manually creating a database

3.2.4.3.1 Setting the database properties for the username/password authentication on SQL Server

Procedure

1. Go to *install.properties*.
2. Provide the appropriate values for the following properties:
   - wolterskluwer.system.db.username (schema manipulation user)
   - wolterskluwer.system.db.password
   - wolterskluwer.application.db.username (data manipulation user)

- wolterskluwer.application.db.password
- wolterskluwer.db.connection.url
- wolterskluwer.db.sqlserver.name - this

  property should concatenate the host name and the SQL Server instance name in case you want to use a named instance (for example, *localhost\SQLSERVER_CI*). For the default instance, it should contain the host name and the port (such as *localhost,1433*). This configuration should be provided from the UI during the creation of a workspace.
- wolterskluwer.sqlserver.default.driver
- wolterskluwer.datasource.jndi.prefix - for WebSphere Liberty, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*.

3. Create a privileged user (schema manipulation user) login: this must be declared at wolterskluwer.system.db.username, with password wolterskluwer.system.db.password. This is a user who has the rights needed to create a new schema, users, grant permissions to users.

   Warning:
   If the password contains symbol "\", issues may occur during an engine upgrade process. It is recomended to not use the "\" symbol in the wolterskluwer.system.db.password definition.

   The privileged user login can be created in several ways; one method is using the “sqlcmd” utility.

   The following script is an example of creating a privileged user login via sqlcmd, using the name "RE_ADMIN" and the password "P@ssw0rd", and granting all privileges:

```
CREATE LOGIN [RE_ADMIN] WITH PASSWORD='P@ssw0rd', DEFAULT_DATABASE=[master],
CHECK_EXPIRATION=OFF, CHECK_POLICY=ON
ALTER SERVER ROLE [bulkadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [dbcreator] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [diskadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [processadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [securityadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [serveradmin] ADD MEMBER [RE_ADMIN
ALTER SERVER ROLE [setupadmin] ADD MEMBER [RE_ADMIN]
ALTER SERVER ROLE [sysadmin] ADD MEMBER [RE_ADMIN]
```

4. Create a less privileged user (data manipulation user) login – this must be declared at wolterskluwer.application.db.username, with password wolterskluwer.application.db.password. This is a user who has rights to read and write data from tables. This user will be further used by the application to access the database.

   The less privileged user login can be created in several ways; one method is using the “sqlcmd” utility.

   The following script is an example of creating a less privileged user login, as an example with the name "RE_USER" and password "P@ssw0rd":

```
CREATE LOGIN [RE_USER] WITH PASSWORD='P@ssw0rd', DEFAULT_DATABASE=[master],
CHECK_EXPIRATION=OFF, CHECK_POLICY=ON
```

What to do next

Choose to proceed with one of the following options:

- The database will be created automatically for you - refer to topic [Automatic database creation using user/

password authentication.

- Create a database manually - refer to topic Manually create a database using username/password authentication.

3.2.4.3.1.1 Automatic database creation using user/password authentication

The database is created automatically during the OneSumX Regulatory Engine installation. In case you choose to manually create a database, refer to topic Manually create a database using username/password authentication.

The OneSumX Regulatory Engine installer uses system user credentials to automatically create the database and the user. The database name is obtained from wolterskluwer.db.connection.URL.

3.2.4.3.1.2 Manually create a database using username/password authentication

Prerequisites

Set the database properties for the username/password authentication. For details, refer to topic Setting the database properties for the username/password authentication on SQL Server.

Context

File *db_scripts/mssql/CREATE_SCHEMA.sql* contains a SQL script to create the database.

File *db_scripts/mssql/CREATE_USER_FOR_LOGIN.sql* contains a SQL script to create the user for the login and to grant permissions to read and write to the database.

Procedure

1. Copy the *CREATE_SCHEMA.sql* script to the database client executor.
2. Provide the actual values for the placeholders.
3. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

   ```
   CREATE DATABASE [${databaseName}];
   ```

   databaseName

   Name of the database where the tables are created by the reporting installer.

4. Copy the *CREATE_USER_FOR_LOGIN.sql* script to the database client executor.
5. Provide the actual values for the placeholders.
6. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

   ```
   use ${databaseName};
   go
   CREATE USER [${userName}] FOR LOGIN [${userName}];
   go
   -- granting required privileges
   EXEC sp_addrolemember 'db_datareader', [${userName}];
   ```

```
EXEC sp_addrolemember 'db_datawriter', [${userName}];
```

databaseName

Name of the database where the tables are created by the reporting installer.

userName

Name of the less privileged database user (that will be used by the application to access the database).

3.2.4.3.2 Setting the database properties for the integrated authentication on SQL Server

Procedure

1. Open *install.properties* and add values to the following properties:
   - wolterskluwer.sqlserver.integrated.security (by default it is set to false. To use integrated security, the property value needs to be set to true.)
   - wolterskluwer.db.connection.url
   - wolterskluwer.sqlserver.default.driver
   - wolterskluwer.db.sqlserver.name - this

     property should concatenate the host name and the SQL Server instance name in case you want to use a named instance (for example, *localhost\SQLSERVER_CI*). For the default instance, it should contain the host name and the port (such as *localhost,1433*). This configuration should be provided from the UI during the creation of a workspace.
   - wolterskluwer.datasource.jndi.prefix - for WebSphere Liberty, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*.
2. Remove the following properties, because they are not needed anymore:
   - wolterskluwer.system.db.username
   - wolterskluwer.system.db.password
   - wolterskluwer.application.db.username
   - wolterskluwer.application.db.password

What to do next

Choose to proceed with one of the following options:

- The database will be created automatically for you - refer to topic Automatic database creation with integrated security on SQL Server.
- Create a database manually - refer to topic Manually create a database with integrated security on SQL Server.

3.2.4.3.2.1 Automatic database creation with integrated security on SQL Server

The database is created automatically during the OneSumX Regulatory Engine installation. In case you choose to manually create a database, refer to topic Manually create a database with integrated security on SQL Server.

The OneSumX Regulatory Engine installer uses system user credentials to automatically create the database and the user. The database name is obtained from wolterskluwer.db.connection.URL.

3.2.4.3.2.2 Manually create a database with integrated security on SQL Server

Prerequisites

Set the database properties for the integrated security. For details, refer to topic Setting the database properties for the integrated authentication on SQL Server.

Context

File *db_scripts/mssql/CREATE_SCHEMA.sql* contains a SQL script to create the database.

Procedure

1. Copy this script into the database client executor.
2. Provide the actual values for the placeholders.
3. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

   ```
   CREATE DATABASE [${databaseName}];
   ```

   databaseName

   Name of the database where the tables are created by the reporting installer.

3.2.4.3.3 Setting the properties for the database creation on Oracle Server

Context

The installer uses system user credentials to create the user.

If a database already exists, then that database will be used during installation.

Procedure

1. Open *install.properties* and add values to the following properties:

   - wolterskluwer.db.connection.url
   - wolterskluwer.oracle.default.driver
   - wolterskluwer.oracle.tnsNames.folder -*tnsnames.ora* file location

- wolterskluwer.system.db.username
- wolterskluwer.system.db.password
- wolterskluwer.datasource.jndi.prefix - for OneSumX Regulatory Engine, external app server package, for WebSphere Liberty, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*.

2. Change the value of property wolterskluwer.db.connection.url according to the database connection (remote or local).
3. Ensure that property wolterskluwer.oracle.tnsNames.folder points to the folder of the Oracle client/server that contains *tnsnames.ora*.

   Note:
   If the Oracle server is on another machine, this path should point to the Oracle client location on the machine where OneSumX Regulatory Engine is being installed.

What to do next

Choose to proceed with one of the following options:

- The database will be created automatically for you - refer to topic Automatic database creation on Oracle Server.
- Create a database manually - refer to topic Manually create a database on Oracle Server.

3.2.4.3.3.1 Automatic database creation on Oracle Server

The database is created automatically during the OneSumX Regulatory Engine installation. In case you choose to manually create a database, refer to topic Manually create a database on Oracle Server.

The OneSumX Regulatory Engine installer uses system user credentials to automatically create the database and the user. The database name is obtained from wolterskluwer.db.connection.URL.

3.2.4.3.3.2 Manually create a database on Oracle Server

Prerequisites

Set the properties for the database creation. For details, refer to topic Setting the properties for the database creation on Oracle Server.

Procedure

1. Open file *db_scripts/oracle/CREATE_SCHEMA.sql*. This file contains a SQL script to create a database.
2. Copy this script into the database client executor.
3. Provide the actual values for the placeholders.
4. Execute the script.

   The following SQL code snippet illustrates the placeholder definitions:

```
CREATE USER ${userName} IDENTIFIED BY "${userPwd}" DEFAULT TABLESPACE users
QUOTA UNLIMITED ON users;
```

userName

name of the database user that will be used by the application to access the database.

userPwd

password of the database user.

3.2.4.4 Security configuration

To make the application secure, update the installation properties in the security section for the applications listed below.

- Admin application
- Reporting hub application
- ReportGenerator application
- Lineage application
- Application database password encryption

Note:
If you didn't generate the service provider files, they will be generated by the installer. The service provider files will be used to configure the SAML clients in the Identity Provider. Refer to topic Configuring the Identity Provider.

3.2.4.4.1 Admin application

admin.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided.

admin.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

admin.keystore.password

Password for the keystore.

admin.certificate.password

Password for the certificate that will be sent to the Identity Provider.

admin.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided.

admin.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider after installation and moved to that path.

admin.auth.entityId

Unique name of the application in the Identity Provider.

admin.defaultuser

The identifier of a default admin user with access to the Admin application. This user will be created automatically during installation.

3.2.4.4.2 Reporting hub application

hub.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided.

hub.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

hub.keystore.password

Password for the keystore.

hub.certificate.password

Password for the certificate that will be sent to the Identity Provider.

hub.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided.

hub.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider after installation and moved to that path.

hub.auth.entityId

Unique name of application in the Identity Provider.

3.2.4.4.3 ReportGenerator application

reportgenerator.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided by the IT Admin.

reportgenerator.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

reportgenerator.keystore.password

Password for the keystore.

reportgenerator.certificate.password

Password for the certificate that will be sent to the Identity Provider.

reportgenerator.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided by IT Admin.

reportgenerator.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider after installation and moved to that path.

reportgenerator.auth.entityId

Unique name of application in the Identity Provider.

3.2.4.4.4 Lineage application

lineage.keystore.file

Keystore file name that will be generated for communication with the Identity Provider when not provided.

lineage.keystore.key

Alias of the key in the keystore that will be used for communication with the Identity Provider.

lineage.keystore.password

Password for the keystore.

lineage.certificate.password

Password for the certificate that will be sent to the Identity Provider.

lineage.serviceprovider.metadata

Path to the file that will contain service provider metadata. This will be generated by the installer when not provided.

lineage.identityprovider.metadata

Path to the file that contains Identity Provider metadata. Should be extracted from the Identity Provider after installation and moved to that path.

lineage.auth.entityId

Unique name of the application in the Identity Provider.

3.2.4.4.5 Application database password encryption

Context

Starting with OneSumX Report Generator Web (TNG) version 2.24.0, the value of property wolterskluwer.application.db.password from the installation properties file needs to be encrypted in the connection.password property, in the *conf/init.xml* file.

The password will be encrypted during the installation or upgrade process, using the following two properties from the *install.properties* file:

- wolterskluwer.db.passphrase.encryption.cipher.algorithm, the default value is AES
- wolterskluwer.db.passphrase.encryption.cipher.securityKey, empty

Follow the procedure to generate the security key.

Procedure

1. Choose a 16 characters long string as a security key.

   ### Example

   16CharSecureKey!

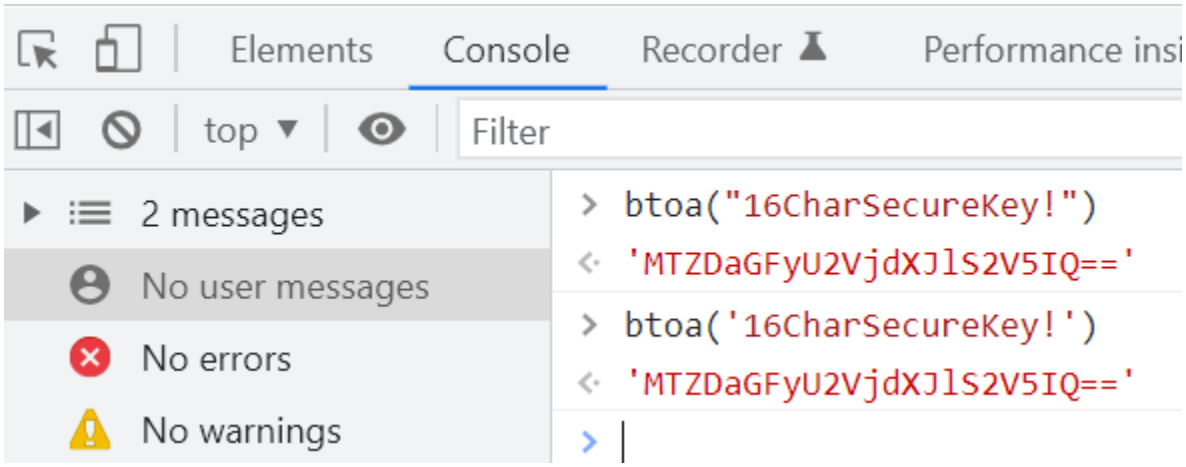
2. Encode it to the Base64 format using any online tool or software that provides this encoding capability.

   ### Example

   a. Open the developer tools (press key F12) in the browser and click the **Console** tab.

   b. Use the `btoa` JavaScript function with your chosen string as parameter
   `btoa('16CharSecureKey!')`

   c. Copy the output `MTZDaGFyU2VjdXJlS2V5IQ==`



3. Save the Base64 encoded value into installation property wolterskluwer.db.passphrase.encryption.cipher.securityKey or provide it in the command line during the upgrade or installation (for example
   `-Dwolterskluwer.db.passphrase.encryption.cipher.securityKey=MTZDaGFyU2VjdXJlS2V5IQ==`
).

3.2.4.5 Setting additional installer properties

Procedure

1. Set the property pointing to the location of the 32 bit Java:

```
rgtng.jre32.path=C:/tools/jre_32-1.8.0_191
```

2. Optionally, change the following properties, which will be used by the installer to create keystores for the trust store and the upload client, described in topic Registering upload .cmd client:

```
hmac.secret=welcome
truststore.path=conf/security/trust_store.keystore
truststore.password=changeit
upload.client.id=client
upload.keystore.file=conf/security/client.keystore
upload.keystore.password=changeit
upload.certificate.alias=client
```

3. Specify the path to the directories that will store the reporting and business files:

```
rgtng.file.browser.root=C:/reporting/
workspaces.legacy.root=C:/reporting/artifacts
workspaces.workdir.root=C:/reporting/shared
```

3.2.4.6 Adding users to OneSumX Regulatory Engine from the Identity Provider

Prerequisites

- Ensure that the Identity Provider mappers are configured correctly as described in topic Configure SAML mappers.

Context

You can enable the automatic user creation in OneSumX Regulatory Engine using the Identity Provider.

Procedure

1. Go to the installation folder of OneSumX Regulatory Engine and open the *install.properties* file.
2. Set property wolterskluwer.automatic.user.assignment to *true*.

Note:
After setting this property to "true", the user roles and resource groups will be in sync with the Identity Provider. This means that each time you add/remove a user from a role/resource group in OneSumX Regulatory Engine, this needs to be reflected in the Identity Provider.

3.2.4.7 Registering upload .cmd client

Context

This keystore will be generated automatically. You can follow the steps in this section to create the client keystore that will be able to connect to the application API.

Note:
If the client keystores are already configured, skip this step.

The generated file will be used to call the following script: *upload client.cmd*.

Procedure

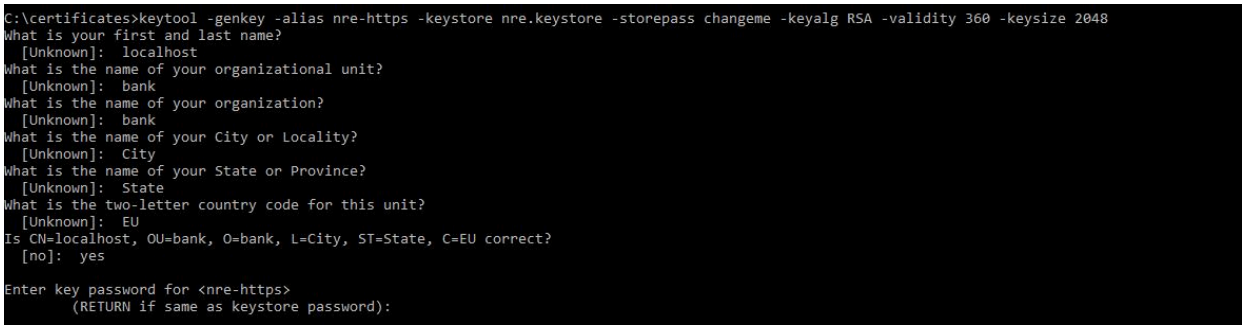
1. Open Command Prompt or PowerShell.
2. Open *install.properties* and add the appropriate values to the following properties:
   - upload.keystore.file = path to keystore
   - upload.keystore.password = keystore password
   - upload.certificate.alias = certificate alias
3. Replace the placeholders with actual values used in *install.properties* for the following properties upload.keystore.file, upload.keystore.password, upload.certificate.alias, and then run the script:

   ```
   keytool -genkey -alias ${upload.certificate.alias} -keyalg RSA -keystore
   ${upload.keystore.file} -keysize 2048 -validity 360 -storepass
   ${upload.keystore.password}
   ```

4. Answer the suite of questions that follow.

   It's important to have the value of the first and last name the same as the value of the host of your application (wolterskluwer.application.host) from the installer properties. For the other settings, provide any values as appropriate.

   To reply to the question “Is ... correct?”, type `yes` and press **ENTER**.



```
C:\certificates>keytool -genkey -alias nre-https -keystore nre.keystore -storepass changeme -keyalg RSA -validity 360 -keysize 2048
What is your first and last name?
  [Unknown]:  localhost
What is the name of your organizational unit?
  [Unknown]:  bank
What is the name of your organization?
  [Unknown]:  bank
What is the name of your City or Locality?
  [Unknown]:  City
What is the name of your State or Province?
  [Unknown]:  State
What is the two-letter country code for this unit?
  [Unknown]:  EU
Is CN=localhost, OU=bank, O=bank, L=City, ST=State, C=EU correct?
  [no]:  yes

Enter key password for <nre-https>
        (RETURN if same as keystore password):
```

3.2.4.8 Installer properties

All the properties that are required for the installation are provided in *install.properties*.

Property	Example	Description
wolterskluwer.applicatio	• ***Localhost***	**URL address of the application. Make sure that**

Property	Example	Description
n.host	• server-01	underscore "_" is NOT used within the machine/alias name. The java.net.URI doesn't support underscores in host names.
wolterskluwer.locally.accessible.host	• *Localhost* • server-01 • server-01.local	URL used for communication between the application's components. For example, when using load balancers, a different value than the one set in wolterskluwer.application.host is needed. The URL needs to be generated using the locally accessible domain name.
wolterskluwer.application.port	8080	Port of the application
wolterskluwer.ssl.keystore.path	*path/to/keystore/application.keystore*	Path to the JKS file that will be used for the HTTPS configuration
wolterskluwer.ssl.keystore.alias	server	Alias in the JKS file for the HTTPS configuration
wolterskluwer.ssl.keystore.password	password	Password from JKS for HTTPS
wolterskluwer.java.cacerts.password	changeit	Password for the Java *cacerts* file
wolterskluwer.jre32.cacerts.password	changeit	Password for JRE 32 bit*cacerts*. The path is configured through this property: rgtng.jre32.path.
wolterskluwer.db.connection.url	• *jdbc:oracle:thin:@//servername:1521/ORACLE*, or • *jdbc:sqlserver://servername:1433;instanceName=Sql2017;databaseName=SYSTEM_DB* • *jdbc:sqlserver://servername:1437;instanceName=Sql2017;databaseName=SYSTEM_DB*	Required database URL of your application. For the SQL server, thedatabaseName property is required at the end of the URL. If an instance is used for the SQL server, use the 2nd variant. If a custom port is used for the SQL server, use the 3rd variant.
wolterskluwer.application.db.username	SYSTEM_DB	Database application user name

Property	Example	Description
wolterskluwer.application.db.password	**P@sSW0rD!**	**Database application password.** **Optionally, you can leave it as parameters when running the install command in the following format:** `install-platform.cmd -Dwolterskluwer.application.db.password=system` ***Note:*** **To ensure that the password is not readable after the installation, refer to section 5.2. from the OneSumX ReportGenerator Administration Guide.**
wolterskluwer.db.sqlserver.name	***localhost\\InstanceName*** ***localhost,1437***	**MSSQL service address. For remote it could be:** `<<MSSQL-HOST>>@<<InstanceName>>` . **In case of default instance, use the 2nd option.**
wolterskluwer.system.db.username	**system**	**A user who has permission to create a schema, users, and grant permissions to users.**
wolterskluwer.system.db.password	**system**	**The password for the user who has permission to create a schema, users, and grant permissions to users.**
wolterskluwer.oracle.tnsNames.folder	***c:/app/oracle/product/19.3.0/client_1/NETWORK/ADMIN/***	**Path to the folder with TNS names of the Oracle database client. This parameter is needed for the configuration of RG TNG.**
wolterskluwer.oracle.default.driver	**Oracle in OraClient19Home1_32bit**	**This property will contain the default driver name for the Oracle 19 database server. This property should be updated in case the default driver name was changed.**
wolterskluwer.sqlserver.default.driver	**ODBC Driver 17 forSQL Server**	**This property will contain the default driver name for the SQL Server 2017 database server. This property should be updated in case the default driver name was changed.**
wolterskluwer.sqlserver.integrated.security	**false**	**Integrated security authentication for SQL Server**

Property	Example	Description
Property	**Example**	**Description**
wolterskluwer.application.host	***Localhost***	**URL address of the application. Make sure that underscore "_" is NOT used within the machine/alias name. The java.net.URI doesn't support underscores in host names.**
wolterskluwer.application.port	**8080**	**Port of the application**
wolterskluwer.ssl.keystore.path	***path/to/keystore/application.keystore***	**Path to the JKS file that will be used for the HTTPS configuration**
wolterskluwer.ssl.keystore.alias	**server**	**Alias in the JKS file for the HTTPS configuration**
wolterskluwer.ssl.keystore.password	**password**	**Password from JKS for HTTPS**
wolterskluwer.java.cacerts.password	**changeit**	**Password for the Java *cacerts* file**
wolterskluwer.jre32.cacerts.password	**changeit**	**Password for JRE 32 bit*cacerts*. The path is configured through this property: rgtng.jre32.path.**
wolterskluwer.db.connection.url	• ***jdbc:oracle:thin:@//servername:1521/ORACLE*, or** • ***jdbc:sqlserver://servername:1433;instanceName=Sql2017;databaseName=SYSTEM_DB*** • ***jdbc:sqlserver://servername:1437;instanceName=Sql2017;databaseName=SYSTEM_DB***	**Required database URL of your application.** **For the SQL server, thedatabaseName property is required at the end of the URL.** **If an instance is used for the SQL server, use the 2nd variant.** **If a custom port is used for the SQL server, use the 3rd variant.**
wolterskluwer.application.db.username	**SYSTEM_DB**	**Database application user name**
wolterskluwer.application.db.password	**P@sSW0rD!**	**Database application password.** **Optionally you can leave it as parameters when running install command in the following format:**

Property	Example	Description
		`install-platform.cmd -Dwolterskluwer.application.db.password=system` ***Note:*** **To ensure the password will not be readable after installation please refer to chapter 5.2. from the OneSumX ReportGenerator Administration Guide.**
wolterskluwer.db.sqlserver.name	***localhost\\InstanceName*** ***localhost,1437***	**MSSQL service address. For remote it could be:** `<<MSSQL-HOST>>@<<InstanceName>>` . **In case of default instance, use the 2nd option.**
wolterskluwer.system.db.username	**system**	**A user who has permission to create a schema, users, and grant permissions to users.**
wolterskluwer.system.db.password	**system**	**The password for the user who has permission to create a schema, users, and grant permissions to users.**
wolterskluwer.oracle.tnsNames.folder	***c:/app/oracle/product/12.1.0/client_1/NETWORK/ADMIN/***	**Path to folder with TNS names of Oracle database client. This parameter is needed for configuration of RG TNG.**
wolterskluwer.oracle.default.driver	**Oracle in OraClient19Home1_32bit**	**This property will contain default driver name for Oracle 19 database server. This property should be updated in case that default driver name was changed.**
wolterskluwer.sqlserver.default.driver	**ODBC Driver 17 for SQL Server**	**This property will contain default driver name for SQL Server 2017 database server. This property should be updated in case that default driver name was changed.**
wolterskluwer.sqlserver.integrated.security	**false**	**Integrated security authentication for SQL Server**
maestro.application.drillback.url	***localhost***	**URL address of the Maestro application. For use in Lineage**
maestro.application.drill	**MAESTRO_DB**	**Maestro database name. For use in Lineage**

Property	Example	Description
back.database.name		
wolterskluwer.db.passphrase.encryption.cipher.algorithm	• **AES** • **AES256**	**The algorithm used for encryption**
wolterskluwer.db.passphrase.encryption.cipher.securityKey		**The key to be used for encryptng the application database password (wolterskluwer.application.db.password property) in the *conf/init.xml* file. It is a required, not an empty parameter, and the value must be a string with the length of 128-bit (16 bytes or characters) for AES or 256-bit (32 bytes) for AES 256, encoded to the Base64 format.** **Refer to topic Application database password encryption.**

3.2.4.8.1 Security part for AdminConsole application

Property	Example	Description
admin.keystore.file	***conf/security/RegulatoryPlatform.jks***	**Path to JKS file used for SAML authentication.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
admin.keystore.key	**wkfs-saml-key**	**Certificate path in keystore.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
admin.keystore.password	**changeme**	**Keystore password**
admin.certificate.password	**changeme**	**Certificate password**
admin.serviceprovider.metadata	***conf/security/regulatory-platform-localhost-sp.xml***	**Path to Service Provider metadata file.**

Property	Example	Description
		Note: **If it doesn't exist, it will be generated by installer.**
admin.identityprovider.metadata	***conf/security/client-tailored-saml-idp-metadata.xml***	**Path to Identity Provider metadata file.** **Must be provided by the user.**
admin.auth.entityId	**wkfs-regapp-sp**	**Specifies ID referenced in URI and tokens**
admin.defaultuser	**Admin**	**Specifies ID of the predefined user with a System Administrator role** ***Note:*** **Property admin.defaultuser support capital letters only from version 4.7.0. Upgrade of version 3.1.0 should be performed using 4.7.0 distribution. Upgrades from version 3.0.0 to version 4.7.0 are not possible.** **Changing the user from one version to another is also not possible.**

3.2.4.8.2 Security part for Reporting Hub application

Property	Example	Description
hub.keystore.file	***conf/security/RegulatoryPlatform.jks***	**Path to JKS file used for SAML authentication.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
hub.keystore.key	**wkfs-saml-key**	**Certificate path inside keystore.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
hub.keystore.password	**changeme**	**Keystore password**
hub.certificate.password	**changeme**	**Certificate password**

Property	Example	Description
hub.serviceprovider.metadata	***conf/security/regulatory-platform-localhost-sp.xml***	**Path to Service Provider metadata file.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
hub.identityprovider.metadata	***conf/security/client-tailored-saml-idp-metadata.xml***	**Path to Identity Provider metadata file.** **Must be provided by user.**
hub.auth.entityId	**rh-regapp-sp**	**Specifies ID referenced in URI and tokens**

3.2.4.8.3 Security part for ReportGenerator application

Property	Example	Description
reportgenerator.keystore.file	***conf/security/rgtng_client.jks***	**Path to JKS file used for SAML authentication.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
reportgenerator.keystore.key	**rgtng_cert_alias**	**Certificate path inside keystore.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
reportgenerator.keystore.password	**12345678**	**Keystore password**
reportgenerator.certificate.password	**12345678**	**Certificate password**
reportgenerator.serviceprovider.metadata	***Reportgenerator-localhost-sp.xml***	**Path to Service Provider metadata file.** ***Note:*** **If it doesn't exist, it will be generated by installer.**
reportgenerator.identityp	***conf/security/client-***	**Path to Identity Provider metadata file.**

Property	Example	Description
rovider.metadata	*tailored-saml-idp-metadata.xml*	Must be provided by user.
reportgenerator.auth.entityId	rgtng-client	Specifies ID referenced in URI and tokens
hmac.secret	welcome	This parameter is one of the parameters used to create a new Security Token.
truststore.path	*conf/security/trust_store.jks*	Path to the trust store that stores all the components certificates
truststore.password	changeme	Password to trust store
upload.client.id	upload-client	Client ID for *install-repository-client.cmd* that uploads reporting modules
upload.keystore.file	*conf/security/upload_client.jks*	Path to keystore for *install-repository-client.cmd*
upload.keystore.password	changeme	Password to keystore for *install-repository-client.cmd*
upload.certificate.alias	upload	Alias certificate name that will be uploaded in the trust store
rgtng.file.browser.root	*C:/WK*	Path where files trusted to be used are stored, its default value is equal to the value of workspaces.legacy.root.
workspaces.legacy.root	*C:/WK/Artifacts*	Path where reporting modules files will be placed after installation
workspaces.workdir.root	*C:/WK/SHAREDRWDIRECTORY*	Path where all artifact files generated in runtime will be stored
reportgenerator.default.user.role	TemplateRole	Reportgenerator default user role. This property is used in case of automatic user provisioning.
remove.sec.roles.links	True/False	In case this property is set to "True", when a user is removed from the Resource Group, the link between the user and the role from reportgenerator will be removed. In case the property is set to "False", the

Property	Example	Description
		link remains.
rgtng.jre32.path	*C:/Program Files/Java/jre_32-1.8.0_191*	Path to 32 bit java Runtime Environment
install.upgrade.from.folder.with.previous.installation	*C:/path/to/previous/installation*	Property to be defined when the upgrade workflow will be used and you have a folder with previous installation
install.upgrade.from.zip.with.config	*C:/path/to/file/configuration.zip*	Property to be defined when the upgrade workflow will be used, and you have a ZIP file with configuration. This file is generated by the uninstall workflow.
install.upgrade.from.zip.with.content.store	*C:/path/to/file/content_store.zip*	Property to be defined when the upgrade workflow will be used and you have a ZIP file with content store. This file is generated by the uninstall workflow.

3.2.4.8.4 Security part for Lineage application

Property	Example	Description
lineage.keystore.file	*conf/security/Lineage.jks*	Path to the JKS file used for SAML authentication. If it doesn't exist, it will be generated by installer.
lineage.keystore.key	lineage-alias	Certificate path inside keystore. If it doesn't exist, it will be generated by installer.
lineage.keystore.password	changeme	Keystore password
lineage.certificate.password	changeme	Certificate password
lineage.serviceprovider.metadata	*lineage-localhost-sp.xml*	Path to the Service Provider metadata file. If it doesn't exist, it will be generated by installer.
lineage.identityprovider.metadata	*conf/security/client-tailored-saml-idp-metadata.xml*	Path to Identity Provider metadata file. Must be provided by the user.
lineage.auth.entityId	lineage	Specifies ID referenced in URI and tokens

3.2.4.8.5 Default properties that can be redefined

Default properties that can be redefined by inserting them in *install.properties:*

Property	Example	Description
webAppFolder	***deploy/webapps***	**Path to folder that contains WAR files to be deployed onto application server**
outputConfigurationPath	***conf/wkfs-external.properties***	**Path where properties file needed for application launch will be generated**
processingEngineConfigurationPath	***conf/wkfs-processing-engine.properties***	**Path where processing engine properties file will be generated**
standalone.path	***conf/standalone.xml***	**Path where *standalone.xml* file for standalone installation will be generated**
yaml.path	***conf/config.yml***	**Path where *config.yml* file for standalone installation will be generated**
contentStoreFolder	***etc/contentstore/repository***	**Folder where content store files will be stored**
admin-webapp.context	**admin-webapp**	**Web context for admin console**
reporting-webapp.context	**reporting-webapp**	**Web context for reporting hub**
rgtng-webapp.context	**reportgenerator**	**Web context for ReportGenerator**
lineage-webapp.context	**lineage-webapp**	**Web context for lineage**
rgtng.max.number.of.instances	**30**	**Maximum number of Report Generator instances that can be active at a given time** ***Warning:*** **This property needs to be adapted based on the available resources of the machine where the engine is installed.**
rgtng.file.browser.root	**C:/WK/SharedRWDirectory**	**Can be redefined to any path including the path defined by the workspaces.legacy.root property.** **This parameter is needed for the configuration of RG**

Property	Example	Description
		TNG.
auth.adminGroup	Admin	Security group name for administrators
token.validity	PT8H	This parameter is one of the parameters used to create a new Security Token.
token.algorithm	HMAC	This parameter is one of the parameters used to create a new Security Token. Only HMAC is supported.
hmac.size	512	This parameter is one of the parameters used to create a new Security Token. The recommended value is 512, higher values are not supported.
wolterskluwer.datasource .jndi.prefix	java:/wolterskluwer/ datasources	Prefix of JNDI name ***Warning:*** In case of the WebSphere application server, this property is mandatory and must be set to the value *jdbc/wolterskluwer/datasources*
wolterskluwer.datasource .jndi.datasource.name	SYSTEM_DB	Data source name for System database
rgtng.install.location	*deploy/legacy/WK ReportGenerator*	Folder where ReportGenerator exe should be deployed. If left empty, the default location will be: *deploy/ legacy/WK ReportGenerator*
apikey.validity	P1Y	Time of validity of the token for external systems. By default, 1 Year
access-control.cache.poll-interval	300	Time interval between refreshes of access control information in seconds.
container.tmpDir	*%LOCALAPPDATA%/Wolte rsKluwer/ temp_regulatory_engine*	Path to the temporary folder used by the solution.
processing.engine.memor y.size	1G, 512M, 4G, etc	Off-heap memory size used by in-memory processing.

Property	Example	Description
		It is not present by default in *install.properties*, it's default value 8G.
wkfs.saml.max.authentication.age	**7200, 28800, etc.**	**Configures the client saml maximum authentication age that defaults to 7200 secs(2 hrs). The value should be the same as the value for the session expiration in the IdP settings. For example, the ADFS server token is valid by default for 8 hrs.**

Note:
In addition, the configuration properties can also be passed on to the installer as command line parameters.

Note:
In case you encounter performance issues during specific processes (enrichment, transmission etc.), verify that property `com.wolterskluwer.reg.processing.system.offheap.memory.size` is set to an appropriate value (for example, 30G or higher if needed) in startup-swarm, for the embedded container.

Please perform any needed changes for this property for the external app server setup, if required.

3.2.5 Installing OneSumX Regulatory Engine

Prerequisites

Uninstall all Wolters Kluwer ReportGenerator applications that are installed on the machine where you are performing the installation.

- Run *appwiz.cpl* (Programs and Features) and then uninstall all ReportGenerator applications from the list.

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Go to the installation directory by typing
   `cd [installation drive:]\[RE installation folder]\bin`.
3. Generate the security key as described in topic Application database password encryption.
4. Execute *install-platform.cmd*.

Results

The progress will be shown in a dialog box.

5. Ensure that the installation is complete by checking whether the following files are present in the related (sub)folders:

Folder	Filename
conf	***wkfs-external.properties***
	wkfs-processing-engine.properties
conf/security	***AdminConsole.jks***
	AdminConsole-localhost-sp.xml

Note:
The log files are found in the */logs* folder.

For WebSphere Liberty, logs are located at: *C:\server\wlp/usr/servers/defaultServer/logs*.

For JBoss EAP, logs are located at: *C:\server\jboss-eap-7.3/logs*.

Warning:
Ensure that you don't change the value of the admin.defaultuser property from *install.properties* after the *install-platform.cmd* script was run.

Warning:
If the publicly accessible domain name of the solution differs from the locally accessible domain name of the application server, then take the following action:

- In the *wkfs-external.properties* file, update the domain contained in the URLs set up for the following properties from the publicly accessible domain name to the locally accessible domain name:
  - wkfs.content-store.uri
  - wkfs.component.lineage.provider.processing-engine.uri
  - com.wolterskluwer.reg.reporting.uri
  - com.wolterskluwer.reg.reportgenerator.uri
  - com.wolterskluwer.reg.admin.uri
  - com.wolterskluwer.common.processing-engine.uri
  - wkfs.component.lineage.application.context.uri

3.2.5.1 Time synchronization

If OneSumX Regulatory Engine and the Identity Provider server are running on different computers, then the time settings of the computers must be synchronized. If the time settings are not synchronized, and they differ by more than 60 seconds, then the authentication process may fail. For details, refer to topic Authentication process fails if OneSumX Regulatory Engine and the Identity Provider server are not synced.

3.2.6 Configuring the application server

Depending on your implementation, choose one of the following options:

- Configuring JBoss
- Configuring WebSphere Liberty

3.2.6.1 Configuring JBoss

1. Copy deployments (WAR) files from */deploy/drivers* to the application server.
2. Copy the JDBC drivers from the */deploy/drivers* folder to the application server.
3. Configure the Java options and parameters
4. Enable HTTPS by providing the path and credentials to the keystore described in topic Configuring the HTTPS protocol.
5. Define the datasource in the application server configuration for the system database generated by the installation script.

3.2.6.1.1 Copy deployment files for JBoss EAP

Context

This task contains the steps for copying the deployment files from the RE distribution into the JBoss EAP folder.

Procedure

1. Copy the WAR files from *[installation drive:]\[RE installation folder]\deploy\webapps*.
2. Paste the copied files to *C:\server\jboss-eap-7.3\standalone\deployments*.

3.2.6.1.2 Copying the JDBC drivers

Depending on your implementation, choose one of the following options:

- Copy JDBC drivers for Oracle
- Copy JDBC drivers for SQL Server

3.2.6.1.2.1 Copy JDBC drivers for Oracle

Procedure

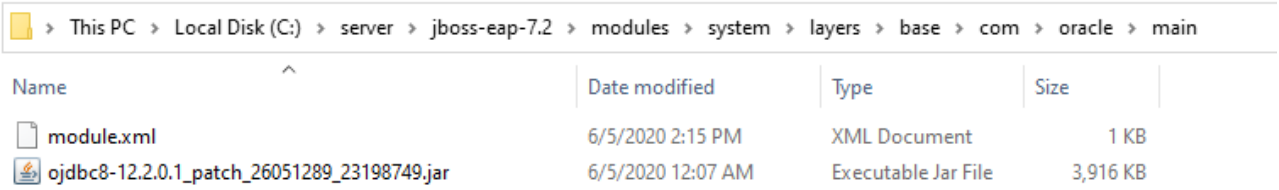
1. Go to *C:\server\jboss-eap-7.3\modules\system\layers\base\com\oracle* and create the *main* folder.
2. Go to folder *C:\new-reporting-distribution\deploy\drivers* and copy the *ojdbc8-12.2.0.1_patch_26051289_23198749.jar* driver into the newly created folder.
3. Create *module.xml* at *C:\server\jboss-eap-7.3\modules\system\layers\base\com\oracle\main* with the following content:

```
<module xmlns="urn:jboss:module:1.1" name="com.oracle">
```

```
  <resources>
    <resource-root path="ojdbc8-12.2.0.1_patch_26051289_23198749.jar"/>
  </resources>
  <dependencies>
    <module name="javax.api"/>
    <module name="javax.transaction.api"/>
  </dependencies>
</module>
```

Results

Figure 1. Oracle driver for JBoss EAP



3.2.6.1.2.2 Copy JDBC drivers for SQL Server

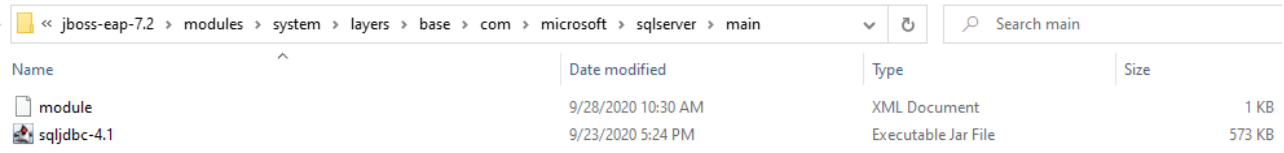
Procedure

1. Go to *C:\server\jboss-eap-7.3\modules\system\layers\base\com\microsoft\sqlserver* and create the *main* folder.
2. Go to folder *C:\new-reporting-distribution\deploy\drivers* and copy the *mssql-jdbc-9.4.1.jre8.jar* driver into the newly created folder.
3. Create *module.xml* at *C:\server\jboss-eap-7.3\modules\system\layers\base\com\microsoft\sqlserver\main* with the following content:

```
<module xmlns="urn:jboss:module:1.3" name="com.microsoft.sqlserver">
     <resources>
          <resource-root path="mssql-jdbc-9.4.1.jre8.jar"/>
     </resources>
     <dependencies>
          <module name="javax.api"/>
          <module name="javax.transaction.api"/>
     </dependencies>
</module>
```

Results

Figure 1. SQL Server driver for JBoss EAP



3.2.6.1.3 Configure the Java options and parameters

Procedure

1. Go to *C:\server\jboss-eap-7.3\standalone\configuration\standalone.xml*.
2. Add system properties after the 'extensions' element:

```
<system-properties>
 <property name="wkfs.commons.configuration.file.local" value="C:\new-
reporting-distribution\conf\wkfs-external.properties"/>
 <property name="file.encoding" value="UTF-8"/>
 <property name="reportgenerator.init.file.location" value="C:\newreporting-
distribution\conf\init.xml"/>
 <property name="reportgenerator.initAuthentication.file.location"
value="C:\new-reporting-distribution\conf\init-authentication.xml"/>
 <property name="com.wolterskluwer.configuration" value="C:\newreporting-
distribution\conf\wkfs-processing-engine.properties"/>
 <property name="java.library.path" value="C:\newreporting-distribution\deploy\
drivers\dll"/>
</system-properties>
```

For the system properties details, see [System properties](#).

3. Edit *[installation drive:]\[RE installation folder]\jboss-eap-7.3\bin\standalone.conf.bat* by replacing the Java options:

```
set "JAVA_OPTS=-Xms1G -Xmx1G -XX:MetaspaceSize=96M XX:MaxMetaspaceSize=256m"
```

The options above must be replaced by the following item:

```
set "JAVA_OPTS=-Xms3G -Xmx6G -XX:MetaspaceSize=1G"
```

3.2.6.1.3.1 System properties

Note:
The following system properties must be specified and passed to every web application.

Property	Example	Description
file.encoding	**UTF-8**	**File system encoding, must be equal to UTF-8.**
java.io.tmpdir	***%LOCALAPPDATA%/WoltersKluwer/temp_regulatory_engine***	**Path to the temporary folder used by the application.**
wkfs.commons.configuration.file.local	***C:\reporting\conf\wkfs-external.properties***	**Absolute path to wkfs-external.properties generated by the installer.**

Property	Example	Description
reportgenerator.init.file.location	***C:\reporting\conf\init.xml***	**Absolute path to *init.xml* generated by the installer.**
reportgenerator.initAuthentication.file.location	***C:\reporting\conf\init-authentication.xml***	**Absolute path to *init-authentication.xml* generated by the installer.**
com.wolterskluwer.configuration	***C:\reporting\conf\wkfs-processing-engine.properties***	**Absolute path to wkfs-processing-engine.properties generated by the installer.**
java.library.path	***C:\reporting\deploy\drivers\dll***	**Absolute path to the *deploy\drivers\dll* folder generated by the installer.**

3.2.6.1.4 Configure HTTPS

Procedure

1. Go to *C:\server\jboss-eap-7.3\standalone\configuration\standalone.xml*.
2. Edit the 'keystore' subelement of the ApplicationRealm:

```
<security-realm name="ApplicationRealm">
 <server-identities>
  <ssl>
   <keystore path="${keystore.path}" keystore-password="${keystore.password}"
alias="${server.alias}" key-password="${keystore.password}" generate-self-
signed-certificate-host="localhost"/ >
  </ssl>
 </server-identities>
 ...
</security-realm>
```

3. Properly set the following parameters:
   - *${server.alias}* (for example, nre-https) - replace this with the keystore alias.
   - *${keystore.path}* - replace this with the keystore file path (for example,*conf/security/server.keystore*).
   - *${keystore.password}*- replace this with the keystore password.
4. Find the following subsystem with communication configuration. Ensure that https-listener is enabled and its max-post-size is set to 0.

```
<subsystem xmlns="urn:jboss:domain:undertow:10.0" defaultserver="default-
server" default-virtual-host="default-host" defaultservlet-container="default"
default-security-domain="other">
 <buffer-cache name="default"/>
 <server name="default-server">
  <http-listener name="default" socket-binding="http" max-post-size="0"
redirect-socket="https" enable-http2="true"/>
  <https-listener name="https" socket-binding="https" max-post-size="0"
```

```
security-realm="ApplicationRealm" enable-http2="true"/>
  <host name="default-host" alias="localhost">
   <location name="/" handler="welcome-content"/>
   <http-invoker security-realm="ApplicationRealm"/>
  </host>
 </server>
 ...
</subsystem>
```

5. For security reasons, the TLSv1.2 protocol with strong ciphers should be enforced. To ensure that, add the enabled-protocols and enabled-cipher-suites attributes on the engine element to *standalone.xml*:

```
<server-identities>
    <ssl>
        <engine enabled-protocols="TLSv1.2" enabled-cipher-
suites="TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256:TLS_AES_128_GCM_SHA
256:TLS_AES_128_CCM_SHA256:TLS_AES_128_CCM_8_SHA256"/>
        <keystore path=.../>
    </ssl>
</server-identities>
```

6. Adjust the https port in the socket-binding-group section. By default, it is set to 8443; it must reflect the property wolterskluwer.application.port from *install.properties*:

```
<socket-binding-group name="standard-sockets" default-interface="public" port-
offset="${jboss.socket.binding.port-offset:0}">
...
    <socket-binding name="https" port="${jboss.https.port:8443}"/>
...
</socket-binding-group>
```

3.2.6.1.5 Configuring the datasources

Configure one of the following datasources, depending on your implementation:

- Configure the datasource for Oracle
- Configure the datasource for SQL Server

3.2.6.1.5.1 Configure the datasource for Oracle

Procedure

1. Go to *C:\server\jboss-eap-7.3\standalone\configuration\standalone.xml*.
2. Find `<subsystem xmlns="urn:jboss:domain:datasources:5.0">`
3. Add a driver datasource inside the `drivers` section with following properties:

```
  <driver name="oracle.jdbc.OracleDriver"
module="com.oracle.ojdbc8-12-2-0-1_patch_26051289_23198749">
   <driver-class>oracle.jdbc.OracleDriver </driver-class>
  </driver>
```

4. Add a non-transactional datasource for the system database.

Datasource parameters

jndi-name

Concatenated value of properties wolterskluwer.datasource.jndi.prefixand wolterskluwer.datasource.jndi.datasource.name from *install.properties*.

pool-name

jta

false

properties

See the table below.

Properties

connection-url

Derived from propertywolterskluwer.db.connection.url from *install.properties*.

driver-class

Points to JDBC library, depending on database vendor.

security>user-name

Derived from propertywolterskluwer.application.db.username from *install.properties*.

security>password

Derived from propertywolterskluwer.application.db.password from *install.properties*.

Note:
If custom datasource JNDI name is set in install.properties (wolterskluwer.datasource.jndi.datasource.name property) then system database datasource configuration must have it set in attribute jndi-name instead of default SYSTEM_DB.

```
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/
CUSTOM_DATASOURCE_NAME" pool-name="SYSTEM_DB">
```

Datasource subsytem section after the configuration

```
<subsystem xmlns="urn:jboss:domain:datasources:5.0">
 <datasources>
  <datasource jndi-name="java:jboss/datasources/ExampleDS" pool-name="ExampleDS"
enabled="true" use-java-context="true" statistics-
enabled="${wildfly.datasources.statistics-enabled:${wildfly.statistics-
enabled:false}}">
   <connection-
url>jdbc:h2:mem:test;DB_CLOSE_DELAY=-1;DB_CLOSE_ON_EXIT=FALSE</connection-url>
   <driver>h2</driver>
   <security>
    <user-name>sa</user-name>
    <password>sa</password>
   </security>
```

```
  </datasource>
  <datasource jta="false" jndi-name="java:/wolterskluwer/datasources/SYSTEM_DB"
pool-name="SYSTEM_DB">
   <connection-url>jdbc:oracle:thin:@//localhost:1521/ORACLE</connection-url>
   <driver-class>oracle.jdbc.OracleDriver</driver-class>
   <driver>ojdbc8-12.2.0.1_patch_26051289_23198749.jar</driver>
   <security>
    <user-name>SYSTEM_DB</user-name>
    <password>welcome</password>
   </security>
   <validation>
    <valid-connection-checker class-
name="org.jboss.jca.adapters.jdbc.extensions.oracle.OracleValidConnectionChecker"/
>
    <background-validation>true</background-validation>
    <stale-connection-checker class-
name="org.jboss.jca.adapters.jdbc.extensions.oracle.OracleStaleConnectionChecker"/
>
    <exception-sorter class-
name="org.jboss.jca.adapters.jdbc.extensions.oracle.OracleExceptionSorter"/>
   </validation>
  </datasource>
  <drivers>
   <driver name="h2" module="com.h2database.h2">
    <xa-datasource-class>org.h2.jdbcx.JdbcDataSource</xa-datasource-class>
   </driver>
   <driver name="ojdbc8-12.2.0.1_patch_26051289_23198749.jar" module="com.oracle">
    <driver-class>oracle.jdbc.OracleDriver</driver-class>
   </driver>
  </drivers>
 </datasources>
</subsystem>
```

3.2.6.1.5.2 Configure the datasource for SQL Server

Procedure

1. Go to *C:\server\jboss-eap-7.3\standalone\configuration\standalone.xml*.
2. Find `<subsystem xmlns="urn:jboss:domain:datasources:5.0">`
3. Add a driver datasource inside the `drivers` section with following properties:

```
 <driver name="com.microsoft.sqlserver.jdbc.SQLServerDriver"
module="sqlserver.jdbc.driver">
    <driver-class>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver-class>
 </driver>
```

4. Add a non-transactional datasource for the system database.

   Datasource parameters

   jndi-name

   Concatenated value of properties wolterskluwer.datasource.jndi.prefixand wolterskluwer.datasource.jndi.datasource.name from *install.properties*.

   pool-name

jta

false

properties

See the table below.

Properties

connection-url

Derived from propertywolterskluwer.db.connection.url from *install.properties*.

driver-class

Points to JDBC library, depending on database vendor.

security>user-name

Derived from propertywolterskluwer.application.db.username from *install.properties*.

security>password

Derived from propertywolterskluwer.application.db.password from *install.properties*.

Note:
If custom datasource JNDI name is set in install.properties (wolterskluwer.datasource.jndi.datasource.name property) then system database datasource configuration must have it set in attribute jndi-name instead of default SYSTEM_DB.

```
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/
CUSTOM_DATASOURCE_NAME" pool-name="SYSTEM_DB">
```

Note:
For SQL Server with integrated security, the datasource configuration should look as follows: (note that `connection-url` property ends with `integratedSecurity=true` and no credentials used).

5. Optionally, for an integrated authentication, declare the JDBC driver as global module. To do that, find the element `<subsystem xmlns="urn:jboss:domain:ee:4.0">` and add the following lines inside this element, if they are missing:

```
<global-modules>
   <module name="sqlserver.jdbc.driver" slot="main"/>
</global-modules>
```

Datasource configuration examples

SQL Server with an integrated authentication

```
<subsystem xmlns="urn:jboss:domain:datasources:5.0">
<datasources>
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/SYSTEM_DB"
pool-name="SYSTEM_DB">
```

```
  <connection-
url>jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB;integratedSecurity=true
</connection-url>   <driver>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver>
</datasource>
<drivers>
  <driver module="sqlserver.jdbc.driver"
name="com.microsoft.sqlserver.jdbc.SQLServerDriver">
    <driver-class>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver-class>
  </driver>
</drivers>
</datasources>
</subsystem>
```

SQL Server with a login password authentication

```
<subsystem xmlns="urn:jboss:domain:datasources:5.0">
<datasources>
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/SYSTEM_DB"
pool-name="SYSTEM_DB">
  <connection-
url>jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB</connection-url>
<security>
  <user-name>SYSTEM_DB</user-name>
  <password>welcome</password>
</security>
<driver>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver>
</datasource>
<drivers>
  <driver module="sqlserver.jdbc.driver"
name="com.microsoft.sqlserver.jdbc.SQLServerDriver">
    <driver-class>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver-class>
  </driver>
</drivers>
</datasources>
</subsystem>
```

3.2.6.2 Configuring WebSphere Liberty

1. Copy deployments (WAR) files from */deploy/drivers* to the application server.
2. Configure the Java options and parameters
3. Enable HTTPS by providing the path and credentials to the keystore described in topic [Configuring the HTTPS protocol](#).
4. Define the datasource in the application server configuration for the system database generated by the installation script.
5. Create the logger configuration file.

3.2.6.2.1 Copy deployment files for WebSphere Liberty

Context

This task contains the steps for copying the deployment files from the RE distribution into the WebSphere Liberty folder.

Procedure

1. Copy the WAR files from *[installation drive:]\[RE installation folder]\deploy\webapps*.
2. Paste the copied files to *[installation drive:]\[RE installation folder]\wlp\usr\servers\defaultServer\dropins*.

3.2.6.2.2 Configure the Java options and parameters

Procedure

1. Go to *C:\server\wlp\usr\servers\defaultServer*.
2. Create a file called *jvm.options* with the following content:

```
-Xms3G
-Xmx6G
-Dwkfs.commons.configuration.file.local="[installation drive:]\[RE installation
folder]/conf/wkfs-external.properties"
-Dreportgenerator.init.file.location="[installation drive:]\[RE installation
folder]/conf/init.xml"
-Dreportgenerator.initAuthentication.file.location="[installation drive:]\[RE
installation folder]/conf/init-authentication.xml"
-Dcom.wolterskluwer.configuration="[installation drive:]\[RE installation
folder]/conf/wkfs-processing-engine.properties"
-Dlog4j.configuration=file:[installation drive:]\[RE installation folder]/wlp/
usr/servers/defaultServer/logConfiguration.xml
-Djava.library.path="[installation drive:]\[RE installation folder]/deploy/
drivers/dll"
```

For the system properties details, see System properties.

Note:
Each property should be located on a new line.

3.2.6.2.2.1 System properties

Note:
The following system properties must be specified and passed to every web application.

Property	Example	Description
file.encoding	**UTF-8**	**File system encoding, must be equal to UTF-8.**
java.io.tmpdir	***%LOCALAPPDATA%/WoltersKluwer/temp_regulatory_engine***	**Path to the temporary folder used by the application.**
wkfs.commons.configuration.file.local	***C:\reporting\conf\wkfs-external.properties***	**Absolute path to wkfs-external.properties generated by the installer.**

Property	Example	Description
reportgenerator.init.file.location	***C:\reporting\conf\init.xml***	**Absolute path to *init.xml* generated by the installer.**
reportgenerator.initAuthentication.file.location	***C:\reporting\conf\init-authentication.xml***	**Absolute path to *init-authentication.xml* generated by the installer.**
com.wolterskluwer.configuration	***C:\reporting\conf\wkfs-processing-engine.properties***	**Absolute path to wkfs-processing-engine.properties generated by the installer.**
java.library.path	***C:\reporting\deploy\drivers\dll***	**Absolute path to the *deploy\drivers\dll* folder generated by the installer.**

3.2.6.2.3 Configure HTTPS

Procedure

1. Go to *C:\server\wlp\usr\servers\defaultServer*.
2. In the *server.xml* file, add the `ssl-1.0` feature into the `featureManager` section:

```
    <featureManager>
        <feature>javaee-7.0</feature>
        <feature>ssl-1.0</feature>
    </featureManager>
```

3. Find `httpEndpoint` section inside the `server` tag and add the `httpsPort` attribute with the appropriate value from the property wolterskluwer.application.port from *install.properties* file.

```
<httpEndpoint id="defaultHttpEndpoint" host="localhost" httpPort="8080"
httpsPort="8443"/>
```

4. Next to the `httpEndoint` configuration, add a new section called `keyStore`, and add your keystore path, alias and password from install.properties

```
<keyStore id="defaultKeyStore" type="JKS" location="C:/new-reporting-
distribution/conf/security/application.keystore" password="changeit"/>
```

5. For HttpOnly and Secure flags on cookies the following configuration should be added inside `server` section:

```
<httpSession cookieSecure="true" cookieHttpOnly="true"/>
```

6. For security reasons, the TLSv1.2 protocol with strong ciphers should be enforced. To ensure that, add before the `keyStore` section the following configuration:

```
<ssl id="defaultSSLConfig"
         sslProtocol="TLSv1.2"

enabledCiphers="TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256:TLS_AES_128
_GCM_SHA256:TLS_AES_128_CCM_SHA256:TLS_AES_128_CCM_8_SHA256"
         keyStoreRef="defaultKeyStore"
         trustStoreRef="defaultTrustStore"
         clientAuthenticationSupported="true" />

<ssl id="controllerConnectionConfig"
         sslProtocol="TLSv1.2"

enabledCiphers="TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256:TLS_AES_128
_GCM_SHA256:TLS_AES_128_CCM_SHA256:TLS_AES_128_CCM_8_SHA256"/>

<ssl id="memberConnectionConfig"
         sslProtocol="TLSv1.2"

enabledCiphers="TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256:TLS_AES_128
_GCM_SHA256:TLS_AES_128_CCM_SHA256:TLS_AES_128_CCM_8_SHA256"/>
```

3.2.6.2.4 Configuring the datasources

Procedure

1. Go to *C:\server\wlp\usr\servers\defaultServer*.
2. Modify the configurations inside the *server.xml* file as follows:

   - For Oracle - see Create datasource configuration for Oracle.
   - For SQL Server - see Configure the datasource for SQL Server.

3.2.6.2.4.1 Create datasource configuration for Oracle

Procedure

1. Add the following section to create a shared library.

```
<library id="databaseLib">
    <fileSet dir="[installation drive:]\[RE installation folder]\wlp\usr\
servers\defaultServer" includes="*.jar"/>
</library>
```

2. Enumerate all 4 applications and add the reference to the shared library from the previous step.

```
<application type="war" id="admin-webapp" name="admin-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\admin-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="lineage-webapp" name="lineage-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\lineage-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
```

```
  <application type="war" id="reportgenerator" name="reportgenerator"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reportgenerator.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="reporting-webapp" name="reporting-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reporting-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
```

3. Add EclipseLink JAR as a shared library across all applications. Add one more fileSet pointing to *[installation drive:]\[RE installation folder]/wlp/dev/api/third-party* to the shared library that you previously created.

```
<library id="databaseLib">
   <fileSet dir="[installation drive:]\[RE installation folder]\wlp\usr\
servers\defaultServer" includes="*.jar"/>

    <fileSet dir="[installation drive:]\[RE installation folder]\wlp\dev\api\
third-party"
includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar" />
  </library>
```

4. Specify transaction timeouts (in seconds); the recommended value is 20 minutes (1200 s), and it may be increased if needed.

Example

```
<transaction totalTranLifetimeTimeout="1200s"
propogatedOrBMTTranLifetimeTimeout="1200s"/>
```

5. Add a datasource for system database with corresponding parameters previously set in *install.properties*.

   Datasource parameters

   jndiName

   Concatenated value of propertieswolterskluwer.datasource.jndi.prefix and wolterskluwer.datasource.jndi.datasource.name from *install.properties*.

   type

   java.sql.Driver

   transactional

   false

   Properties

   URL

   Derived from property wolterskluwer.db.connection.url from *install.properties*.

   jdbcDriver

   Points to library id, previously configured as shared across applications.

databaseName

Depends on database vendor. On Oracle it is derived from property wolterskluwer.application.db.username from *install.properties*.

user

Derived from propertywolterskluwer.application.db.username from *install.properties*.

password

Derived from propertywolterskluwer.application.db.password from *install.properties*.

Note:

If custom datasource JNDI name is set in install.properties (wolterskluwer.datasource.jndi.datasource.name property) then system database datasource configuration must have it set in attribute jndiName instead of default SYSTEM_DB.

```
<dataSource jndiName="jdbc/wolterskluwer/datasources/
CUSTOM_DATASOURCE_NAME" type="java.sql.Driver" transactional="false">
```

Example of server configuration for an Oracle datasource:

```
    [...]
    <transaction totalTranLifetimeTimeout="1200s"
propogatedOrBMTTranLifetimeTimeout="1200s"/>

    <jpa
defaultPersistenceProvider="org.eclipse.persistence.jpa.PersistenceProvider"/>

  <library id="databaseLib">
    <fileSet dir="[installation drive:]\[RE installation folder]\wlp\usr\
servers\defaultServer" includes="*.jar"/>
    <fileSet dir="[installation drive:]\[RE installation folder]\wlp\dev\api\
third-party"
includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar" />
  </library>

  <application type="war" id="admin-webapp" name="admin-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\admin-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="lineage-webapp" name="lineage-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\lineage-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="reportgenerator" name="reportgenerator"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reportgenerator.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="reporting-webapp" name="reporting-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reporting-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <dataSource jndiName="jdbc/wolterskluwer/datasources/SYSTEM_DB"
type="javax.sql.DataSource" transactional="false">
```

```
    <jdbcDriver libraryRef="databaseLib"/>
    <properties URL="jdbc:oracle:thin:@localhost:1521/ORACLE"
databaseName="SYSTEM_DB" user="SYSTEM_DB" password="welcome"/>
    <containerAuthData user="SYSTEM_DB" password="welcome" />
</dataSource>
  [...]
 </server>
```

Note:
The connection parameters should reflect the actual values from *install.properties*.

3.2.6.2.4.2 Configure the datasource for SQL Server

Procedure

1. Add the following section to create a shared library:

```
<library id="databaseLib">
    <file name="[installation drive:]\[RE installation folder]\deploy\drivers\
mssql-jdbc-9.4.1.jre8.jar" />
</library>
```

2. Enumerate all four applications and add the reference to the shared library from the previous step.

```
    <application type="war" id="admin-webapp" name="admin-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\admin-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="lineage-webapp" name="lineage-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\lineage-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="reportgenerator" name="reportgenerator"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reportgenerator.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="reporting-webapp" name="reporting-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reporting-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
```

3. Add EclipseLink JAR as a shared library across all applications. Library is located in *[installation drive:]\[RE installation folder]/wlp/dev/api/third-party* under directory where the websphere was unzipped.

```
<library id="databaseLib">
    <file name="[installation drive:]\[RE installation folder]\deploy\drivers\
mssql-jdbc-9.4.1.jre8.jar" />
    <fileSet dir="[installation drive:]\[RE installation folder]\wlp\dev\api\
third-party"
includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar" />
  </library>
```

4. Specify transaction timeouts (in seconds); the recommended value is 20 minutes (1200 s), and it may be increased if needed.
5. Add a datasource for system database with corresponding parameters previously set in *install.properties*.

   Datasource parameters

   jndiName

   Concatenated value of propertieswolterskluwer.datasource.jndi.prefixand wolterskluwer.datasource.jndi.datasource.name from *install.properties*.

   type

   java.sql.Driver

   transactional

   false

   Properties

   URL

   Derived from propertywolterskluwer.db.connection.url from *install.properties*.

   jdbcDriver

   Points to library id, previously configured as shared across applications.

   databaseName

   Depends on database vendor. It is derived from the ending of the connection URL.

   user

   Derived from property wolterskluwer.application.db.username from *install.properties*.

   password

   Derived from property wolterskluwer.application.db.password from *install.properties*.

   integratedSecurity

   Optional parameter, derived from property wolterskluwer.sqlserver.integratedSecurity from *install.properties*.

   Note:
   If custom datasource JNDI name is set in install.properties (wolterskluwer.datasource.jndi.datasource.name property) then system database datasource configuration must have it set in attribute jndiName instead of default SYSTEM_DB.

   ```
   <dataSource jndiName="jdbc/wolterskluwer/datasources/
   CUSTOM_DATASOURCE_NAME" type="java.sql.Driver" transactional="false">
   ```

Example

Example of a server configuration for the datasource with username/password authentication:

```
    [...]
    <transaction totalTranLifetimeTimeout="1200s"
propogatedOrBMTTranLifetimeTimeout="1200s"/>

    <jpa
defaultPersistenceProvider="org.eclipse.persistence.jpa.PersistenceProvider"/>

  <library id="databaseLib">
    <file name="[installation drive:]\[RE installation folder]\deploy\drivers\
mssql-jdbc-9.4.1.jre8.jar" />
    <fileSet dir="[installation drive:]\[RE installation folder]\wlp\dev\api\
third-party" includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar"
/>
  </library>

  <application type="war" id="admin-webapp" name="admin-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\admin-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="lineage-webapp" name="lineage-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\lineage-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="reportgenerator" name="reportgenerator"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reportgenerator.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
  <application type="war" id="reporting-webapp" name="reporting-webapp"
location="[installation drive:]\[RE installation folder]\container\servers\
defaultServer\dropins\reporting-webapp.war">
    <classloader commonLibraryRef="databaseLib" />
  </application>
    <dataSource jndiName="jdbc/wolterskluwer/datasources/SYSTEM_DB"
type="java.sql.Driver" transactional="false">
    <jdbcDriver libraryRef="databaseLib" />
    <properties URL="jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB"
databaseName="SYSTEM_DB" user="SYSTEM_DB" password="welcome" />
    <containerAuthData user="SYSTEM_DB" password="welcome" />
  </dataSource>
  [...]
 </server>
```

Example of a server configuration for the datasource with integrated security authentication:

```
<dataSource jndiName="jdbc/wolterskluwer/datasources/SYSTEM_DB"
type="javax.sql.DataSource" transactional="false">
<jdbcDriver libraryRef="databaseLib"/> <properties.microsoft.sqlserver
URL="jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB"
integratedSecurity="true" databaseName="SYSTEM_DB" serverName="localhost"
instanceName="SQLSERVER"/>
</dataSource>
```

3.2.6.2.5 Create the logger configuration file

Procedure

1. Go to *C:\server\wlp\usr\servers\defaultServer*.
2. Create a file called *logConfiguration.xml* with the following content:

```
<?xml version="1.0" encoding="UTF-8" ?>
<!DOCTYPE log4j:configuration SYSTEM "log4j.dtd">
<log4j:configuration xmlns:log4j='http://jakarta.apache.org/log4j/'>

    <appender name="console" class="org.apache.log4j.ConsoleAppender">
        <layout class="org.apache.log4j.PatternLayout">
            <param name="ConversionPattern"
                   value="%d{yyyy-MM-dd HH:mm:ss} %-5p %c{1}:%L - %m%n" />
        </layout>
    </appender>

    <root>
        <level value="INFO" />
        <appender-ref ref="console" />
    </root>

</log4j:configuration>
```

3.3 Configuring the Identity Provider

This section describes how to configure the Identity Providers for the OneSumX Regulatory Engine installation. Depending on your configuration, refer to one of the following topics:

- Configuring Keycloak
- Configuring Active Directory

3.3.1 Configuring Keycloak

3.3.1.1 Create SAML clients

Context

For Keycloak, you need to configure the SAML authentication for users via the Keycloak administration console.

Procedure

1. If an error occurs and the message `{{notification.header}} {{notification.message}}` is shown, mark the website as trusted. To do so, in Internet Explorer 11:
   a. From the (**Tools**) menu, select **Internet Options**.
   b. Select the **Security** tab.
   c. Select zone **Trusted Sites**.
   d. Select the **Sites** button.
   e. Select the **Add** button to add a site and close the window.
   f. Select **OK** in the **Internet Options** dialog.

g. Refresh the page.

2. In the Keycloak administrator console, select **Clients** and click **Create**.
3. Select **Select file**.

Results

This will allow you to select the file generated during the installation as a file with service provider metadata. This file represents the service provider configuration.

4. Choose the file generated or provided by you in *${ admin.serviceprovider.metadata }* and select **Save**.
5. Repeat steps 1 to 4 for *${ hub.serviceprovider.metadata }*, *${lineage.serviceprovider.metadata}* and *${reportgenerator.serviceprovider.metadata}*.

3.3.1.1.1 Configure SAML mappers

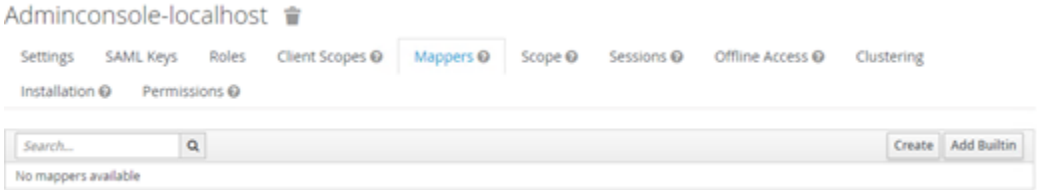
Context

Configuring SAML mappers is an optional step.

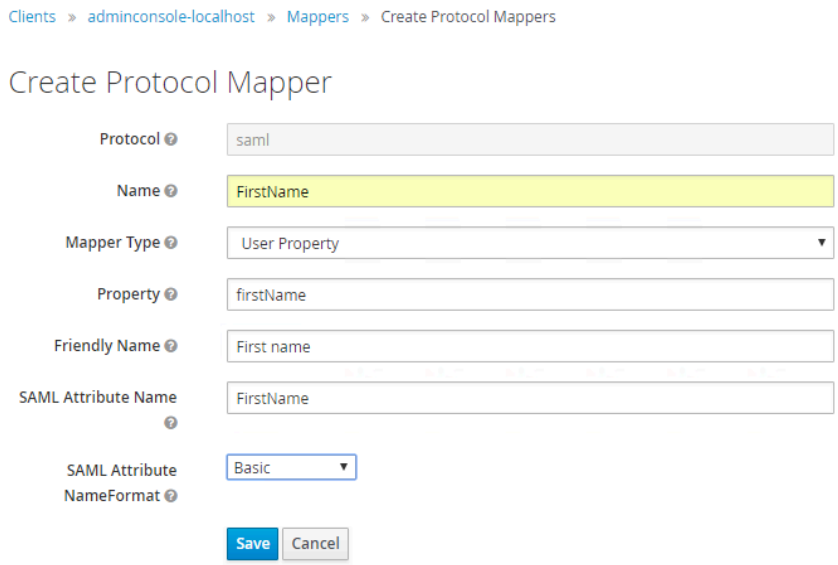
Next to the login details of the user, Keycloak can also contain some additional information such as first and last name and the email name. To propagate this information in the application, some additional mappers on Keycloak must be configured.

Procedure

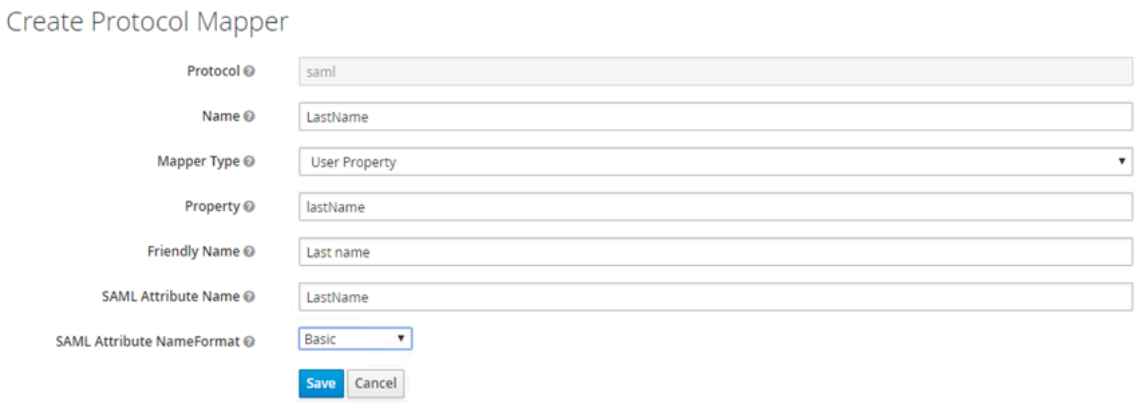
1. In the Keycloak administrator console, go to the **Clients** page.
2. Choose a client, for example, **Adminconsole-localhost**, and select the **Mappers** tab.



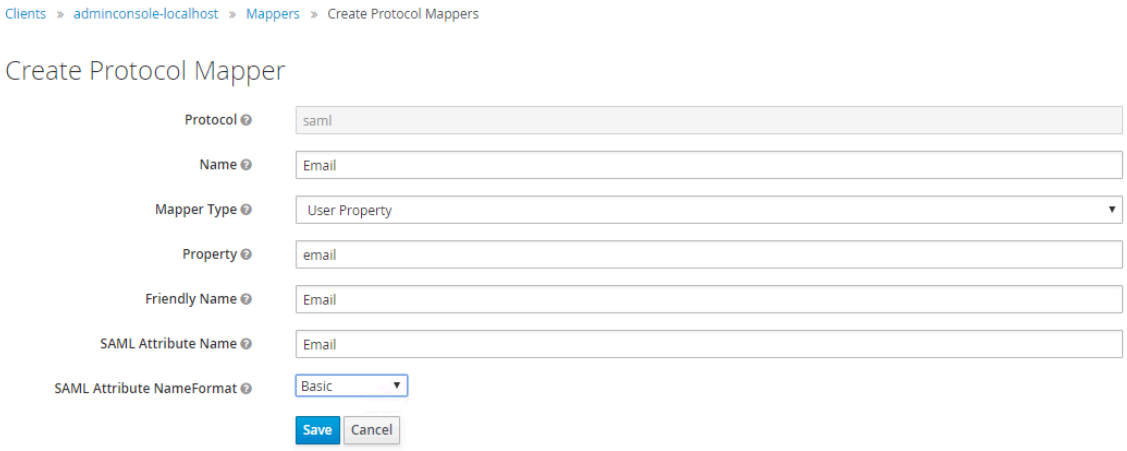
3. To propagate the first name of the user to the application, select **Create**, provide the information in the form as illustrated here and select **Save** to store the settings.



4. To propagate the last name of the user to the application, select **Create**, provide the information in the form as illustrated here and select **Save** to store the settings.



5. To propagate the email name of the user to the application, select **Create**, provide the information in the form as illustrated here and select **Save** to store the settings.



6. Depending on the properties you want to propagate to the application, repeat steps 1-5 for the Reporting

hub, Lineage, and ReportGenerator clients.
7. Leave Keycloak running.

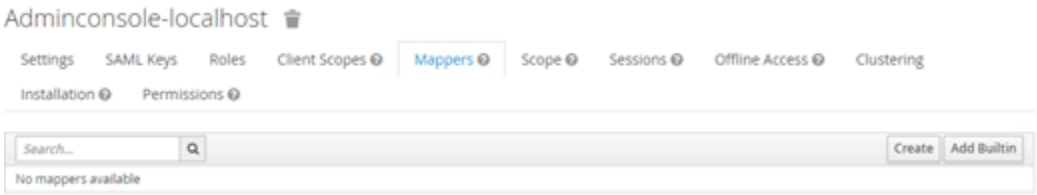
3.3.1.1.1.1 Configure a SAML mapper for retrieving groups information

Context

To assign a user to a resource group, Keycloak needs to send information about groups. Therefore, a mapper configuration is needed.

Procedure

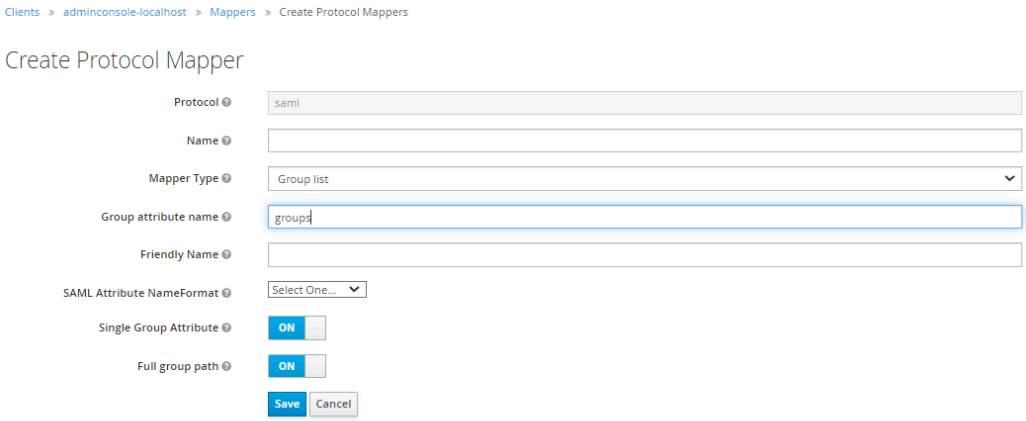
1. In the Keycloak administrator console, go to the **Clients** page.
2. Choose a client, for example, **Adminconsole-localhost**, and click the **Mappers** tab.



3. In the **Mappers** tab, click **Create**.

Results

The **Create Protocol Mapper** page is displayed.



4. In the **Mapper Type** list, select **Group list**.
5. Add the following details:

   - In the **Group attribute name** box, enter **groups**.
   - In the **Friendly Name** box, enter **groups**.

Warning:
The fields are case sensitive.

All the other details should be left as is.

6. Click **Save**.

3.3.1.2 Export the identity provider metadata

Procedure

1. In the Keycloak console, select **Realm Settings** and then select *SAML 2.0 Identity Provider Metadata*.
2. Save the opened file.
3. Rename the file manually to be in line with property *${ admin.identityprovider.metadata }* from the *install.properties* file.
4. Copy the file to the *conf/security* folder.
5. One common identity provider file can be used for admin, reporting hub, lineage and ReportGenerator. If you want to use different ones, repeat the previous steps for reporting hub, lineage and ReportGenerator applications, that is:
   a. Select **Realm Settings** and select *SAML 2.0 Identity Provider Metadata*. Save the opened file.
   b. Rename the file(s) manually to be in line with property *${ hub.identityprovider.metadata }* for reporting-hub and\or *${lineage.identityprovider.metadata}* for Lineage, and\or *${reportgenerator.identityprovider.metadata}* for ReportGenerator from the *install.properties* file.
   c. Copy the file(s) to the *conf/security* folder.

3.3.2 Configuring Active Directory

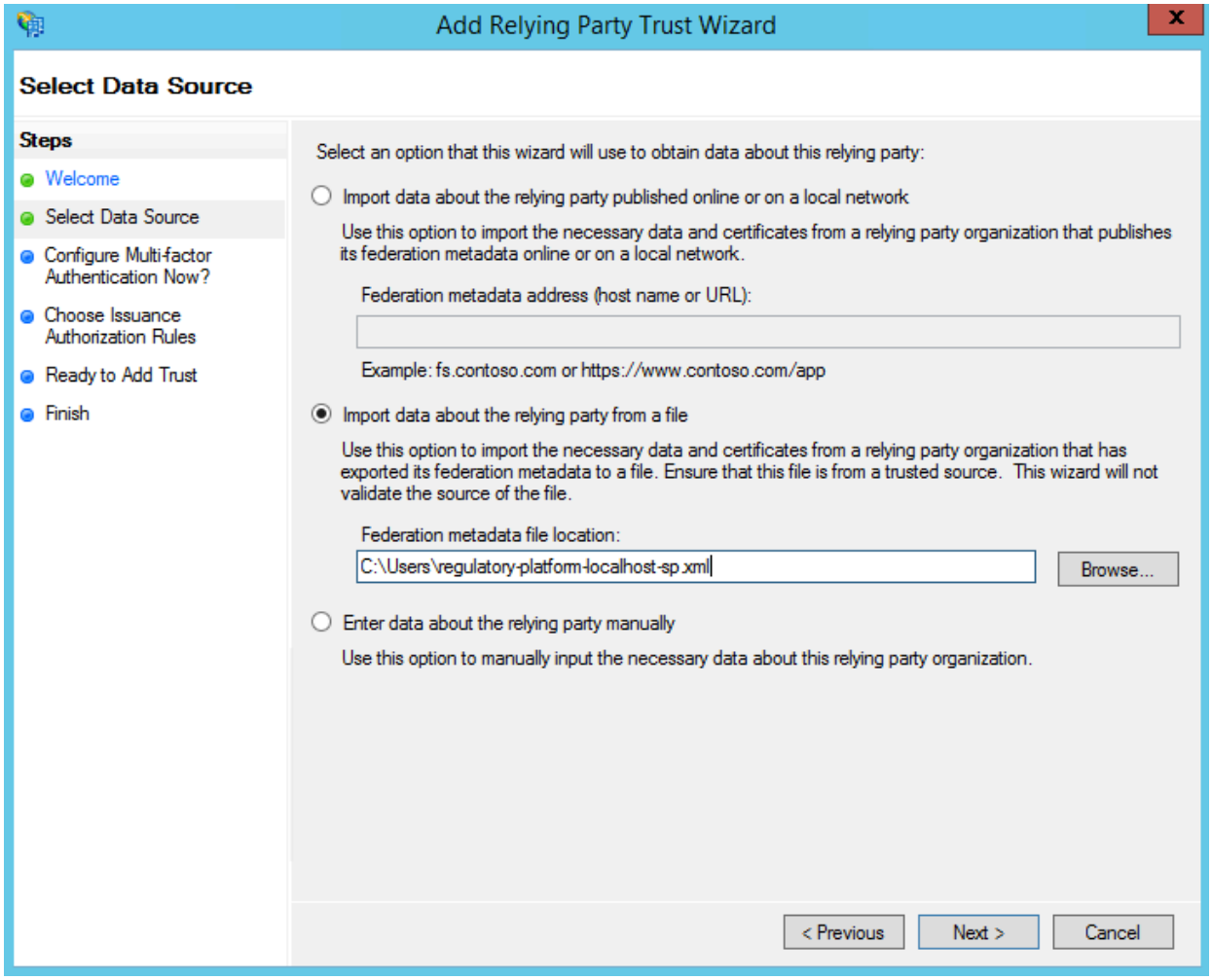
3.3.2.1 Add a Relying Party Trust

Context

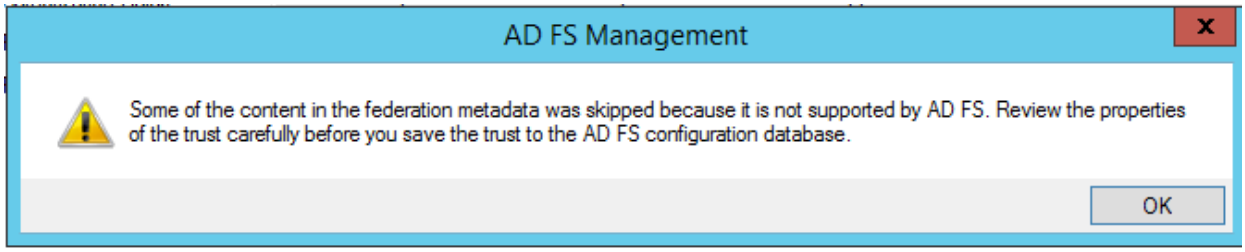
The connection between AD FS and Regulatory Platform is defined using a Relying Party Trust.

Procedure

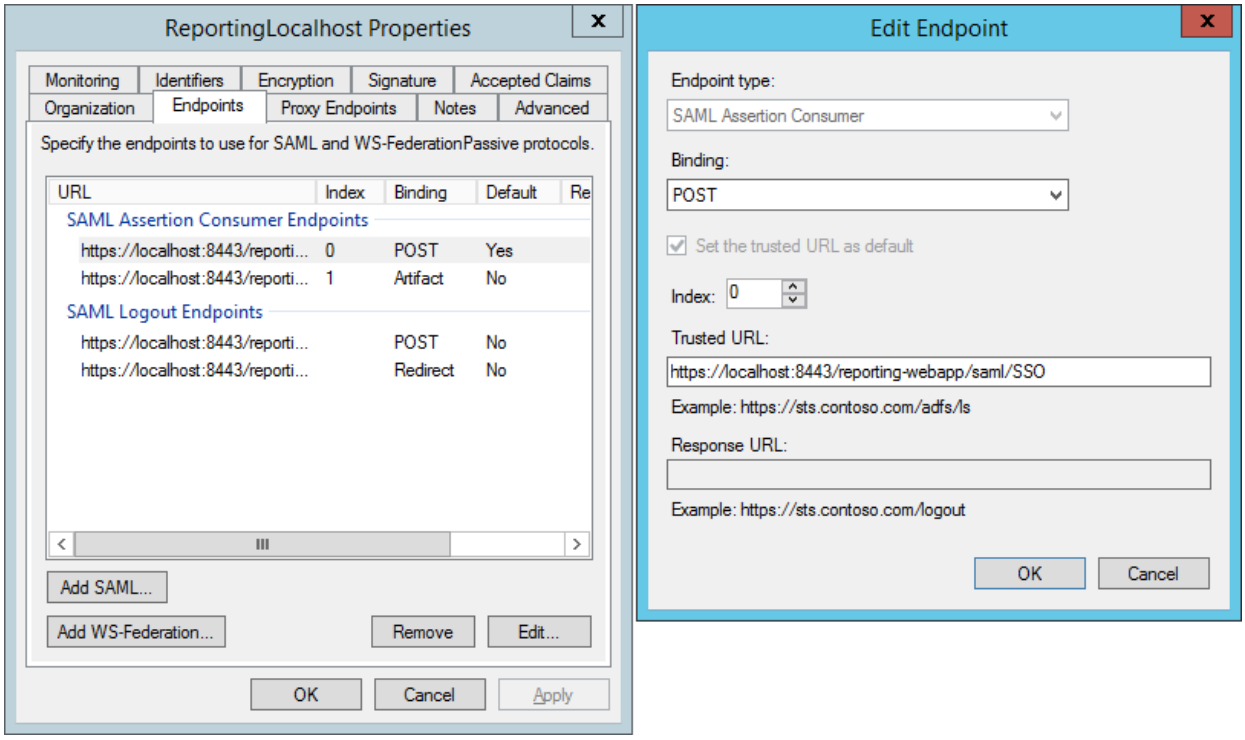
1. Open AD FS Management, right-click the *Relying Party Trust* folder and, in the **context** menu, select **Add Relying Party Trust**.
2. In the **Welcome** screen of the activated new trust configuration wizard, select **Start** to proceed.
3. In the **Select Data Source** screen, select option **Import data about the relying party from a file**.
4. To set the **Federation metadata file location**, browse for the *regulatory-platform-localhost-sp.xml* file generated during the installation and select **Next**. This file is located at *INSTALLATION_PATH/conf/security*.

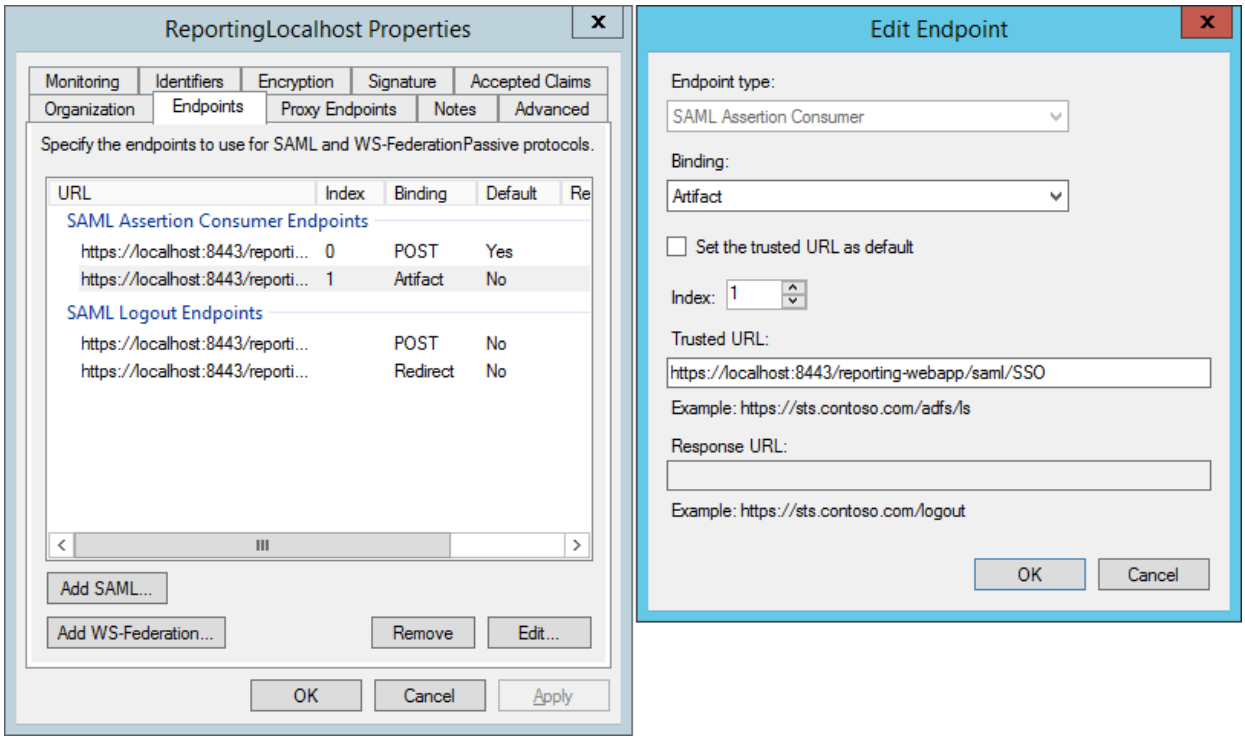


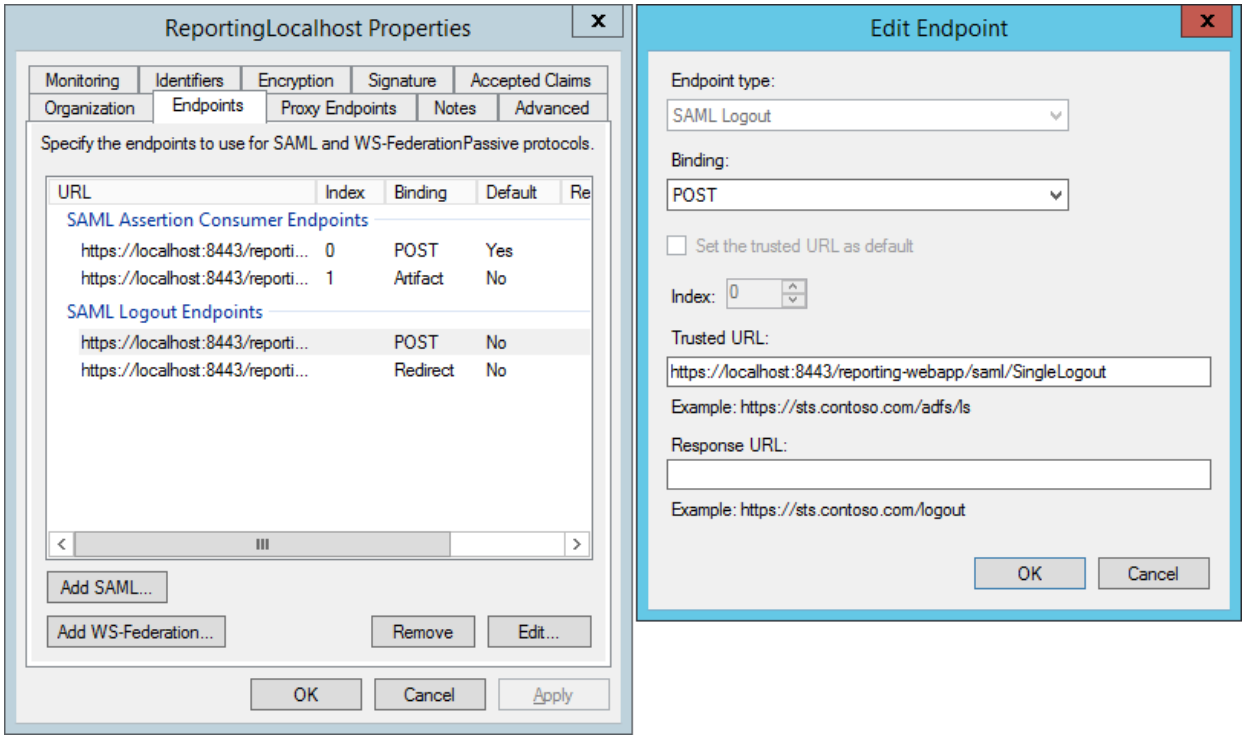
5. Depending on the AD FS version, the content of the service provider metadata file might not be correctly imported, in which case a warning may appear. Follow the instructions of the warning and check the imported data, completing it where data is missing.

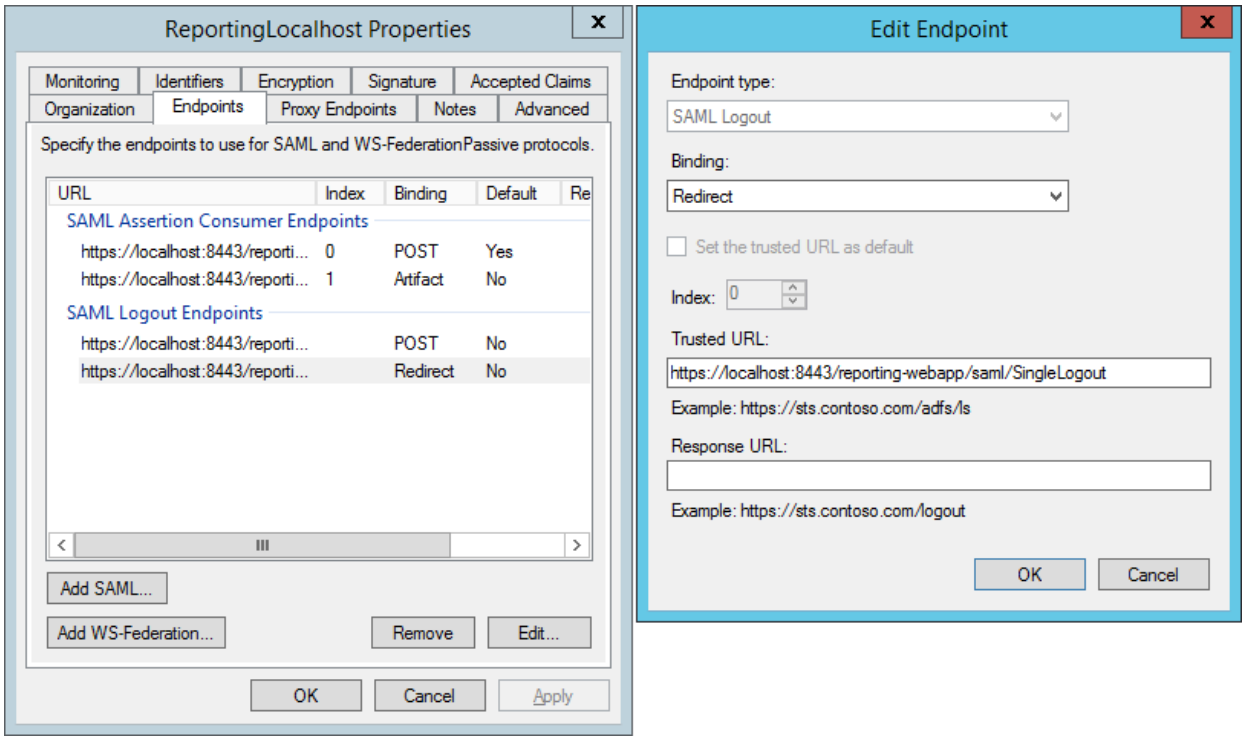


6. In the **Specify Display Name** screen, enter a Display Name for the relying party and select **Next**.
7. In the **Multi-factor Authentication** screen, leave the default **I do not want to configure multi-factor authentication settings for this relying party trust**, at this time of selection and select **Next**.
8. In the **Authorization Rules Issuance** screen, leave the default selection: **Permit all users to access this relying party selection** and select **Next**.
9. Check all the data in the tabs and add any missing information if applicable. Endpoints may have to be added manually:


ReportingLocalhost Properties
Monitoring
Identifiers
Encryption
Signature
Accepted Claims
Organization
Endpoints
Proxy Endpoints
Notes
Advanced
Specify the endpoints to use for SAML and WS-FederationPassive protocols.
URL
Index
Binding
Default
Re
SAML Assertion Consumer Endpoints
https://localhost:8443/reporti... 0 POST Yes
https://localhost:8443/reporti... 1 Artifact No
SAML Logout Endpoints
https://localhost:8443/reporti... POST No
https://localhost:8443/reporti... Redirect No
Add SAML...
Add WS-Federation...
Remove
Edit...
OK
Cancel
Apply
Edit Endpoint
Endpoint type:
SAML Assertion Consumer
Binding:
POST
Set the trusted URL as default
Index: 0
Trusted URL:
https://localhost:8443/reporting-webapp/saml/SSO
Example: https://sts.contoso.com/adfs/ls
Response URL:
Example: https://sts.contoso.com/logout
OK
Cancel


ReportingLocalhost Properties
Monitoring
Identifiers
Encryption
Signature
Accepted Claims
Organization
Endpoints
Proxy Endpoints
Notes
Advanced
Specify the endpoints to use for SAML and WS-FederationPassive protocols.
URL
Index
Binding
Default
Re
SAML Assertion Consumer Endpoints
https://localhost:8443/reporti... 0 POST Yes
https://localhost:8443/reporti... 1 Artifact No
SAML Logout Endpoints
https://localhost:8443/reporti... POST No
https://localhost:8443/reporti... Redirect No
Add SAML...
Add WS-Federation...
Remove
Edit...
OK
Cancel
Apply
Edit Endpoint
Endpoint type:
SAML Assertion Consumer
Binding:
Artifact
Set the trusted URL as default
Index: 1
Trusted URL:
https://localhost:8443/reporting-webapp/saml/SSO
Example: https://sts.contoso.com/adfs/ls
Response URL:
Example: https://sts.contoso.com/logout
OK
Cancel





Note:

AD FS may require the use of HTTPS when the Endpoints are configured.

3.3.2.1.1 Create all Relying Party Trusts and Adding Federation metadata file

Procedure

1. Execute the steps from section Add a Relying Party Trust and section Manage the claim rule settings for the other two service provider files: *Reportgenerator-localhost-sp.xml*, *Lineage-localhost-sp.xml* and *AdminConsole-localhost-sp.xml*.
2. After the three relying party trusts have been created, download the *federationmetadata.xml* file from the following URL and save it on the disk: https://localhost/federationmetadata/2007-06/federationmetadata.xml.

   Note:
   Identity Provider file (*federationmetadata.xml*) must be present on the disk. URLs that point to this file are not supported!

3. In the *install.properties* file, you must provide the path to the federation metadata file for the following properties:

   - admin.identityprovider.metadata = *PATH_TO_FEDERATION_METADATA_FILE/federationmetadata.xml*
   - hub.identityprovider.metadata = *PATH_TO_FEDERATION_METADATA_DILE/federationmetadata.xml*
   - reportgenerator.identityprovider.metadata = *PATH_TO_FEDERATION_METADATA_FILE/federationmetadata.xml*
   - *lineage.identityprovider.metadata = PATH_TO_FEDERATION_METADATA_FILE/federationmetadata.xml*

   Note:
   Due to US export restrictions, Java JDK comes with a limited set of cryptography capabilities. Download Java Cryptography Extension (JCE) Unlimited Strength Jurisdiction Policy Files 8 and follow the instructions to remove these limitations.

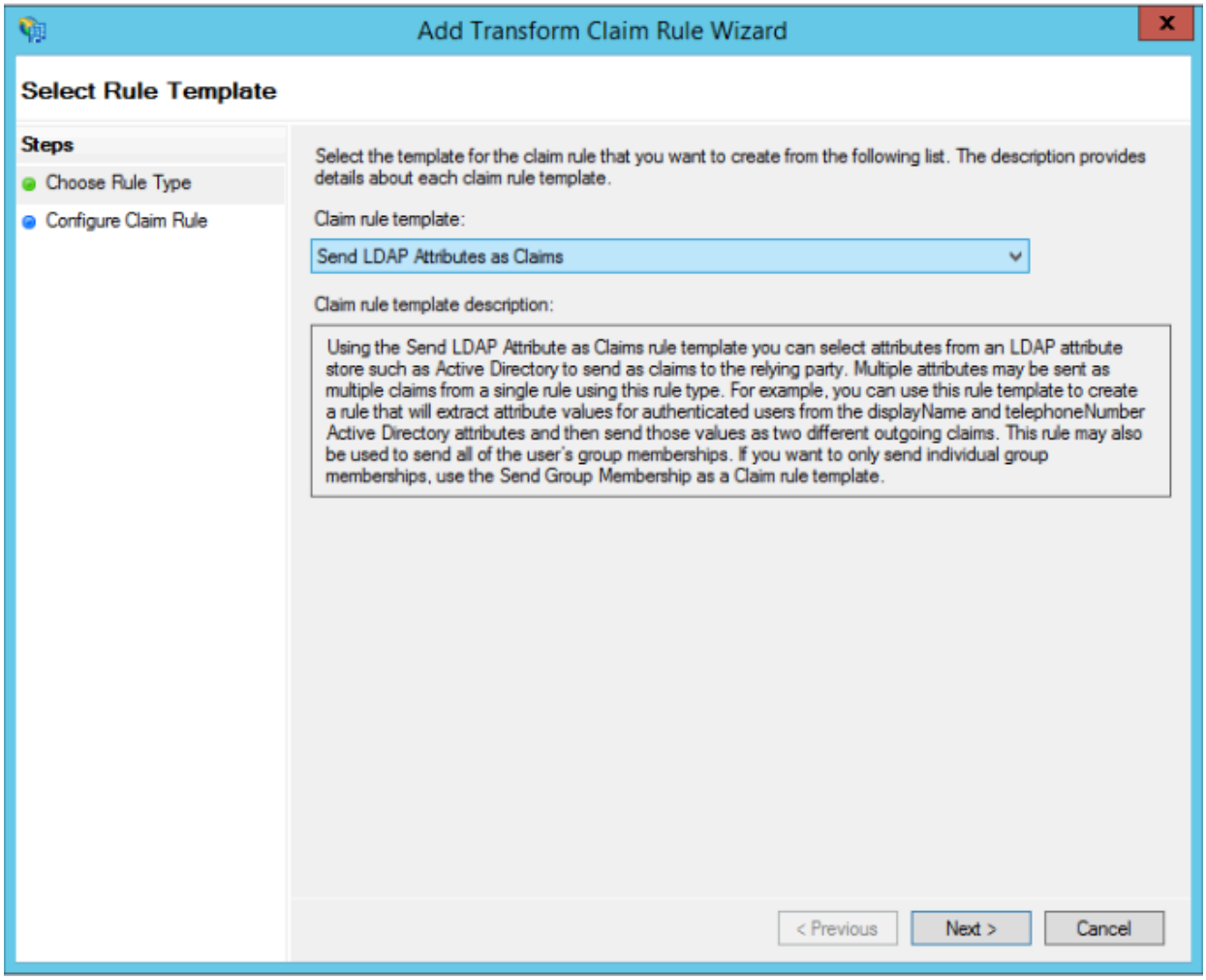
3.3.2.2 Manage the claim rule settings

Context

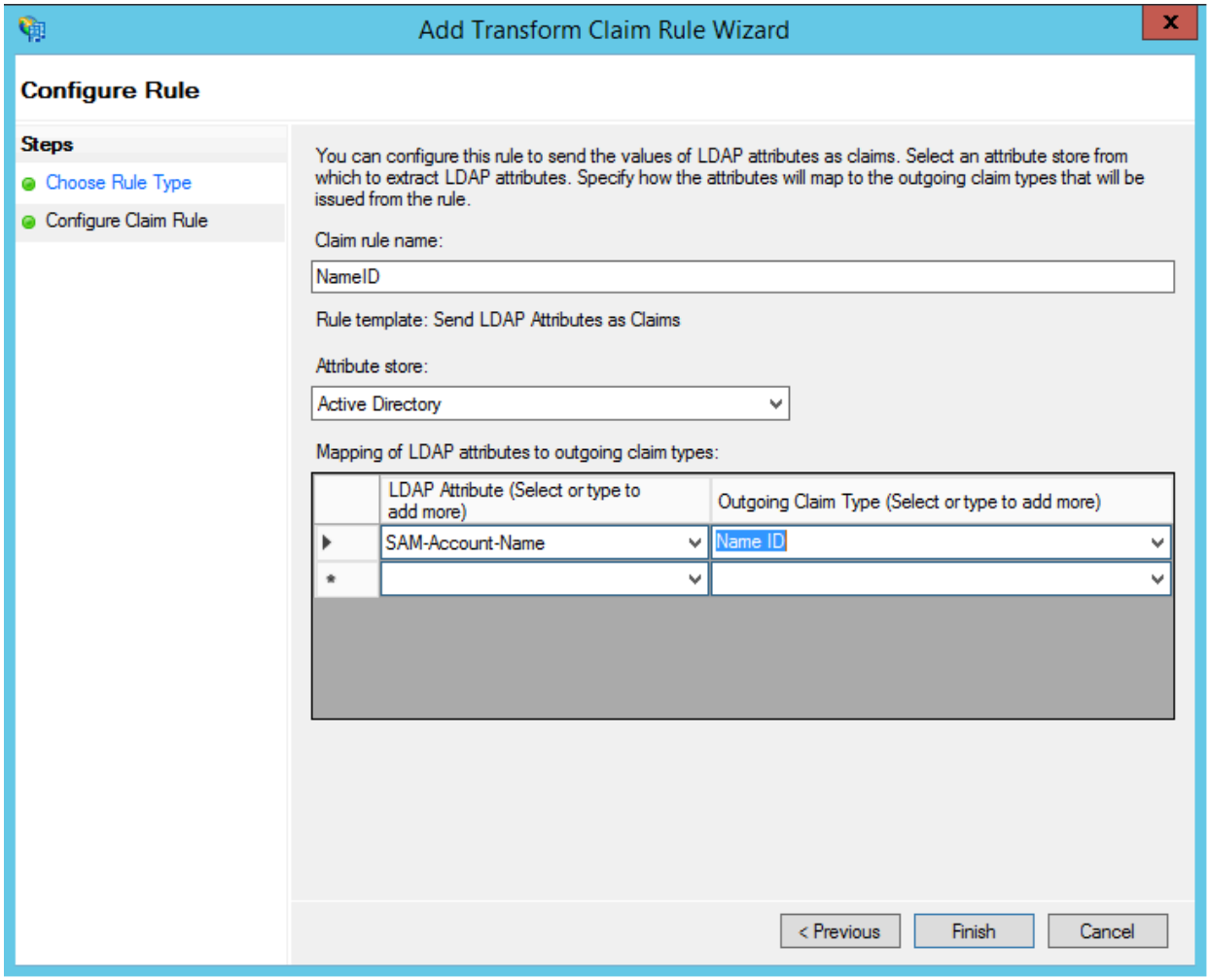
Next, you will also have to manage the claim rule settings.

Procedure

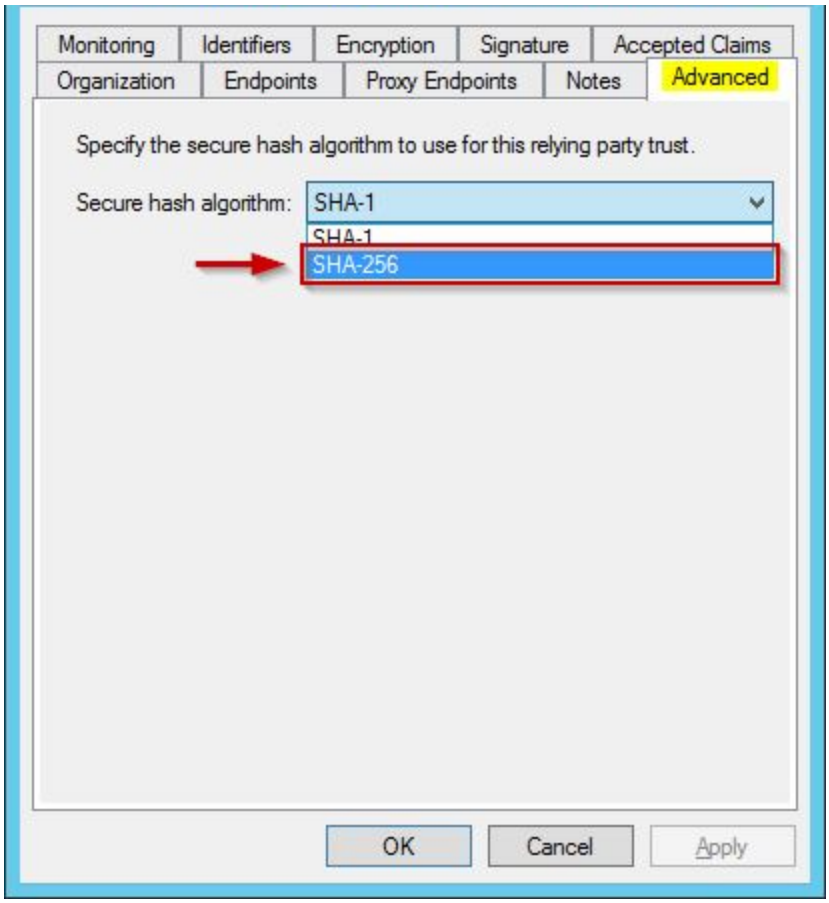
1. In AD FS Management, from the **Action** menu, option **Device Registration Service**, choose **Edit Claim Rules**.
2. Select **Add Rule** to open the **Add Transform Claim Rule** wizard.
3. In the **Select Rule Template** screen, under **Claim rule template**: choose **Send LDAP Attributes as Claims** and select **Next**.



4. In the **Configure Rule** screen, under **Claim rule name**: add **NameID**.
5. Choose **Active Directory** as **Attribute store**.
6. Choose **SAM-Account-Name** as **LDAP Attribute** and **Name ID** as **Outgoing claim type**.
7. Select **Finish** to complete the wizard configuration.



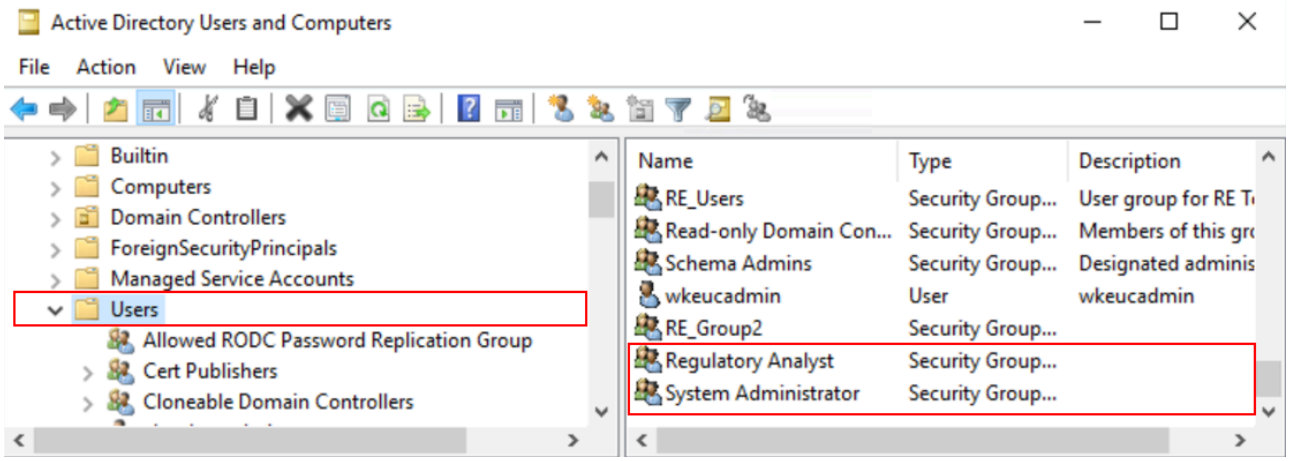
8. Right-click on the created relying party trust, select **Properties**, go to the **Advanced** tab and select **SHA-256** as **Secure hash algorithm**.



3.3.2.2.1 Configure the Claim Rules for groups

Prerequisites

- Create the necessary user groups in AD FS: the default groups (Regulatory Analyst, System Administrator) and any other needed group.
- Add users to these groups. Add the admin user of OneSumX Regulatory Engine (admin.defaultuser) to the System Administrator group.



Context

In AD FS Management, you need to configure the Claim Rules for groups to receive the information in the SAML response.

You need to create a rule for the predefined groups as well as for the customized groups.

Procedure

1. Open AD FS Management, click the *Relying Party Trusts* folder, then select the Relying Party Trust for Admin Console. For details, refer to topic Add a Relying Party Trust.
2. On the **Actions** sidebar, click **Edit Claim Issuance Policy**.

   ## Results

   The **Edit Claim Issuance Policy** dialog box is displayed.

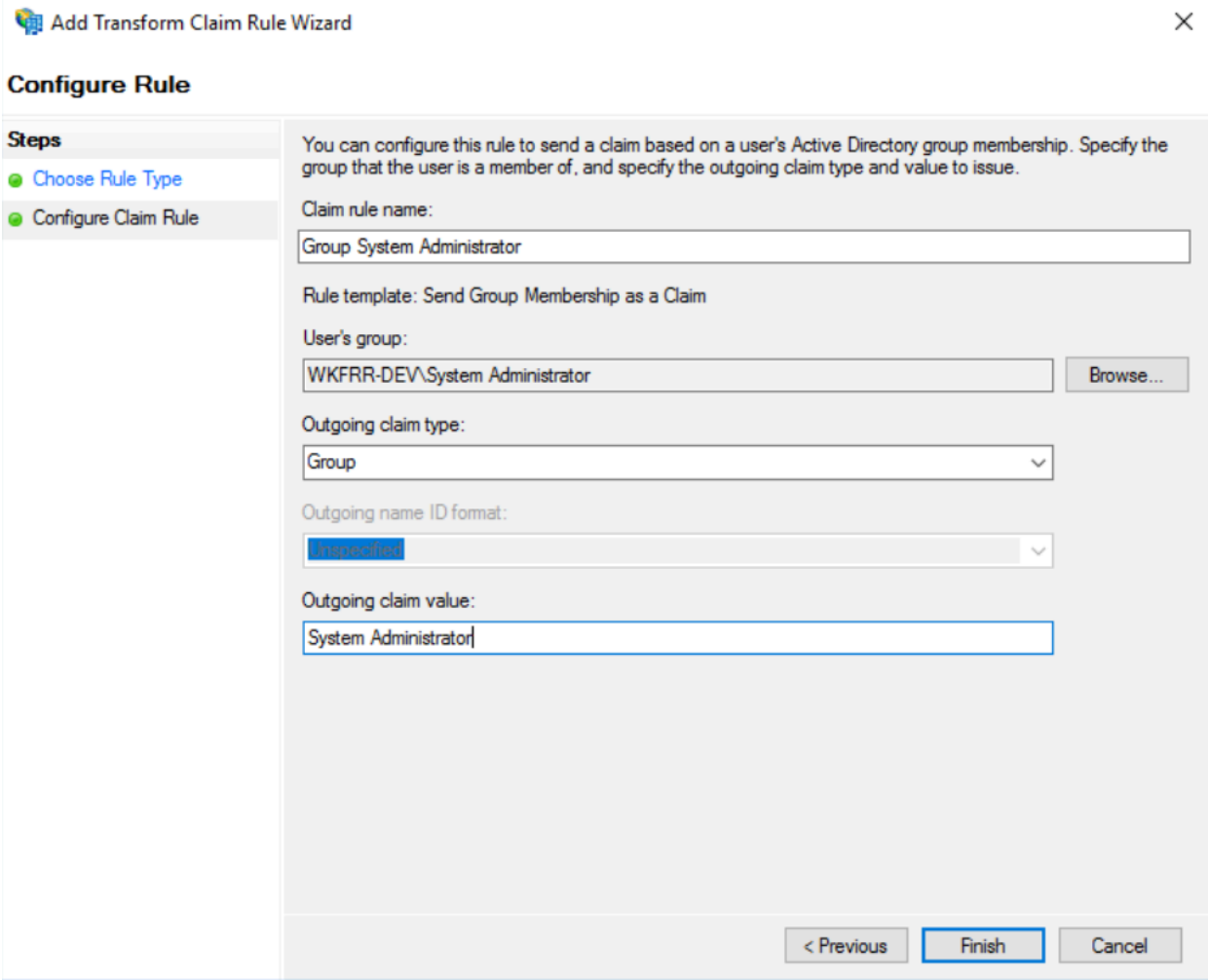
3. On the **Issuance Transform Rules** tab, click **Add Rule**.

   ## Results

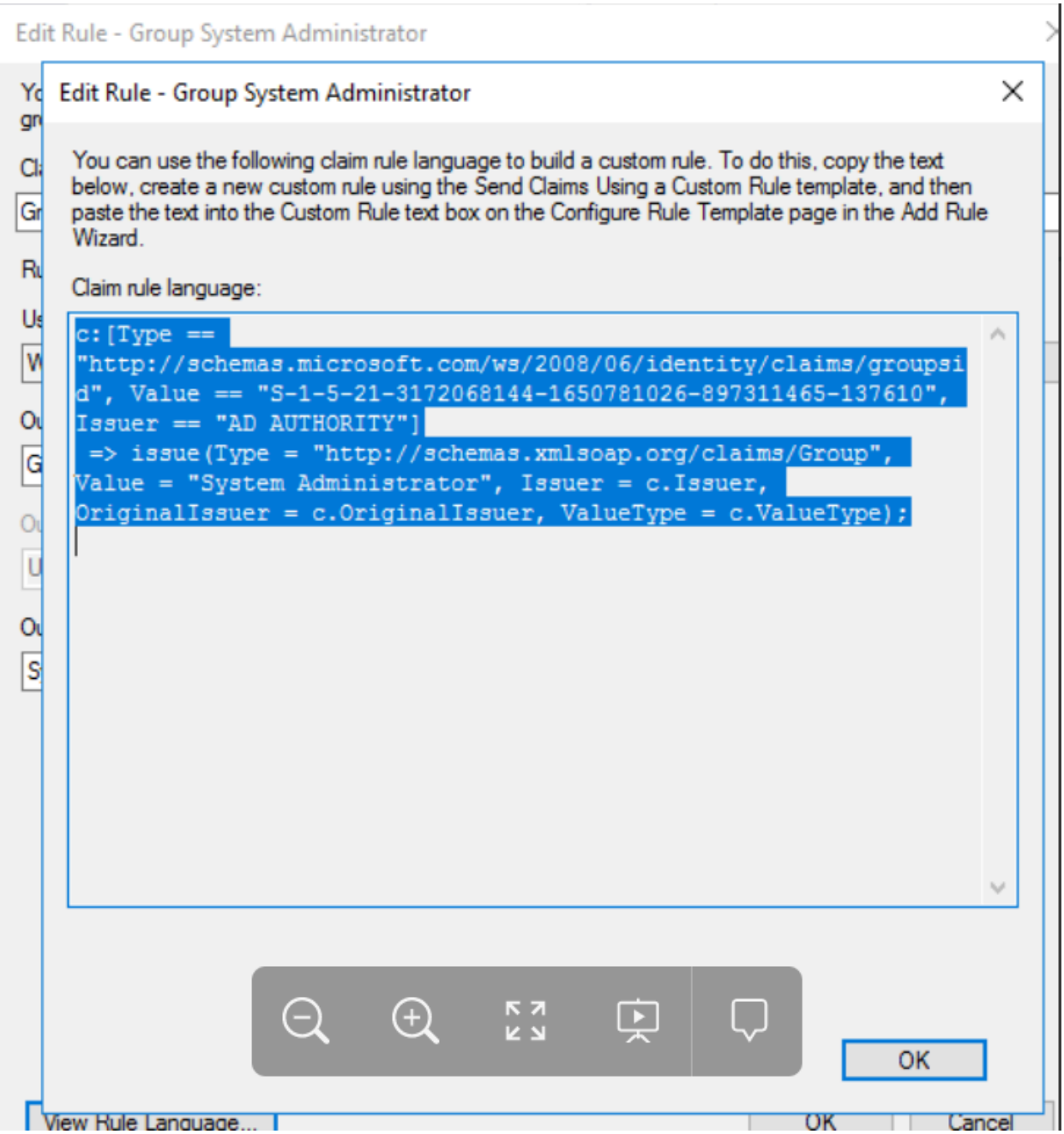
   The **Add Transform Claim Rule Wizard** is displayed.

4. On the **Select Rule Template** page, in the **Claim rule template** list, select the **Send Group Membership as a Claim** option, then click **Next**.
5. On the **Configure Rule** page, perform the following actions:

   - In the **Claim rule name** box, add a name for the rule.
   - Next to the **User's group** box, click **Browse** to select the user group that you want to map to the group rule, for example, System Administrator.
   - In the **Outgoing claim type** list, select **Group**.
   - In the **Outgoing claim value** box, enter the group's name.



6. Click **Finish** to save the group rule.
7. In the **Edit Claim Issuance Policy** dialog box, on the **Issuance Transform Rules** tab, select the group rule created at steps 3 to 6 in this procedure and click **Edit Rule**.
8. In the **Edit Rule** dialog box, click **View Rule Language**, then copy the text from the **Claim rule language** box.

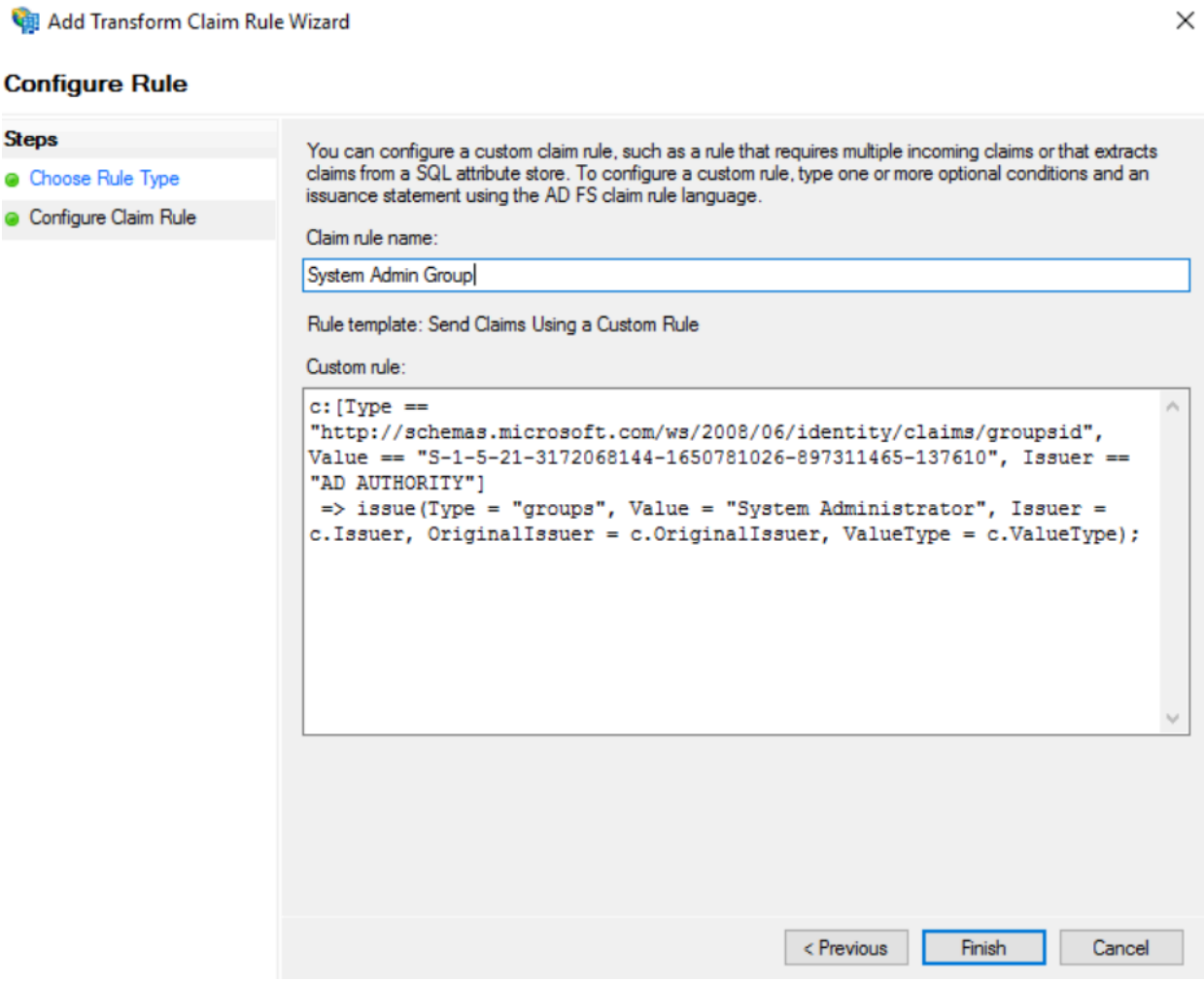


9. Click **OK** to return to the **Edit Rule** dialog box.
10. In the **Edit Claim Issuance Policy** dialog box, on the **Issuance Transform Rules** tab, select the rule that you created at steps 3 to 6 in this procedure and click **Remove Rule**.
11. In the **Edit Claim Issuance Policy** dialog box, on the **Issuance Transform Rules** tab, click **Add Rule**.

Results

The **Add Transform Claim Rule Wizard** is displayed.

12. On the **Select Rule Template** page, in the **Claim rule template** list, select the **Send Claims Using a Custom Rule** option, and click **Next**.
13. On the **Configure Rule** page, in the **Claim rule name** box, add a name for the rule.
14. In the **Custom rule** box, paste the text that you copied at step 8, and then delete the "http://schemas.xmlsoap.org/claims/" string from the **Type** field. This should leave only the word **Group**. Change it to **groups**.



15. Click **Finish** to save the group rule.

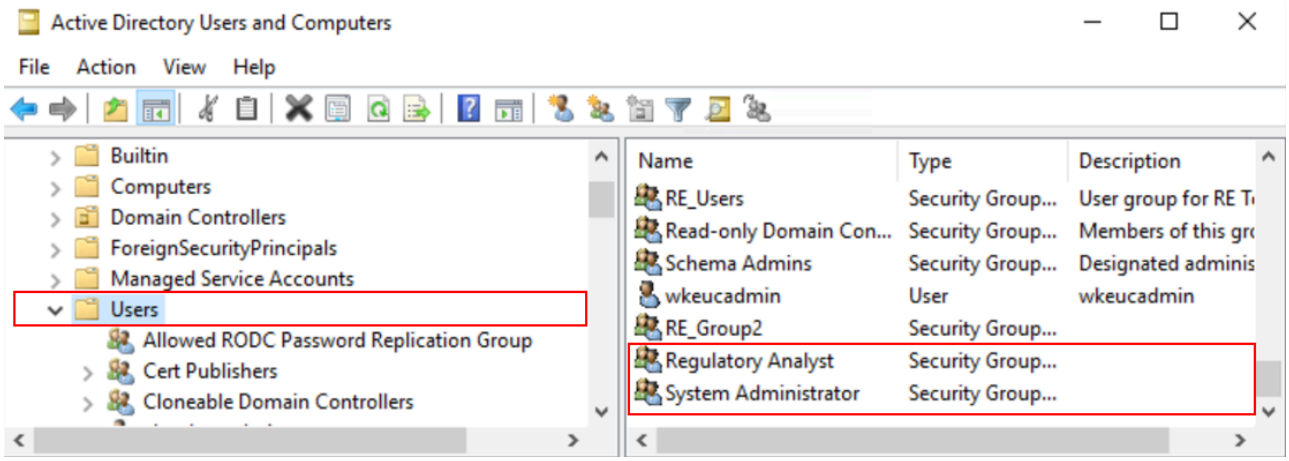
What to do next

In OneSumX Regulatory Engine, you need to create a resource group or role with the exact name of the user group created in AD FS, to have the automatic user assignment in place.

3.3.2.2.2 Configure the Claim Rules for roles

Prerequisites

- Create the necessary user groups in AD FS: the default groups (Regulatory Analyst, System Administrator) and any other needed group.
- Add users to these groups. Add the admin user of OneSumX Regulatory Engine (admin.defaultuser) to the System Administrator group.



Context

In AD FS Management, you need to configure the Claim Rules for roles to receive the information in the SAML response.

You need to create a rule for the predefined groups as well as for the customized groups.

Procedure

1. Open AD FS Management, click the *Relying Party Trusts* folder, then select the Relying Party Trust for Admin Console. For details, refer to topic Add a Relying Party Trust.
2. On the **Actions** sidebar, click **Edit Claim Issuance Policy**.

   # Results

   The **Edit Claim Issuance Policy** dialog box is displayed.

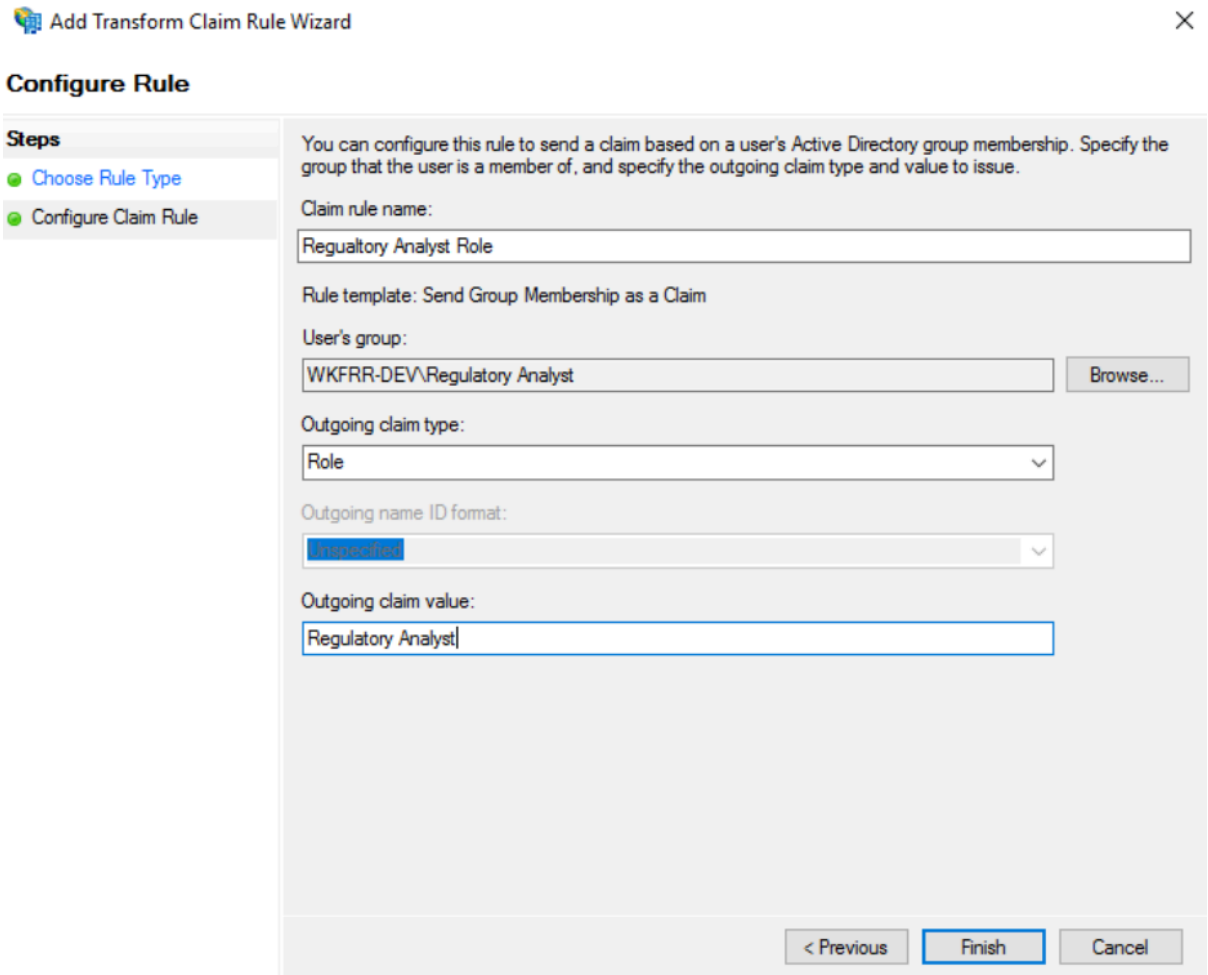
3. On the **Issuance Transform Rules** tab, click **Add Rule**.

   # Results

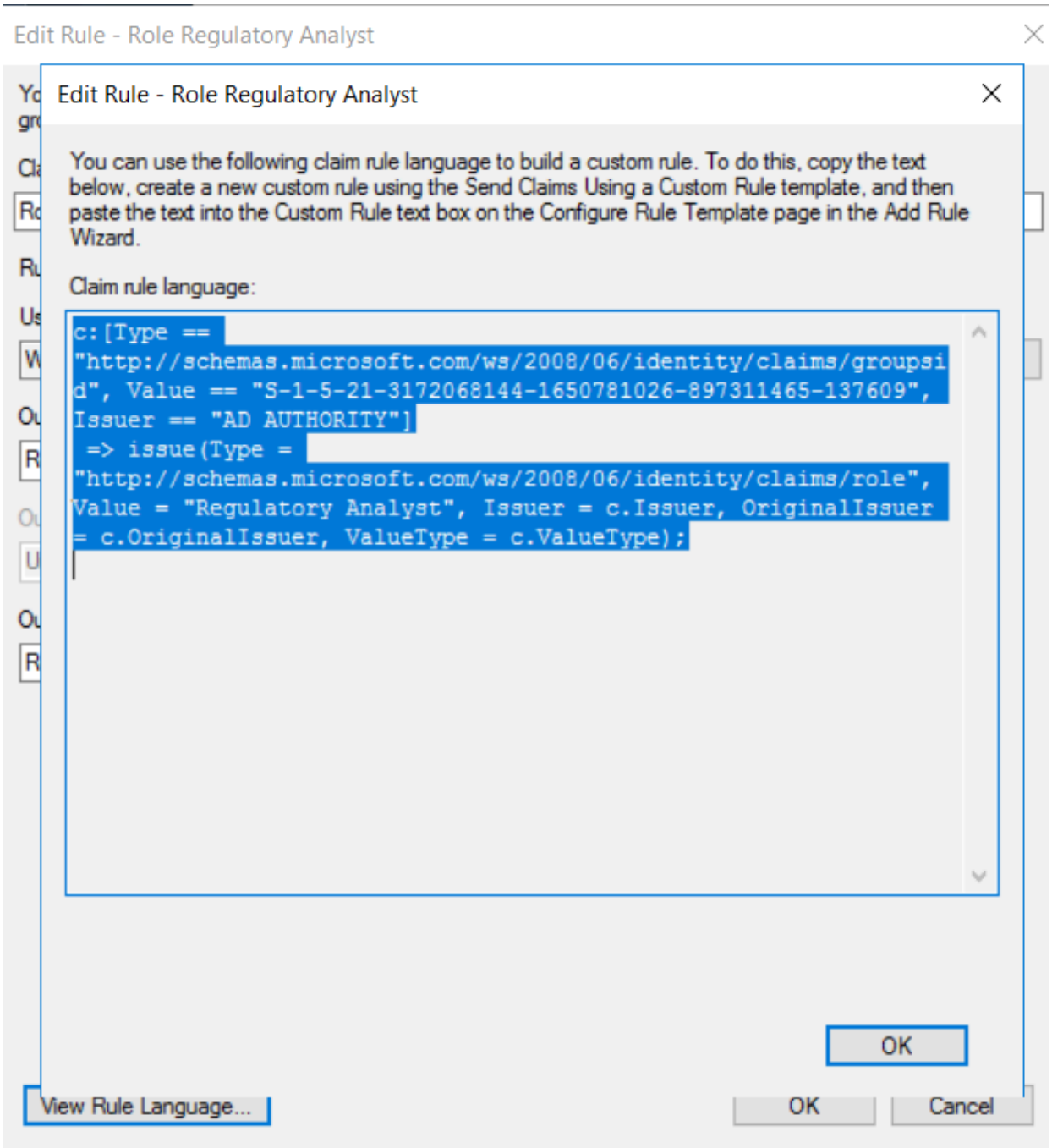
   The **Add Transform Claim Rule Wizard** is displayed.

4. On the **Select Rule Template** page, in the **Claim rule template** list, select the **Send Group Membership as a Claim** option, then click **Next**.
5. On the **Configure Rule** page, perform the following actions:

   - In the **Claim rule name** box, add a name for the rule.
   - Next to the **User's group** box, click **Browse** to select the user group that you want to map to the role rule, for example, Regulatory Analyst.
   - In the **Outgoing claim type** list, select **Role**.
   - In the **Outgoing claim value** box, enter the group's name.



6. Click **Finish** to save the role rule.
7. In the **Edit Claim Issuance Policy** dialog box, on the **Issuance Transform Rules** tab, select the role rule created at steps 3 to 6 in this procedure and click **Edit Rule**.
8. In the **Edit Rule** dialog box, click **View Rule Language**, then copy the text from the **Claim rule language** box.



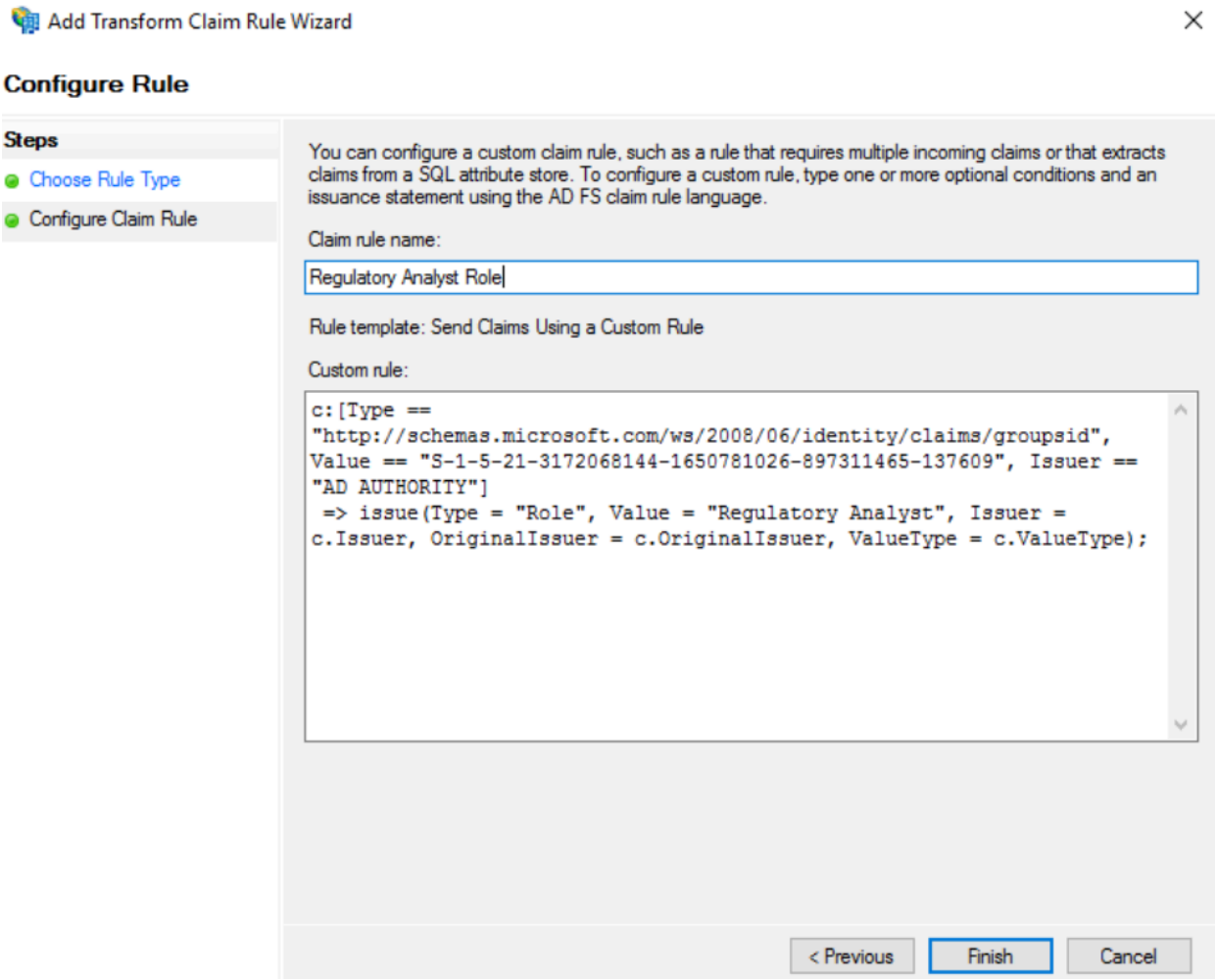
9. Click **OK** to return to the **Edit Rule** dialog box.
10. In the **Edit Claim Issuance Policy** dialog box, on the **Issuance Transform Rules** tab, select the rule that you created at steps 3 to 6 in this procedure and click **Remove Rule**.
11. In the **Edit Claim Issuance Policy** dialog box, on the **Issuance Transform Rules** tab, click **Add Rule**.

Results

The **Add Transform Claim Rule Wizard** is displayed.

12. On the **Select Rule Template** page, in the **Claim rule template** list, select the **Send Claims Using a Custom Rule** option, and click **Next**.
13. On the **Configure Rule** page, in the **Claim rule name** box, add a name for the rule.
14. In the **Custom rule** box, paste the text that you copied at step 8, and then delete the "http://schemas.microsoft.com/ws/2008/06/identity/claims/" string from the **Type** field. This should leave

only the word **role**. Change the first letter to uppercase: **Role**.



15. Click **Finish** to save the role rule.

What to do next

In OneSumX Regulatory Engine, you need to create a resource group or role with the exact name of the user group created in AD FS, to have the automatic user assignment in place.

4 Starting OneSumX Regulatory Engine

After installing/upgrading *OneSumX Regulatory Engine*, you will be able to start it.

To start OneSumX Regulatory Engine, you need to start the app server.

To start the app server, refer to one of the following topics, depending on your OneSumX Regulatory Engine implementation:

- Starting the application server for the embedded app server package
- Starting the app server for the external app server package

4.1 Starting the application server for the embedded app server package

You can start the application server for the embedded app server package using one of the following methods:

- Starting the app server as a service
- Start the app server via a script

We highly recommend running OneSumX Regulatory Engine as a service instead of launching it via a script (*startup-swarm.cmd*).

4.1.1 Starting the app server as a service

Prerequisites

- Register the OneSumX Regulatory Engine application as a Windows Service as described in topic Installing embedded app server package as a Windows Service.

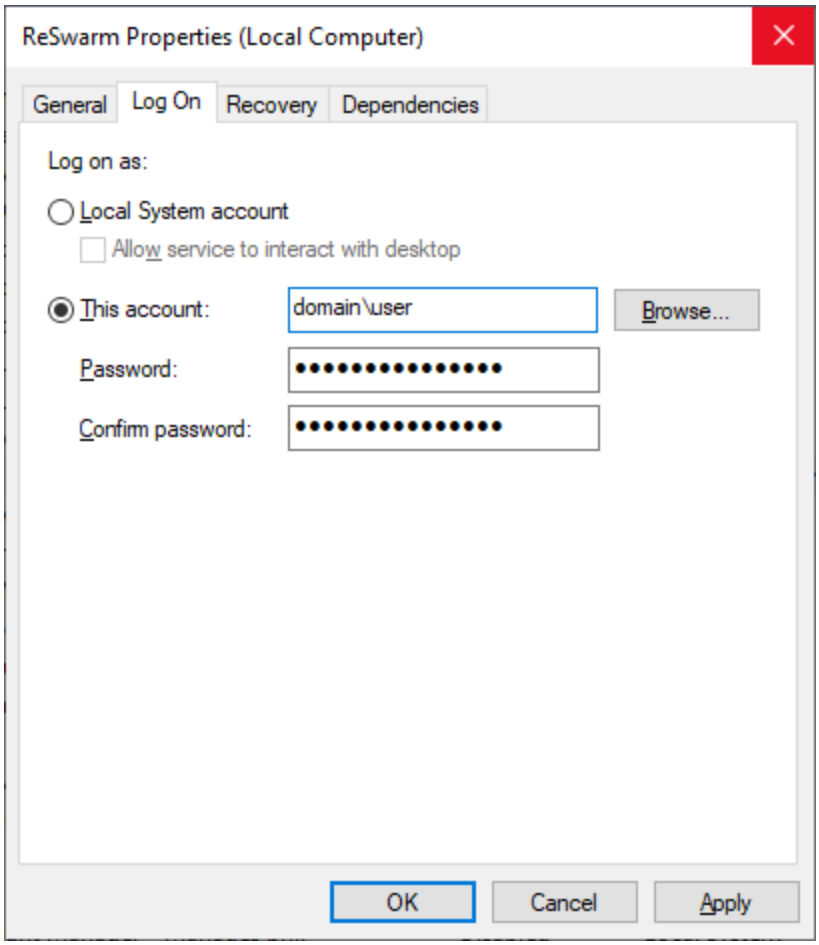
Procedure

1. Navigate to **Windows Services**.
2. Start the service.

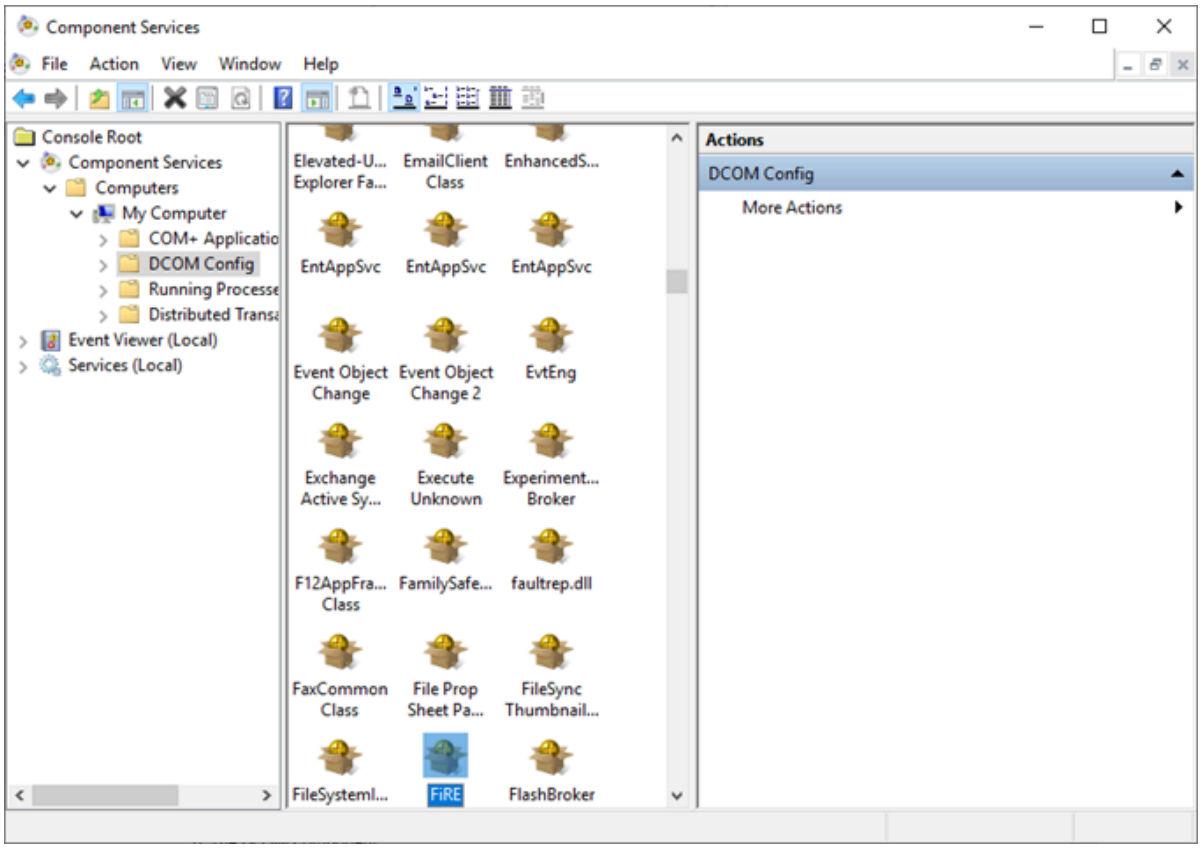
   Note:
   It may take about 2 minutes for the service to start up.

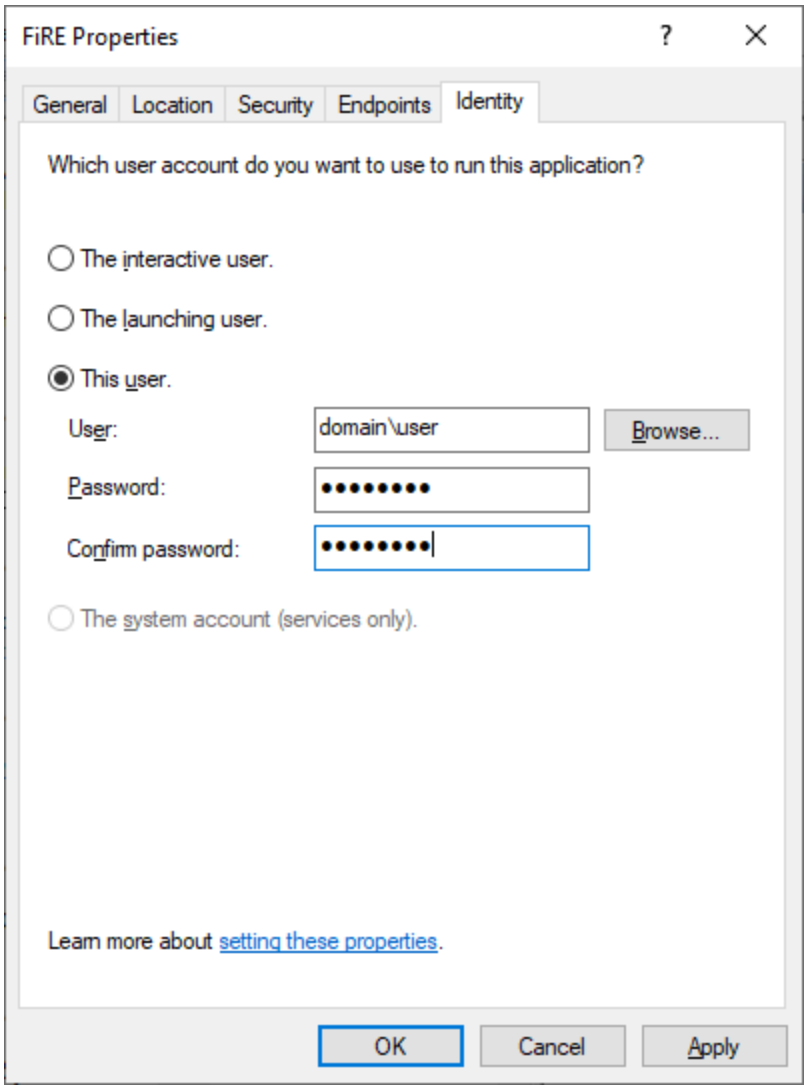
   If the application is started as a service, the user with which the service is started should have interactive logon enable if export of reports or workcollections(from Report Generator application) is performed while using LEGACY_REPORT_TO_EXCEL flag

3. OneSumX Regulatory Engine has no restrictions regarding the number of concurrent users, but the “Local system account” of the Microsoft Windows OS is restricted to 10 background processes. When the web server is running as a service under an account, to avoid this restriction, there are two steps to follow:
   a. Use a specific user account for the startup service:



b. Use a specific user account for the DCOM Component: in **Control Panel** > **Administrative Tools** > **Component Services**, set the FRS Fire object to **This User**.


Component Services
File Action View Window Help
Console Root
Component Services
Computers
My Computer
COM+ Applicatio
DCOM Config
Running Processe
Distributed Transa
Event Viewer (Local)
Services (Local)
Elevated-U... Explorer Fa...
EmailClient Class
EnhancedS...
EntAppSvc
EntAppSvc
EntAppSvc
Event Object Change
Event Object Change 2
EvtEng
Exchange Active Sy...
Execute Unknown
Experiment... Broker
F12AppFra... Class
FamilySafe...
faultrep.dll
FaxCommon Class
File Prop Sheet Pa...
FileSync Thumbnail...
FileSystemI...
FiRE
FlashBroker
Actions
DCOM Config
More Actions



Results

The app server containing OneSumX Regulatory Engine will be up and running and you can access it via your browser, at the following entry points:

- The admin console will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(admin-webapp.context)*, for example,https://machine-name:8080/admin-webapp.
- The reporting hub will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(reporting-webapp.context)*, for example, https://machine-name:8080/reporting-webapp.
- The ReportGenerator will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(rgtng-webapp.context)*, for example, https://machine-name:8080/reportgenerator.

4.1.1.1 Installing embedded app server package as a Windows Service

Procedure

1. Start a new command line in Administrator mode in directory *bin/service*.
2. Run the following command, after replacing `<service_name>` with the name of the service:

   ```
   service install /name <service_name>
   ```

   Ensure that the service name does not contain any space characters.

4.1.1.2 Uninstalling OneSumX Regulatory Engine with WildFly as a Windows Service

Procedure

1. Start a new command line in Administrator mode in directory *bin/service*.
2. Run the following command, after replacing `<service_name>` with the name of the service:

   ```
   service uninstall /name <service_name>
   ```

4.1.2 Start the app server via a script

Procedure

1. Start Command Prompt or PowerShell with administrator privileges.
2. Type *[installation drive:]\[RE installation folder]\bin* by typing:
   `cd [installation drive:]\[RE installation folder]\bin`.
3. Execute the *startup-swarm.cmd* batch file.

Results

Warning:
Starting with OneSumX Regulatory Engine 4.4.0, *startup-swarm.cmd* is not using WildFly Swarm anymore, but WildFly Full. The old naming will be removed in the future versions.

Upon execution of the batch file, the progress of the application deployment can be followed, and the successful deployment result and application startup are shown as illustrated here:

```
: /admin
10:09:00,782 INFO  [org.jboss.as.server] (main) WFLYSRV0010: Deployed "reporting-swarm-1.7.0-SNAPSHOT.war" (runtime-name
 : "reporting-swarm-1.7.0-SNAPSHOT.war")
10:09:00,793 INFO  [org.wildfly.swarm] (main) WFSWARM99999: WildFly Swarm is Ready
```

Note:
Starting the application may take between 5 and 10 minutes, depending on the computer.

The app server containing OneSumX Regulatory Engine will be up and running and you can access it via your browser, at the following entry points:

- The admin console will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(admin-webapp.context)*, for example,https://machine-name:8080/admin-webapp.
- The reporting hub will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(reporting-webapp.context)*, for example, https://machine-name:8080/reporting-webapp.
- The ReportGenerator will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(rgtng-webapp.context)*, for example, https://machine-name:8080/reportgenerator.

4.2 Starting the app server for the external app server package

To start the app server, refer to one of the following topics, depending on your implementation:

- Start WebSphere Liberty
- Start JBoss EAP

4.2.1 Start WebSphere Liberty

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Type `Cd [installation drive:]\[RE installation folder]\wlp\bin`.
3. Type server start.

Note:
Starting the application may take between 5 and 10 minutes, depending on the computer.

Results

The app server containing OneSumX Regulatory Engine will be up and running and you can access it via your browser, at the following entry points:

- The admin console will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(admin-webapp.context)*, for example,https://machine-name:8080/admin-webapp.
- The reporting hub will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(reporting-webapp.context)*, for example, https://machine-name:8080/reporting-webapp.
- The ReportGenerator will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(rgtng-webapp.context)*, for example, https://machine-name:8080/reportgenerator.

4.2.2 Start JBoss EAP

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Type `Cd [installation drive:]\[RE installation folder]\jboss-eap7.3`.
3. Type `standalone.bat`.

Note:
Starting the application may take between 5 and 10 minutes, depending on the computer.

Results

The app server containing OneSumX Regulatory Engine will be up and running and you can access it via your browser, at the following entry points:

- The admin console will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(admin-webapp.context)*, for example,https://machine-name:8080/admin-webapp.
- The reporting hub will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(reporting-webapp.context)*, for example, https://machine-name:8080/reporting-webapp.
- The ReportGenerator will be accessible via *https://${ wolterskluwer.application.host }:${ wolterskluwer.application.port }/ $(rgtng-webapp.context)*, for example, https://machine-name:8080/reportgenerator.

5 Stopping OneSumX Regulatory Engine

To stop OneSumX Regulatory Engine, you need to stop the app server.

To stop the app server, refer to one of the following topics, depending on your OneSumX Regulatory Engine implementation:

- Stopping the app server for the embedded app server package
- Stopping the app server for the external app server package

5.1 Stopping the app server for the embedded app server package

You can stop the application server for the embedded app server package using one of the following options:

- Stop the app server as a service
- Stop the app server via script

5.1.1 Stop the app server as a service

Context

Warning:
Stopping the service immediately after starting it will result in errors. Wait a few minutes for the WildFly to finish loading and then stop the service.

Procedure

1. Navigate to **Windows Services**.
2. Find the RE service.
3. Stop the service.

5.1.2 Stop the app server via script

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Type `cd [installation drive:]\[RE installation folder]/bin`.
3. Type `shutdown-swarm.cmd`.

5.2 Stopping the app server for the external app server package

To stop the app server, refer to one of the following topics, depending on your implementation:

- Stop WebSphere Liberty
- Stop JBoss EAP

5.2.1 Stop WebSphere Liberty

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Type `Cd [installation drive:]\[RE installation folder]\wlp\bin`.
3. Type server stop.

5.2.2 Stop JBoss EAP

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Type `Cd [installation drive:]\[RE installation folder]\jboss-eap-7.3`.
3. Type `./jboss-cli.sh --connect --command=:shutdown`

6 Upgrading OneSumX Regulatory Engine

Before upgrading OneSumX Regulatory Engine, you need to download and extract the Semansys package, as described in topic Download the Semansys package.

To upgrade OneSumX Regulatory Engine, refer to one of the following options, depending on your implementation:

- Upgrading OneSumX Regulatory Engine using an embedded app server package
- Upgrading OneSumX Regulatory Engine using an external app server package

Warning:
After the upgrade, in case an SQL Server user with the *Public* role is used in the workspace connection, refer to topic SQL Server user missing execute grant.

6.1 Download the Semansys package

Prerequisites

- You downloaded and extracted the distribution package for OneSumX Regulatory Engine.

Context

Starting from OneSumX Report Generator Web (TNG) version 2.24.0, the Semansys libraries are not distributed along with the engine.

These libraries are required if you need to perform a XBRL delivery.

Procedure

1. Go to the Product Download Center and download the following files: *RG_Semansys_artifacts* and *RG_Semansys_license*.
2. Extract them in the following location: *[installation drive:]\[RE installation folder]\deploy\legacy\WK ReportGenerator\bin*

6.2 Upgrading OneSumX Regulatory Engine using an embedded app server package

To upgrade OneSumX Regulatory Engine using an embedded app server package, you need to prepare the environment first, and then start the upgrading process, as follows:

1. Prepare the environment as follows:
   - Stop any running instance of OneSumX Regulatory Engine.
   - Stop any running instances of ReportGenerator by opening **Windows Task Manager** and discarding any existing ReportGenerator references.
   - Remove the previous installations of ReportGenerator via the **Windows Control Panel**.

Note:

- For the upgrade process for the configurations that use SQL Server, the wolterskluwer.db.connection.url property should contain the instance name, for example,

```
jdbc:sqlserver://localhost:1436;instanceName=SQLSERVER;databaseName=databaseName
```

- For the upgrade process for the configurations that use ORACLE, check that the wolterskluwer.oracle.tnsNames.folder property contains the path towards the new Oracle 19 Client, for example, *C:/app/oracle/product/19.3.0/client_1/network/admin*. Ensure that the path is updated in both *install.properties* files (the existing one and the one in the new version). The following path must be updated in the *init.xml* file as well: <tns-name url="*C:/app/oracle/product/19.3.0/client_1/network/admin*"/>

- Set a path outside the previously installed application folder for the following properties: wolterskluwer.ssl.keystore.path, workspaces.legacy.root, and workspaces.workdir.root. This enables you to avoid losing any of the previously created artifacts and to be able to use the updated application.

- Unregister the previous OneSumX Regulatory Engine Windows Service. For details, refer to Uninstalling OneSumX Regulatory Engine with WildFly as a Windows Service.

2. If you have an SQL Server with a username/password authentication and you want to switch to an integrated security, then follow the procedures described in topic Switching to integrated security on SQL Server for the embedded app server package.

3. Start the upgrading process, using one of the following options:

   - Upgrading OneSumX Regulatory Engine from folder

   - Upgrading OneSumX Regulatory Engine based on an archive

6.2.1 Switching to integrated security on SQL Server for the embedded app server package

Context

Note:
The following feature works on versions 4.1.0 and later.

The following steps need to be executed before launching *the upgrade-platform.cmd* script in order to switch to the integrated security for the SQL Server configuration. For this section, we assume that the previous installation folder is *C:\previous-reporting-distribution*, the new installation folder is *C:\new-reporting-distribution* and the system database name is SYSTEM_DB.

Procedure

1. Modify the *C:\previous-reporting-distribution\conf\install.properties*:
   a. Set thewolterskluwer.sqlserver.integrated.security property to true.
   b. Remove the following properties, as they will no longer be used:

      - wolterskluwer.system.db.username

      - wolterskluwer.system.db.password

      - wolterskluwer.application.db.username

- wolterskluwer.application.db.password

2. Modify the *C:\previous-reporting-distribution\conf\init.xml*:
   a. If properties `<userDataBaseRepository hibernateConnectionId="SYSTEM_DB"/>` `<HibernateConfiguration name="SYSTEM_DB">` and `<db-connection-id value="SYSTEM_DB"/>` do not refer to the system database name, then change them to match.
   b. Modify the property `<property name="connection.url">` by adding ;integratedSecurity=true at the end of the value
   c. Delete two properties named `<property name="connection.username">` and `<property name="connection.password">`. The final configuration of the session-factory part should look as follows:

Example

```
<session-factory>
<property
name="connection.driver_class">com.microsoft.sqlserver.jdbc.SQLServerDriver<
/property>
<property
name="connection.url">jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB
;integratedSecurity=true</property>
<property
name="dialect">org.hibernate.dialect.SQLServer2012Dialect</property>
<property name="hibernate.jdbc.use_get_generated_keys">true</property>
<property name="hibernate.c3p0.min_size">5</property>
<property name="hibernate.c3p0.max_size">20</property>
<property name="hibernate.c3p0.timeout">300</property>
<property name="hibernate.c3p0.max_statements">0</property>
<property name="hibernate.c3p0.idle_test_period">300</property>
<property name="hibernate.current_session_context_class">thread</property>
<property
name="hibernate.connection.provider_class">org.hibernate.connection.C3P0Conn
ectionProvider</property>
<property name="hibernate.show_sql">false</property>
</session-factory>
```

3. Modify *C:\previous-reporting-distribution\conf\init-authentication.xml*:
   a. If the property `<userDataBaseRepository hibernateConnectionId="SYSTEM_DB"/>` does not refer to the system database name, then change it to match.
4. Modify *C:\previous-reporting-distribution\conf\standalone.xml*:
   a. In the datasources section, find the configuration of the system datasource:

Example

```
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/
SYSTEM_DB" pool-name="SYSTEM_DB">
    <connection-
url>jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB</connection-url>
<driver>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver>
    <security>
                <user-name>RE_USER</user-name>
                <password>P@ssw0rd</password>
```

```
        </security>
</datasource>
```

Note:
The datasource section may contain extra properties such as validation and pool settings. If these properties are present, they should be left unchanged.

b. Modify the property `<connection-url>` by adding ;integratedSecurity=true at the end of the value.

c. Remove the entire `<security>` section with `<user-name>` and `<password>` properties inside.

The final configuration should look as follows:

Example

```
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/
SYSTEM_DB" pool-name="SYSTEM_DB">
    <connection-
url>jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB;integratedSecurit
y=true</connection-url>
<driver>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver>
</datasource>
```

5. Repeat step 4, but for the LDM and GDM datasources.
6. Copy the following SQL script into the database client executor, provide the actual values for the placeholders, and execute:

   The following SQL code snippet illustrates the placeholder definitions:

```
USE ${systemDBName};
GO
DELETE FROM AUTH_DATABASE_CONFIG
  FROM AUTH_DATABASE_CONFIG config
  JOIN AUTH_DATABASE_CONF_PROPERTY property
 ON config.PROPERTY_ID = property.ID
  JOIN workspace_databases wdb ON config.DATABASE_ID = wdb.database_id
  JOIN workspace ON workspace.id = wdb.workspace_id
WHERE property.NAME in ('connection.username', 'connection.password')
  AND workspace.name = '${workspaceName}'
GO
UPDATE AUTH_DATABASE_CONFIG
        SET VALUE = CONCAT(VALUE, ';integratedSecurity=true')
        FROM AUTH_DATABASE_CONFIG config JOIN AUTH_DATABASE_CONF_PROPERTY
property
        ON config.PROPERTY_ID = property.ID
        JOIN workspace_databases wdb on config.DATABASE_ID = wdb.database_id
        JOIN workspace on workspace.id = wdb.workspace_id
                WHERE property.NAME in ('connection.url')
                AND workspace.name = '${workspaceName}'
```

systemDBName

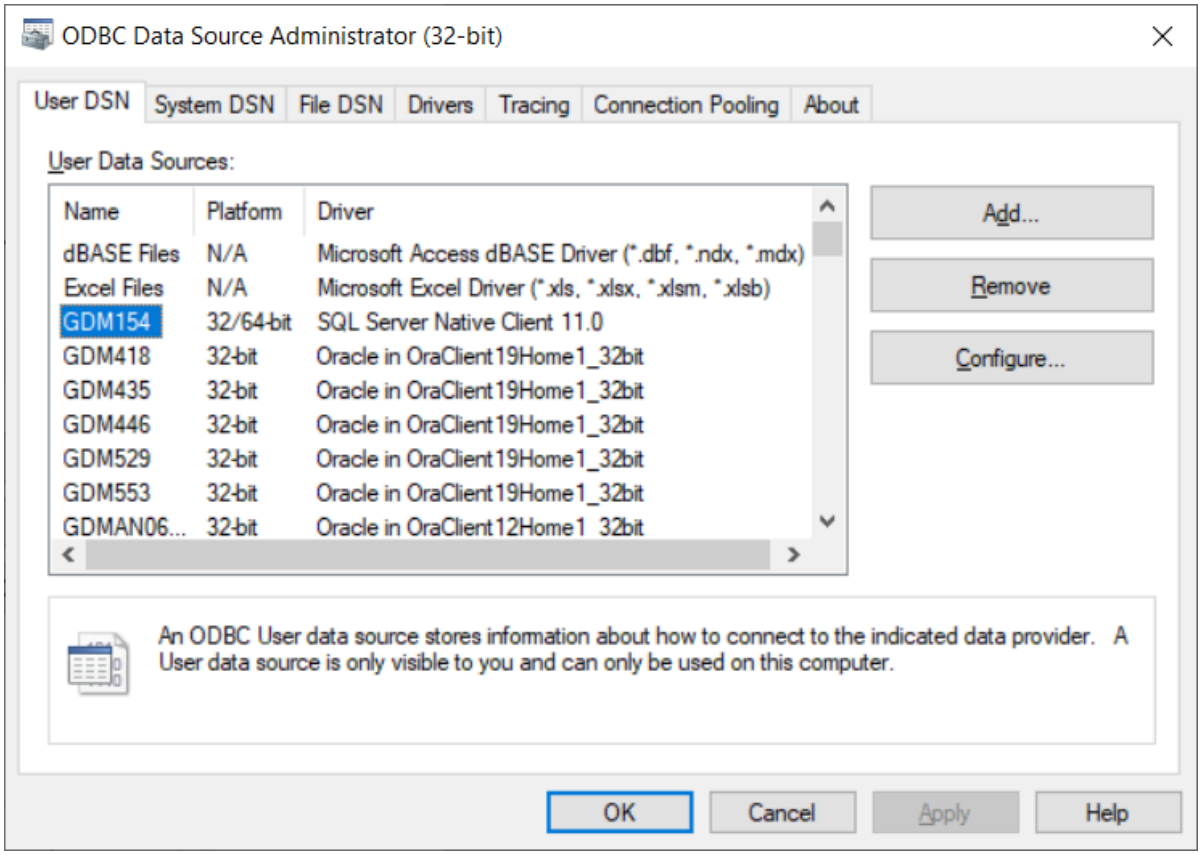
System database name

workspaceName

Workspace name

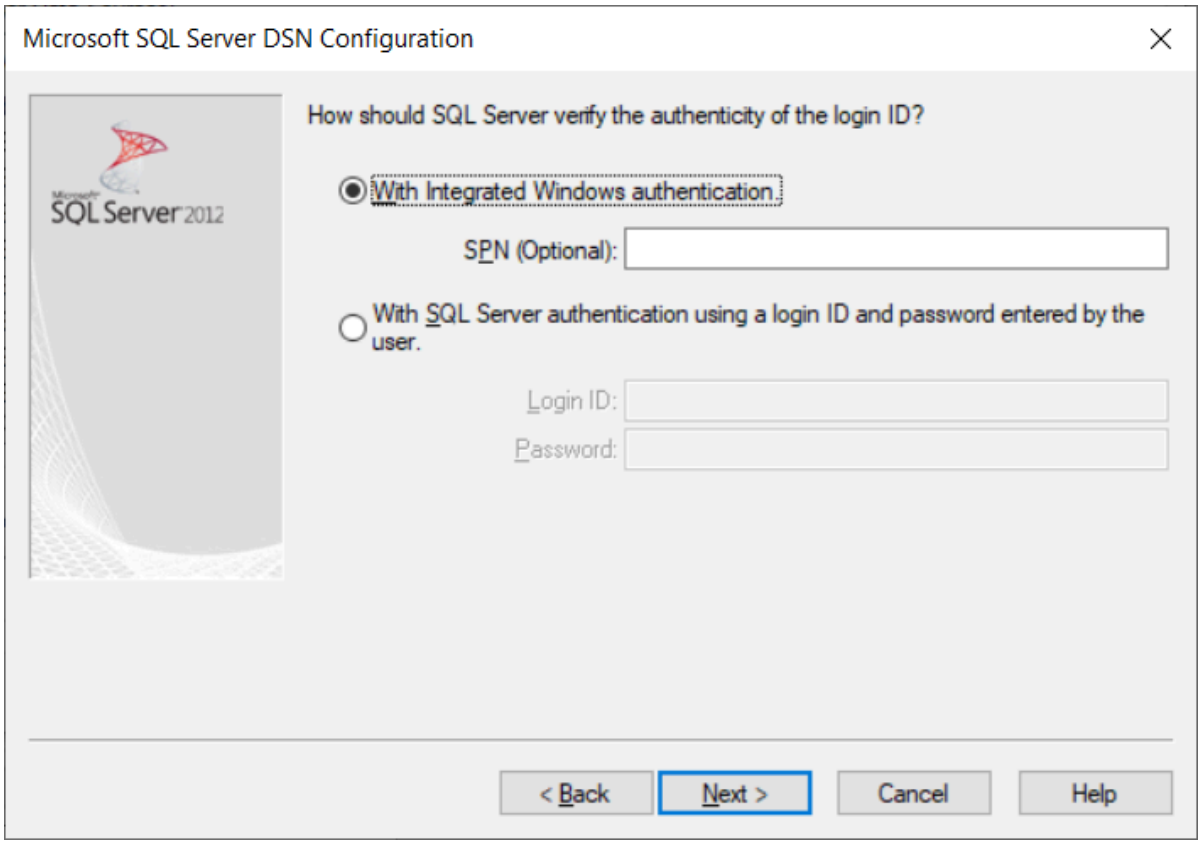
7. Change the ODBC data source configuration by running odbcad32.exe - "ODBC Data Sources":
   a. On the **User DSN** tab, select the data source, and click **Configure**.

Example



   b. Click **Next**.
   c. On the following screen, select the **With integrated Windows authentication** authenticity verification method.

Example



d. Click **Next**, leave the rest as default and complete the configuration.

8. Repeat step 7, but for the LDM datasource.
9. Proceed to the upgrade process.

Note:

Upon successful completion of the upgrade process, ensure that *C:\new-reporting-distribution\conf\install.properties* does not contain the following properties, otherwise, remove them manually:

- wolterskluwer.system.db.username
- wolterskluwer.system.db.password
- wolterskluwer.application.db.username
- wolterskluwer.application.db.password

6.2.2 Upgrading OneSumX Regulatory Engine from folder or from an archive

You can upgrade OneSumX Regulatory Engine, using one of the following options:

- Upgrading OneSumX Regulatory Engine from folder
- Upgrading OneSumX Regulatory Engine based on an archive

6.2.2.1 Upgrading OneSumX Regulatory Engine from folder

Context

Take the steps in this procedure if you meet both of the following conditions::

- You have an existing installation in *C:\previous-reporting-distribution*
- You want to install a new version in *C:\new-reporting-distribution*.

Procedure

1. Open file *[new installation drive:]\[new RE installation folder]\conf\install.properties* using your preferred text editor.
2. Provide the applicable location for the following property:

```
install.upgrade.from.folder.with.previous.installation=C:/path/to/previous/
installation
```

3. Save the *[new installation drive:]\[new RE installation folder]\conf\install.properties* file.
4. Generate the security key as described in topic Application database password encryption.
5. Perform one of the following actions:

   - If you saved the security key value specified at step 4 in the *\conf\install.properties* file, right-click the script *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* and select the **Run as administrator** option.
   - If you didn't save the security key specified at step 4, then open a Command Prompt window with administrative rights and enter the following path *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* then add

     `-Dwolterskluwer.db.passphrase.encryption.cipher.securityKey=[MTZDaGFyU2VjdXJlS2V5IQ==]`

     , where `[MTZDaGFyU2VjdXJlS2V5IQ==]` represents the generated security key value.

Results

The Installer will merge *install.properties* of the previous installation into the new installation and copy the contents of the *conf/security* and *etc/content-store/repository* folders.

6. Refer to topic (Datasource connection improvements for existing datasources) before starting the application.

   Note:
   The contents of the folder from the previous installation will be removed completely.

Warning:
To avoid losing any of the previously created artifacts and to be able to use the updated application, ensure that the wolterskluwer.ssl.keystore.path, workspaces.legacy.root, and workspaces.workdir.root properties are set to a path outside the previously installed application folder.

In case of an engine upgrade from OneSumX Regulatory Engine 3.0.0 to 3.1.0 and then to 4.7.0, the upgrade might fail if the 3.1.0 *install.properties* parameter admin.defaultuser value was changed before running the *upgrade-platform.cmd* script. Refer to topic Platform upgrade fails with message "Error during database update" to troubleshoot this issue.

6.2.2.1.1 Datasource connection improvements for existing datasources

Starting with OneSumX Regulatory Engine 4.4.3, some properties for the datasource connection will be added automatically in *[installation drive:]\[RE installation folder]/conf/standalone.xml*.

Depending on your system configuration, choose one of the following options:

- Make improvements to the Oracle datasources
- Make improvements for the SQL Server datasources

6.2.2.1.1.1 Make improvements to the Oracle datasources

Procedure

1. Go to *C:\server\jboss-eap-7.3\conf\standalone.xml*.
2. Find the section that contains the GDM datasource configuration:

```
<datasource jndi-name="java:/wolterskluwer/datasources/GDM_DS" pool-
name="GDM_DS" enabled="true">
        <connection-url>jdbc:oracle:thin:@localhost:1521/ORACLE</connection-
url>
        <driver>oracle</driver>
        <security>
                <user-name>GDM_NAME</user-name>
                <password>GDM_PASSWORD</password>
        </security>
</datasource>
```

3. Add the pool configuration under the **<datasource>** tag.

```
<pool>
      <initial-pool-size>10</initial-pool-size>
       <max-pool-size>32</max-pool-size>
       <min-pool-size>10</min-pool-size>
</pool>
```

4. Add the additional connection under the **<datasource>** tag.

```
<connection-property name="oracle.net.networkCompression">on</connection-
property>
<connection-property
name="oracle.net.networkCompressionThreshold">1024</connection-property>
```

5. Add the validation configuration under the **<datasource>** tag.

```
<validation>
                <check-valid-connection-sql>select 1 from dual</check-valid-
connection-sql>
                <background-validation>true</background-validation>
                <background-validation-millis>10000</background-validation-
millis>
        </validation>
```

6. Repeat steps 2 to 5 for all created datasources (GDM, LDM, and system database).

GDM datasource configuration example

```
<datasource jndi-name="java:/wolterskluwer/datasources/GDM_DS" pool-name="GDM_DS"
enabled="true">
        <connection-url>jdbc:oracle:thin:@localhost:1521/ORACLE</connection-url>
        <driver>oracle</driver>
        <pool>
                <initial-pool-size>10</initial-pool-size>
                <max-pool-size>32</max-pool-size>
                <min-pool-size>10</min-pool-size>
        </pool>
        <connection-property name="oracle.net.networkCompression">on</connection-
property>
        <connection-property
name="oracle.net.networkCompressionThreshold">1024</connection-property>
        <security>
                <user-name>GDM_NAME</user-name>
                <password>GDM_PASSWORD</password>
        </security>
        <validation>
                <check-valid-connection-sql>select 1 from dual</check-valid-
connection-sql>
                <background-validation>true</background-validation>
                <background-validation-millis>10000</background-validation-millis>
        </validation>
</datasource>
```

6.2.2.1.1.2 Make improvements for the SQL Server datasources

Procedure

1. Open file*[installation drive:]\[RE installation folder]\conf\standalone.conf*
2. Find the datasource configuration and remove the following properties:

```
<connection-property name="oracle.net.networkCompression">on</connection-
property>
<connection-property
name="oracle.net.networkCompressionThreshold">1024</connection-property>
```

3. Change the value of property `<check-valid-connection-sql>` from `select 1 from dual` to `select 1`.

6.2.2.2 Upgrading OneSumX Regulatory Engine based on an archive

Context

After you unistall OneSumX Regulatory Engine, several archived files are generated. You can use those archived files to upgrade OneSumX Regulatory Engine.

Assuming you have an archive of configurations and content repository:

- *[old installation drive:]\[old RE installation folder]\uninstall\configuration.zip*
- *[old installation drive:]\[old RE installation folder]\uninstall\content-store.zip*

and you want to upgrade OneSumX Regulatory Engine at *[new installation drive:]\[new RE installation folder]*, follow this procedure.

Procedure

1. Open the file *[new installation drive:]\[new RE installation folder]\conf\install.properties.*
2. Provide the path and file names for the following properties:

```
install.upgrade.from.zip.with.config=[old installation drive:]\[old RE
installation folder]/uninstall/configuration.zip
install.upgrade.from.zip.with.content.store=[old installation drive:]\[old RE
installation folder]/uninstall/content-store.zip
```

3. Save the *[new installation drive:]\[new RE installation folder]\conf\install.properties* file.
4. Generate the security key as described in topic Application database password encryption.
5. Perform one of the following actions:

   - If you saved the security key value specified at step 4 in the *\conf\install.properties* file, right-click the script *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* and select the **Run as administrator** option.

   - If you didn't save the security key specified at step 4, then open a Command Prompt window with administrative rights and enter the following path *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* then add

     `-Dwoltersskluwer.db.passphrase.encryption.cipher.securityKey=[MTZDaGFyU2VjdXJlS2V5IQ==]`

     , where `[MTZDaGFyU2VjdXJlS2V5IQ==]` represents the generated security key value.

6. Right-click the script *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* and select the **Run as administrator** option.

Results

The installer will unzip the *configuration.zip* file and merge *install.properties* of the previous installation into the new installation and copy contents of the *conf/security* folder. The contents of the *content-store.zip* file will also be unzipped into the respective folders.

Warning:

If the following message appears, while running the upgrade:

```
[] [main] com.wkfsfrc.platform.installer.Installer [ERROR] (2019-08-14
11:24:25.531)         [level=ERROR step=VALIDATE_UPGRADE_INPUT_PROPERTIES
message=Missing value for property: [new.property.name]]
[] [main] wkfsfrc.platform.installer.InstallerMain [ERROR] (2019-08-14
11:24:25.531) ValidationException while validating installation input
properties. Exiting...
com.wkfsfrc.platform.installer.exception.ValidationException: Errors were
encountered while validating input properties. Please correct the errors
and rerun the installer
```

That means the mentioned property new.property.name was added in the new version and it does not have a value from the previous installation.

It needs to be given a value by editing the *C:\new-reporting-distribution\conf\install.properties* file.

6.3 Upgrading OneSumX Regulatory Engine using an external app server package

To upgrade OneSumX Regulatory Engine using an external container, you need to prepare the environment first, and then start the upgrading process, as follows:

1. Prepare the environment as follows:

   - Stop any running instance of OneSumX Regulatory Engine and the app server (WebSphere Liberty or JBoss EAP).
   - Stop any running instances of ReportGenerator by opening **Windows Task Manager** and discarding any existing ReportGenerator references.
   - Remove the previous installations of ReportGenerator via the **Windows Control Panel**.

   Note:

   - For the upgrade process for the configurations that use SQL Server, the wolterskluwer.db.connection.url property should contain the instance name, for example,

     ```
     jdbc:sqlserver://localhost:1436;instanceName=SQLSERVER;databaseName=databaseName
     ```

   - For the upgrade process for the configurations that use ORACLE, check that the wolterskluwer.oracle.tnsNames.folder property contains the path towards the new Oracle 19 Client, for example, *C:/app/oracle/product/19.3.0/client_1/network/admin*. Ensure that the path is updated in both *install.properties* files (the existing one and the one in the new version). The following path must be updated in the *init.xml* file as well: <tns-name url="*C:/app/oracle/product/19.3.0/client_1/network/admin*"/>
   - Set a path outside the previously installed application folder for the following properties: wolterskluwer.ssl.keystore.path, workspaces.legacy.root, and workspaces.workdir.root. This enables you to avoid losing any of the previously created artifacts and to be able to use the updated application.

2. If you have an SQL Server with a username/password authentication and you want to switch to an integrated security, then follow the procedures described in topic Switching to integrated security on SQL Server for the external app server package.

3. Start the upgrading process by providing a previous installation path, using one of the following options:

- Upgrading OneSumX Regulatory Engine from folder
- Upgrading OneSumX Regulatory Engine based on an archive

4. Perform the platform upgrade, depending on the application server:
   - Upgrade process for WebSphere Liberty
   - Upgrade process for JBoss EAP

6.3.1 Switching to integrated security on SQL Server for the external app server package

If you have an SQL Server with a username/password authentication and you want to switch to an integrated security, then choose one of the following options, depending on your implementation:

- Switching to integrated security for WebSphere Liberty
- Switching to integrated security for JBoss EAP

6.3.1.1 Switching to integrated security for WebSphere Liberty

Context

Note:
The following feature works on versions 4.1.0 and later.

The following steps need to be executed before launching *the upgrade-platform.cmd* script in order to switch to the integrated security for the SQL Server configuration. For this section, we assume that the previous installation folder is *C:\previous-reporting-distribution*, the new installation folder is *C:\new-reporting-distribution* and the system database name is SYSTEM_DB.

Procedure

1. Modify the *C:\previous-reporting-distribution\conf\install.properties*:
   a. Set thewolterskluwer.sqlserver.integrated.security property to true.
   b. Remove the following properties, as they will no longer be used:
      - wolterskluwer.system.db.username
      - wolterskluwer.system.db.password
      - wolterskluwer.application.db.username
      - wolterskluwer.application.db.password
2. Modify the *C:\previous-reporting-distribution\conf\init.xml*:
   a. If properties `<userDataBaseRepository hibernateConnectionId="SYSTEM_DB"/>` `<HibernateConfiguration name="SYSTEM_DB">` and `<db-connection-id value="SYSTEM_DB"/>` do not refer to the system database name, then change them to match.

b. Modify the property `<property name="connection.url">` by adding ;integratedSecurity=true at the end of the value

c. Delete two properties named `<property name="connection.username">` and `<property name="connection.password">`. The final configuration of the session-factory part should look as follows:

Example

```
<session-factory>
<property
name="connection.driver_class">com.microsoft.sqlserver.jdbc.SQLServerDriver<
/property>
<property
name="connection.url">jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB
;integratedSecurity=true</property>
<property
name="dialect">org.hibernate.dialect.SQLServer2012Dialect</property>
<property name="hibernate.jdbc.use_get_generated_keys">true</property>
<property name="hibernate.c3p0.min_size">5</property>
<property name="hibernate.c3p0.max_size">20</property>
<property name="hibernate.c3p0.timeout">300</property>
<property name="hibernate.c3p0.max_statements">0</property>
<property name="hibernate.c3p0.idle_test_period">300</property>
<property name="hibernate.current_session_context_class">thread</property>
<property
name="hibernate.connection.provider_class">org.hibernate.connection.C3P0Conn
ectionProvider</property>
<property name="hibernate.show_sql">false</property>
</session-factory>
```

3. Modify *C:\previous-reporting-distribution\conf\init-authentication.xml*:

   a. If the property `<userDataBaseRepository hibernateConnectionId="SYSTEM_DB"/>` does not refer to the system database name, then change it to match.

4. Modify the *server.xml*:

   a. Find the system datasource configuration:

Example

```
<dataSource jndiName="jdbc/wolterskluwer/datasources/SYSTEM_DB"
type="java.sql.Driver" transactional="false">
    <jdbcDriver libraryRef="SQLDriver" />
    <properties URL="jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB"
databaseName="SYSTEM_DB" user="SYSTEM_DB" password="welcome" />
    <containerAuthData user="SYSTEM_DB" password="welcome" />
</dataSource>
```

   b. Rename the property properties into properties.microsoft.sqlserver.

   c. Add a new attribute *integratedSecurity="true"* to the property properties.microsoft.sqlserver.

   d. If the attributes *serverName* and *instanceName* (if used) are missing, then add them to the property properties.microsoft.sqlserver.

   e. Remove the attributes *user* and *password* from the property properties.microsoft.sqlserver.

   f. Remove the property containerAuthData, so that the final configuration should look as follows:

Example

```
<dataSource jndiName="jdbc/wolterskluwer/datasources/SYSTEM_DB"
type="javax.sql.DataSource" transactional="false">
    <jdbcDriver libraryRef="SQLDriver"/>
    <properties.microsoft.sqlserver
URL="jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB"
integratedSecurity="true" databaseName="SYSTEM_DB" serverName="localhost"/>
</dataSource>
```

5. Repeat the previous step, but for the LDM and GDM datasources.
6. Copy the following SQL script into the database client executor, provide the actual values for the placeholders, and execute:

   The following SQL code snippet illustrates the placeholder definitions:

```
USE ${systemDBName};
GO
DELETE FROM AUTH_DATABASE_CONFIG
  FROM AUTH_DATABASE_CONFIG config
  JOIN AUTH_DATABASE_CONF_PROPERTY property
 ON config.PROPERTY_ID = property.ID
  JOIN workspace_databases wdb ON config.DATABASE_ID = wdb.database_id
  JOIN workspace ON workspace.id = wdb.workspace_id
WHERE property.NAME in ('connection.username', 'connection.password')
  AND workspace.name = '${workspaceName}'
GO
UPDATE AUTH_DATABASE_CONFIG
        SET VALUE = CONCAT(VALUE, ';integratedSecurity=true')
        FROM AUTH_DATABASE_CONFIG config JOIN AUTH_DATABASE_CONF_PROPERTY
property
        ON config.PROPERTY_ID = property.ID
        JOIN workspace_databases wdb on config.DATABASE_ID = wdb.database_id
        JOIN workspace on workspace.id = wdb.workspace_id
                 WHERE property.NAME in ('connection.url')
                 AND workspace.name = '${workspaceName}'
```

   systemDBName

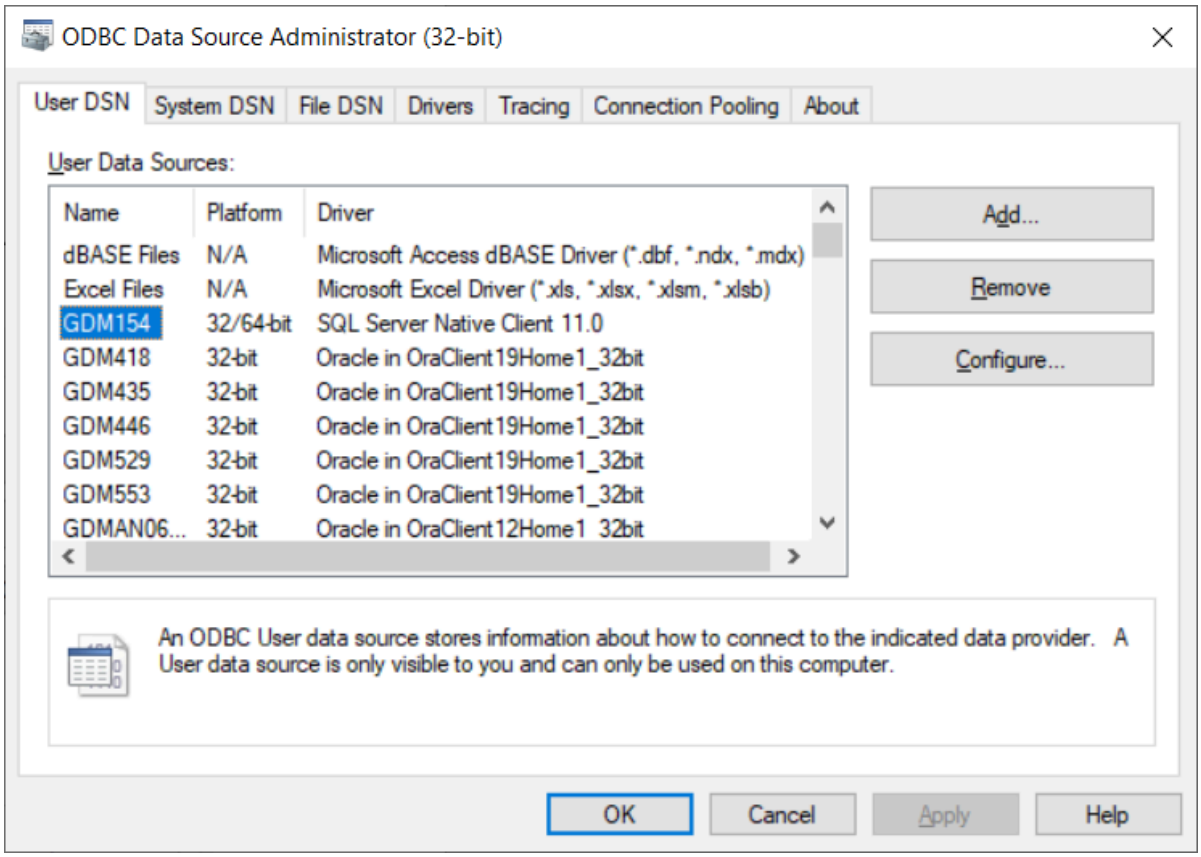
   System database name

   workspaceName

   Workspace name

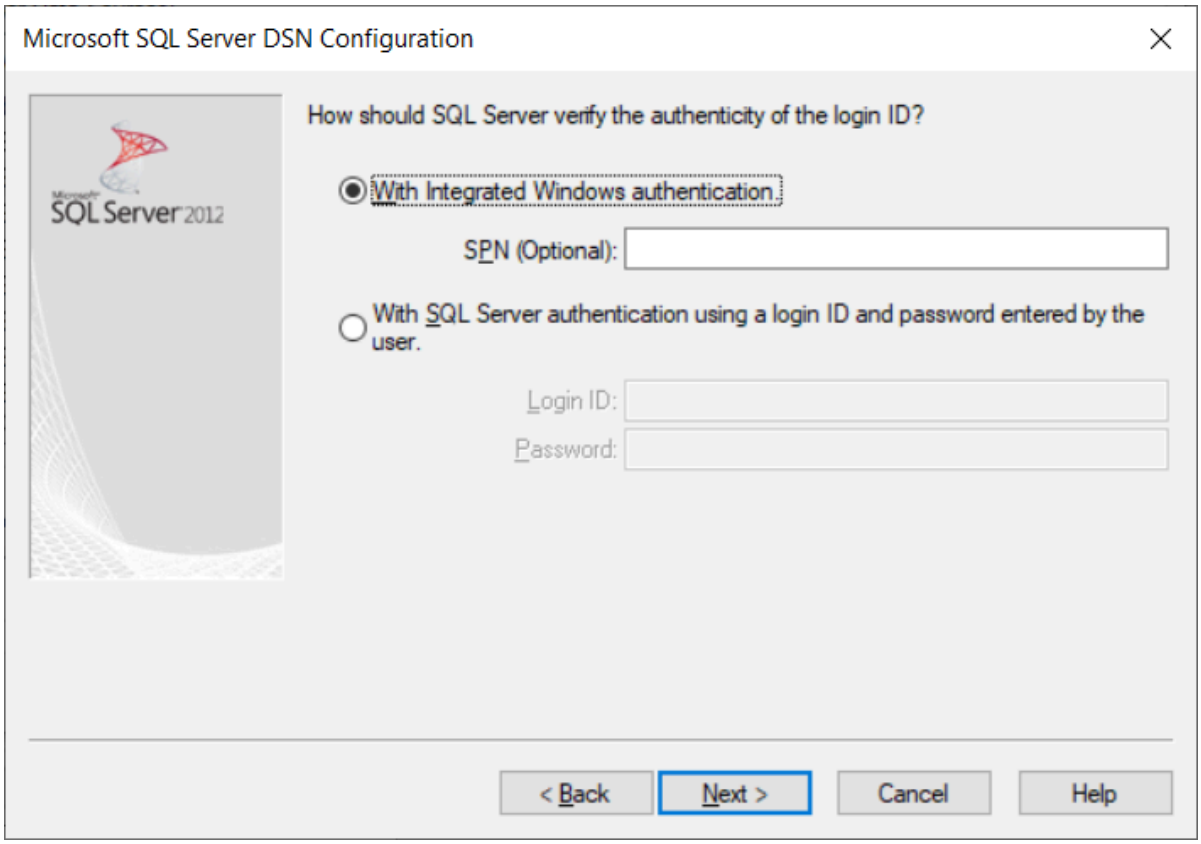
7. Change the ODBC data source configuration by running odbcad32.exe - "ODBC Data Sources":
   a. On the **User DSN** tab, select the data source, and click **Configure**.

Example



b. Click **Next**.
c. On the following screen, select the **With integrated Windows authentication** authenticity verification method.

Example



d. Click **Next**, leave the rest as default and complete the configuration.

8. Repeat step ta_switching_to_integrated_security_on_sql_server_at_the_engine_upgrade.html#id-3033dd23-2913-4d57-8aa6-e8cd67b e3a4154c-bd49-4762-ea19-83286d721ecd, but for LDM datasource.
9. Proceed to the upgrade process.

Note:
Upon successful completion of the upgrade process, ensure that *C:\new-reporting-distribution\conf\install.properties* does not contain the following properties, otherwise, remove them manually:

- wolterskluwer.system.db.username
- wolterskluwer.system.db.password
- wolterskluwer.application.db.username
- wolterskluwer.application.db.password

6.3.1.2 Switching to integrated security for JBoss EAP

Context

Note:
The following feature works on versions 4.1.0 and later.

The following steps need to be executed before launching *the upgrade-platform.cmd* script in order to switch to the integrated security for the SQL Server configuration. For this section, we assume that the previous installation folder is *C:\previous-reporting-distribution*, the new installation folder is *C:\new-reporting-distribution* and the system database name is SYSTEM_DB.

Procedure

1. Modify the *C:\previous-reporting-distribution\conf\install.properties*:
   a. Set thewolterskluwer.sqlserver.integrated.security property to true.
   b. Remove the following properties, as they will no longer be used:
      - wolterskluwer.system.db.username
      - wolterskluwer.system.db.password
      - wolterskluwer.application.db.username
      - wolterskluwer.application.db.password
2. Modify the *C:\previous-reporting-distribution\conf\init.xml*:
   a. If properties `<userDataBaseRepository hibernateConnectionId="SYSTEM_DB"/>` `<HibernateConfiguration name="SYSTEM_DB">` and `<db-connection-id value="SYSTEM_DB"/>` do not refer to the system database name, then change them to match.
   b. Modify the property `<property name="connection.url">` by adding ;integratedSecurity=true at the end of the value
   c. Delete two properties named `<property name="connection.username">` and `<property name="connection.password">`. The final configuration of the session-factory part should look as follows:

Example

```
<session-factory>
<property
name="connection.driver_class">com.microsoft.sqlserver.jdbc.SQLServerDriver<
/property>
<property
name="connection.url">jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB
;integratedSecurity=true</property>
<property
name="dialect">org.hibernate.dialect.SQLServer2012Dialect</property>
<property name="hibernate.jdbc.use_get_generated_keys">true</property>
<property name="hibernate.c3p0.min_size">5</property>
<property name="hibernate.c3p0.max_size">20</property>
<property name="hibernate.c3p0.timeout">300</property>
<property name="hibernate.c3p0.max_statements">0</property>
<property name="hibernate.c3p0.idle_test_period">300</property>
<property name="hibernate.current_session_context_class">thread</property>
<property
name="hibernate.connection.provider_class">org.hibernate.connection.C3P0Conn
ectionProvider</property>
<property name="hibernate.show_sql">false</property>
</session-factory>
```

3. Modify *C:\previous-reporting-distribution\conf\init-authentication.xml*:
   a. If the property `<userDataBaseRepository hibernateConnectionId="SYSTEM_DB"/>` does not refer to the system database name, then change it to match.
4. Modify the file *standalone\configuration\standalone.xml*:
   a. In the datasources section, find the configuration of the system datasource:

Example

```
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/
SYSTEM_DB" pool-name="SYSTEM_DB">
    <connection-
url>jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB</connection-url>
<driver>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver>
    <security>
                <user-name>RE_USER</user-name>
                <password>P@ssw0rd</password>
        </security>
</datasource>
```

Note:
The datasource section may contain extra properties such as validation and pool settings. If these properties are present, they should be left unchanged.

   a. Modify the property `<connection-url>` by adding ;integratedSecurity=true at the end of the value.
   a. Remove the entire `<security>` section with `<user-name>` and `<password>` properties inside.

   The final configuration should look as follows:

Example

```
<datasource jta="false" jndi-name="java:/wolterskluwer/datasources/
SYSTEM_DB" pool-name="SYSTEM_DB">
    <connection-
url>jdbc:sqlserver://localhost:1433;databaseName=SYSTEM_DB;integratedSecurit
y=true</connection-url>
<driver>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver>
</datasource>
```

5. Repeat the previous step, but for the LDM and GDM datasources.
6. Copy the following SQL script into the database client executor, provide the actual values for the placeholders, and execute:

   The following SQL code snippet illustrates the placeholder definitions:

```
USE ${systemDBName};
GO
DELETE FROM AUTH_DATABASE_CONFIG
  FROM AUTH_DATABASE_CONFIG config
  JOIN AUTH_DATABASE_CONF_PROPERTY property
 ON config.PROPERTY_ID = property.ID
  JOIN workspace_databases wdb ON config.DATABASE_ID = wdb.database_id
  JOIN workspace ON workspace.id = wdb.workspace_id
WHERE property.NAME in ('connection.username', 'connection.password')
  AND workspace.name = '${workspaceName}'
GO
UPDATE AUTH_DATABASE_CONFIG
```

```
        SET VALUE = CONCAT(VALUE, ';integratedSecurity=true')
        FROM AUTH_DATABASE_CONFIG config JOIN AUTH_DATABASE_CONF_PROPERTY
property
        ON config.PROPERTY_ID = property.ID
        JOIN workspace_databases wdb on config.DATABASE_ID = wdb.database_id
        JOIN workspace on workspace.id = wdb.workspace_id
                WHERE property.NAME in ('connection.url')
                AND workspace.name = '${workspaceName}'
```

systemDBName

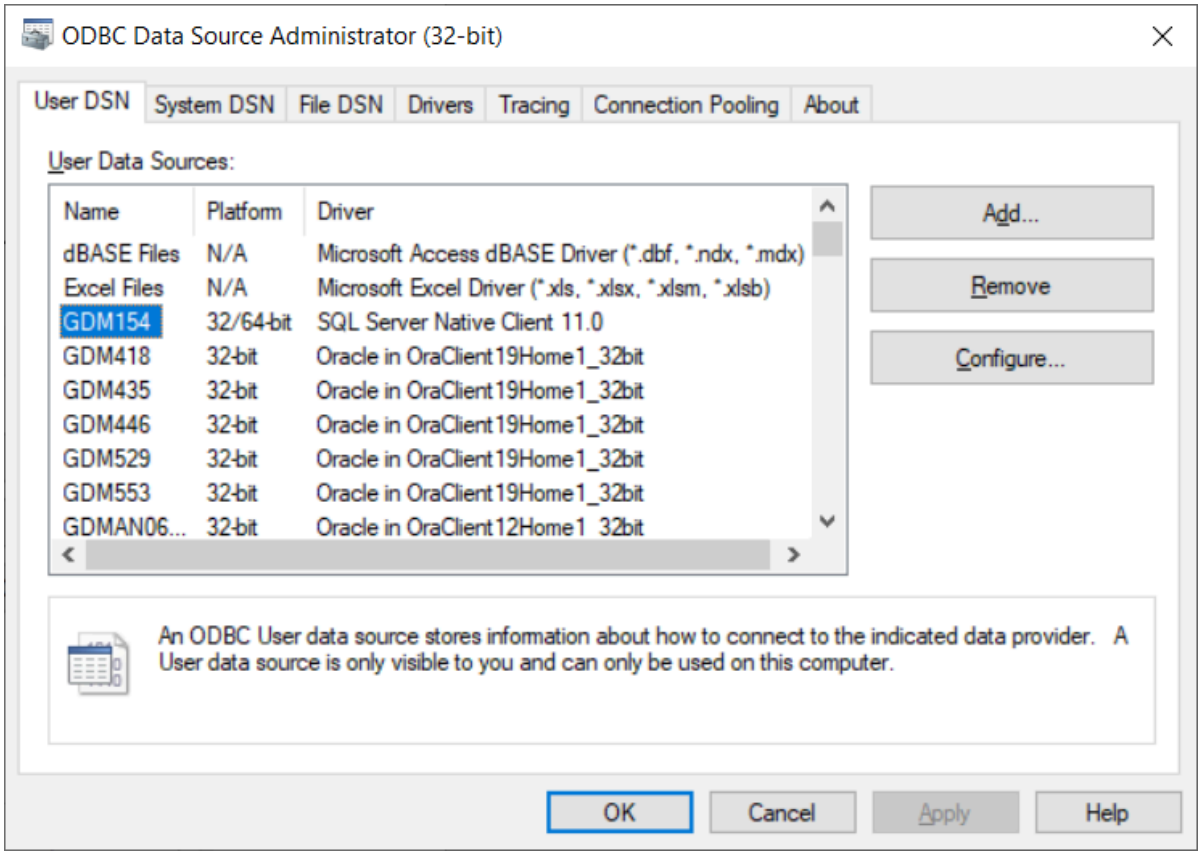
System database name

workspaceName

Workspace name

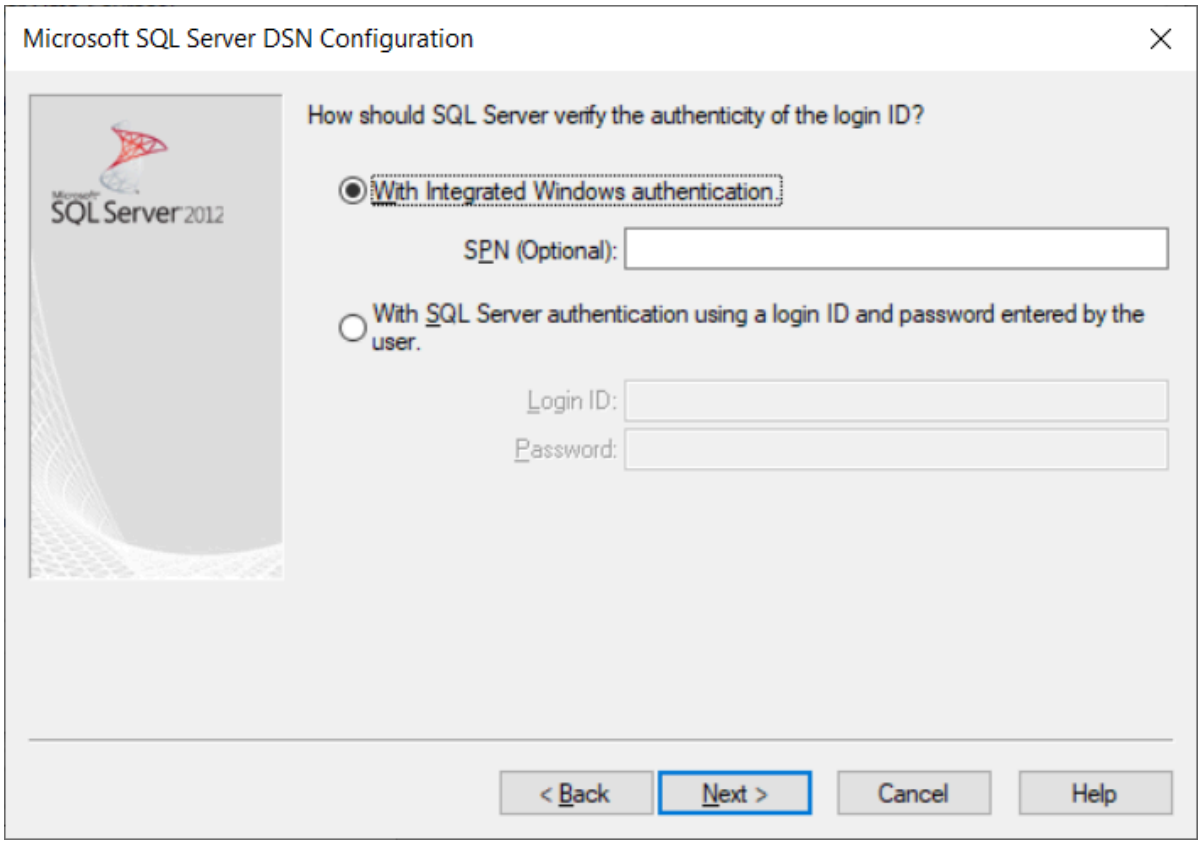
7. Change the ODBC data source configuration by running odbcad32.exe - "ODBC Data Sources":
   a. On the **User DSN** tab, select the data source, and click **Configure**.

Example



   b. Click **Next**.
   c. On the following screen, select the **With integrated Windows authentication** authenticity verification method.

Example



d. Click **Next**, leave the rest as default and complete the configuration.

8. Repeat step 7, but for LDM datasource.
9. Proceed to the upgrade process.

Note:

Upon successful completion of the upgrade process, ensure that *C:\new-reporting-distribution\conf\install.properties* does not contain the following properties, otherwise, remove them manually:

- wolterskluwer.system.db.username
- wolterskluwer.system.db.password
- wolterskluwer.application.db.username
- wolterskluwer.application.db.password

6.3.2 Upgrading from folder or from an archive

You can upgrade OneSumX Regulatory Engine using one of the following options:

- Upgrading OneSumX Regulatory Engine from folder
- Upgrading OneSumX Regulatory Engine based on an archive

6.3.2.1 Upgrading OneSumX Regulatory Engine from folder

Context

Take the steps in this procedure if you meet both of the following conditions:

- You have an existing installation in *C:\previous-reporting-distribution*
- You want to install a new version in *C:\new-reporting-distribution*

Procedure

1. Open file *[new installation drive:]\[new RE installation folder]\conf\install.properties* using your preferred text editor.
2. Provide the applicable location for the following property:

```
install.upgrade.from.folder.with.previous.installation=C:/path/to/previous/
installation
```

3. Save the *[new installation drive:]\[new RE installation folder]\conf\install.properties* file.
4. Generate the security key as described in topic Application database password encryption.
5. Perform one of the following actions:
   - If you have WebSphere Liberty installed, follow the procedure described in topic Upgrade process for WebSphere Liberty.
   - If you have JBoss EAP installed, follow the procedure described in topic Upgrade process for JBoss EAP.

6.3.2.2 Upgrading OneSumX Regulatory Engine based on an archive

Context

After you unistall OneSumX Regulatory Engine, several archived files are generated. You can use those archived files to upgrade OneSumX Regulatory Engine.

Assuming you have an archive of configurations and content repository:

- *[old installation drive:]\[old RE installation folder]\uninstall\configuration.zip*
- *[old installation drive:]\[old RE installation folder]\uninstall\content-store.zip*

and you want to upgrade OneSumX Regulatory Engine at *[new installation drive:]\[new RE installation folder]*, follow this procedure.

Procedure

1. Open the file *[new installation drive:]\[new RE installation folder]\conf\install.properties.*
2. Provide the path and file names for the following properties:

```
install.upgrade.from.zip.with.config=[old installation drive:]\[old RE
installation folder]/uninstall/configuration.zip
install.upgrade.from.zip.with.content.store=[old installation drive:]\[old RE
```

```
installation folder]/uninstall/content-store.zip
```

3. Save the *[new installation drive:]\[new RE installation folder]\conf\install.properties* file.
4. Generate the security key as described in topic Application database password encryption.
5. Perform one of the following actions:

   - If you have WebSphere Liberty installed, follow the procedure described in topic Upgrade process for WebSphere Liberty.

   - If you have JBoss EAP installed, follow the procedure described in topic Upgrade process for JBoss EAP.

6.3.3 Upgrading OneSumX Regulatory Engine depending on the application server

This topic explains the OneSumX Regulatory Engine upgrade process depending on the application server:

- Upgrade process for WebSphere Liberty
- Upgrade process for JBoss EAP

6.3.3.1 Upgrade process for WebSphere Liberty

Procedure

1. Update the legacy properties at *[new installation drive:]\[new RE installation folder]\install.properties*.

```
rgtng.file.browser.root=C:/reporting
workspaces.legacy.root=C:/reporting/artifacts
workspaces.workdir.root=C:/reporting/shared
```

2. Add eclipselink (JPA provider, shipped with WebSphere) as shared library across all applications, if it has not been added yet.

   The library is located at *dev/api/third-party* under the directory where WebSphere was extracted.

   Depending on your implementation, this may differ:

   - For SQL Server:

```
<library id="databaseLib">
    <file name="C:\new-reporting-distribution\deploy\drivers\mssql-
jdbc-9.4.1.jre8.jar" />
    <fileSet dir="C:\server\wlp\dev\api\third-party"
includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar" />
</library>
```

   - For Oracle:

```
<library id="databaseLib">
    <fileSet dir="C:\new-reporting-distribution\deploy\drivers"
includes="*.jar"/>
    <fileSet dir="C:\server\wlp\dev\api\third-party"
includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar" />
</library>
```

3. Perform one of the following actions:

   - If you saved the security key value specified at step 4 in the *\conf\install.properties* file, right-click the script *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* and select the **Run as administrator** option.

   - If you didn't save the security key specified at step 4, then open a Command Prompt window with administrative rights and enter the following path *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* then add

     ```
     -Dwolterskluwer.db.passphrase.encryption.cipher.securityKey=[MTZDaGFyU2VjdXJlS2V5IQ==]
     ```

     , where `[MTZDaGFyU2VjdXJlS2V5IQ==]` represents the generated security key value.

4. For SQL Server, go to *install.properties* and check that the wolterskluwer.db.sqlserver.name property is valid (it has only one backslash "\" between the host and the instance name), for example, *localhost\SQLSERVER_CI*.
5. Replace the WAR files at *C:\server\wlp\usr\servers\defaultServer\dropins* with the content of *[new installation drive:]\[new RE installation folder]\deploy\webapps*.
6. Replace all occurrences of the previous installation folder with the new one at *C:\server\wlp\usr\servers\defaultServer\jvm.options* and *C:\server\wlp\usr\servers\defaultServer\server.xml*.

Results

The upgrade process is complete.

- If you upgraded RE from folder, the Installer will merge *install.properties* of the previous installation into the new installation and copy the contents of the *conf/security* and *etc/content-store/repository* folders.

  The contents of the folder from the previous installation will be removed completely.

- If you upgraded RE from an archive, the installer will unzip the *configuration.zip* file and merge *install.properties* of the previous installation into the new installation and copy contents of the *conf/security* folder. The contents of the *content-store.zip* file will also be unzipped into the respective folders.

Warning:

If the following message appears, while running the upgrade:

```
[] [main] com.wkfsfrc.platform.installer.Installer [ERROR] (2019-08-14
11:24:25.531)         [level=ERROR step=VALIDATE_UPGRADE_INPUT_PROPERTIES
message=Missing value for property: [new.property.name]]
[] [main] wkfsfrc.platform.installer.InstallerMain [ERROR] (2019-08-14
11:24:25.531) ValidationException while validating installation input
properties. Exiting...
com.wkfsfrc.platform.installer.exception.ValidationException: Errors were
encountered while validating input properties. Please correct the errors
and rerun the installer
```

That means the mentioned property new.property.name was added in the new version and it does not have a value from the previous installation.

It needs to be given a value by editing the *C:\new-reporting-distribution\conf\install.properties* file.

6.3.3.2 Upgrade process for JBoss EAP

Context

This section contains an example of an upgrade process from *[old installation drive:]\[old RE installation folder]* to *[new installation drive:]\[new RE installation folder]*.

JBoss EAP is installed at *C:\server\jboss-eap-7.3*.

Procedure

1. Stop the server.
2. Update the legacy properties property at *C:\new-reporting-distribution\install.properties*.

```
rgtng.file.browser.root=C:/reporting
workspaces.legacy.root=C:/reporting/artifacts
workspaces.workdir.root=C:/reporting/shared
```

3. Update the install.upgrade.from.folder.with.previous.installation property at *[new installation drive:]\[new RE installation folder]\install.properties*.

```
install.upgrade.from.folder.with.previous.installation=[old installation
drive:]\[old RE installation folder]
```

4. Copy the new database driver in the appropriate folder: *[new installation drive:]\[new RE installation folder]\reporting-[version number]\deploy\drivers*.

   - For SQL Server:

```
<library id="databaseLib"> <file name="C:\new-reporting-distribution\deploy\
drivers\ mssql-jdbc-9.4.1.jre8.jar" /> <fileSet dir="C:\server\jboss-eap-7.3\
modules\system\layers\base\com\microsoft\sqlserver\main"
includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar" />
</library>
```

   - For Oracle:

```
<library id="databaseLib"> <file name="C:\new-reporting-distribution\deploy\
drivers\ oracle" /> <fileSet dir="C:\server\jboss-eap-7.3\modules\system\
layers\base\com\microsoft\sqlserver\main"
includes="com.ibm.websphere.appserver.thirdparty.eclipselink_*.jar" />
</library>
```

5. Perform one of the following actions:

   - If you saved the security key value specified at step 4 in the *\conf\install.properties* file, right-click the script *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* and select the **Run as administrator** option.

   - If you didn't save the security key specified at step 4, then open a Command Prompt window with administrative rights and enter the following path *[new installation drive:]\[new RE installation folder]\bin\upgrade-platform.cmd* then add

```
-Dwoltersskluwer.db.passphrase.encryption.cipher.securityKey=[MTZDaGFyU2VjdXJlS2V5IQ==]
```

, where `[MTZDaGFyU2VjdXJlS2V5IQ==]` represents the generated security key value.

6. For SQL Server, go to *install.properties* and check that the wolterskluwer.db.sqlserver.name property is valid (it has only one backslash "\" between the host and the instance name), for example, *localhost\SQLSERVER_CI*.
7. Edit *C:\server\jboss-eap-7.3\standalone\configuration\standalone.xml*.

   References to *[old installation drive:]\[old RE installation folder]* must be replaced with the new application's location.

   Add the system properties after the extensions element:

```
    <system-properties>
        <property name="wkfs.commons.configuration.file.local" value="C:\new-
reporting-distribution\conf\wkfs-external.properties"/>
        <property name="file.encoding" value="UTF-8"/>
        <property name="reportgenerator.init.file.location" value="C:\new-
reporting-distribution\conf\init.xml"/>
        <property name="reportgenerator.initAuthentication.file.location"
value="C:\new-reporting-distribution\conf\init-authentication.xml"/>
        <property name="com.wolterskluwer.configuration" value="C:\new-
reporting-distribution\conf\wkfs-processing-engine.properties"/>
 <property name="java.library.path" value="C:\newreporting-distribution\deploy\
drivers\dll"/>
    </system-properties>
```

   Edit the keystore subelement of the ApplicationRealm:

```
            <security-realm name="ApplicationRealm">
                <server-identities>
                    <ssl>
                                                <keystore path="C:/new-reporting-
distribution/conf/security/application.keystore" keystore-password="changeit"
alias="application" key-password="changeit" generate-self-signed-certificate-
host="localhost"/>
                    </ssl>
                </server-identities>
                ...
            </security-realm>
```

8. Copy new deployments (reference to the topic with copy deployments for JBoss)

Results

The upgrade process is complete.

- If you upgraded RE from folder, the Installer will merge *install.properties* of the previous installation into the new installation and copy the contents of the *conf/security* and *etc/content-store/repository* folders.

  The contents of the folder from the previous installation will be removed completely.

- If you upgraded RE from an archive, the installer will unzip the *configuration.zip* file and merge *install.properties* of the previous installation into the new installation and copy contents of the *conf/security* folder. The contents of the *content-store.zip* file will also be unzipped into the respective folders.

Warning:

If the following message appears, while running the upgrade:

```
[] [main] com.wkfsfrc.platform.installer.Installer [ERROR] (2019-08-14
11:24:25.531)         [level=ERROR step=VALIDATE_UPGRADE_INPUT_PROPERTIES
message=Missing value for property: [new.property.name]]
[] [main] wkfsfrc.platform.installer.InstallerMain [ERROR] (2019-08-14
11:24:25.531) ValidationException while validating installation input
properties. Exiting...
com.wkfsfrc.platform.installer.exception.ValidationException: Errors were
encountered while validating input properties. Please correct the errors
and rerun the installer
```

That means the mentioned property new.property.name was added in the new version and it does not have a value from the previous installation.

It needs to be given a value by editing the *C:\new-reporting-distribution\conf\install.properties* file.

7 Uninstalling OneSumX Regulatory Engine

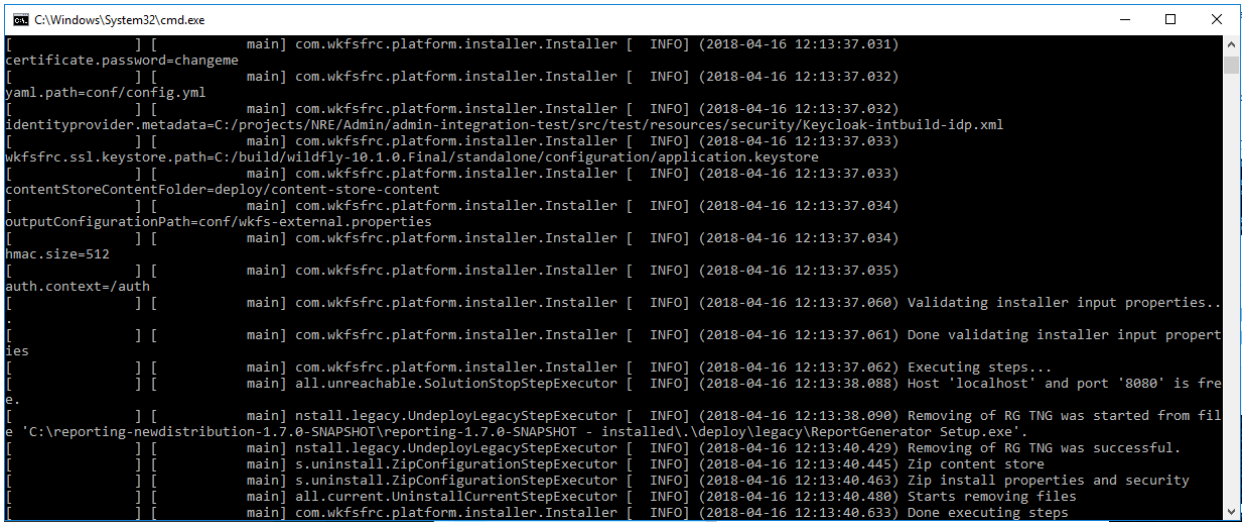
Prerequisites

- Stop OneSumX Regulatory Engine before starting the uninstall process.
- Stop all *ReportGenerator.exe* running instances via TaskManager / Details.
- Remove the previous installations of ReportGenerator via the **Windows Control Panel**.

Procedure

1. Open Command Prompt or PowerShell as administrator.
2. Type `cd \[installation drive:]\[RE installation folder]\bin`.
3. Type `unistall-platform.cmd`.

   Example of an uninstall process:



Results

The results are displayed in the following file structure:

```
├──────lib
│     └──────used libs (jar format)
│
├──────logs
│     └──────RegPlatformInstall_${now date}.log
│
├──────uninstall
│     └──────configuration.zip
│        ├──────AdminConsole.jks
│        ├──────AdminConsole-localhost-sp.xml
│        ├──────install.properties
│        └──────Keycloak-intbuild-idp.xml
│    └──────content_store.zip
│        ├──────whole pack of files from folder \etc\content-store\repository
```

Note:

The *content_store.zip* file contains all uploaded files that were in folder: *C:\reporting-1.7.0-SNAPSHOT\etc\content-store\repository*.

8 Reconfiguring OneSumX Regulatory Engine

8.1 Change the Identity Provider files

Context

This topic describes the procedure you need to follow in case the Identity Providers files were changed.

Procedure

1. Open the *[installation drive:]\[RE installation folder]/conf/install.properties* file and update all the properties that point to a service provider file. You need to take this action for all the components of the solution.

   #### Example

   admin.serviceprovider.metadata=conf/security/AdminConsole-localhost-sp.xml

2. Reconfigure the Identity Provider to use the new service provider files, generate a new Identity Provider file, and update the properties from the *[installation drive:]\[RE installation folder]/conf/install.properties* file to point to the new Identity Provider file.

   #### Example

   ```
   admin.identityprovider.metadata=conf/security/Keycloak-intbuild-idp.xml
   ```

3. Create a backup of the server configuration files that are located under the *conf* folder, except for the *conf/security* subfolder.
4. Install the platform using *install-platform.cmd*. Refer to topic Installation process
5. Paste the files mentioned at step 3.
6. Run OneSumX Regulatory Engine. Refer to topic Starting OneSumX Regulatory Engine.

8.2 Change JRE 32 bit path

Context

This topic describes the procedure to follow in case the JRE 32 bit path needs to change.

Procedure

1. Open the *install.properties* file and change the value of the rgtng.jre32.path property.
2. Create a backup of the server configuration files that are located under the *conf* folder, except for the *conf/security* subfolder.
3. Install the platform using *install-platform.cmd*. Refer to topic Installation process
4. Paste the files mentioned at step 3.
5. Run OneSumX Regulatory Engine. Refer to topic Starting OneSumX Regulatory Engine.

8.3 Manually changing system and application passwords

Use cases for changing application and system database passwords are similar for Oracle/SQL Server (depending on the installation requirements).

This section presents the process of changing these passwords inside OneSumX Regulatory Engine, assuming that the actual database schema passwords have been changed externally.

8.3.1 Updating the system database password manually

Procedure

1. Ensure that the application is stopped.
2. Update the system database password in the server configuration.
3. To be consistent, navigate to *conf/install.properties* and update the wolterskluwer.application.db.password property.

   #### Example

   wolterskluwer.application.db.password=**newpassword**

8.3.2 Manually updating the application database password

Procedure

1. Ensure that the application is stopped.
2. Navigate to *conf/init.xml* and update the `<property name="connection.password">` tag property.

   #### Example

   ```
   <property name="connection.username">SYSTEMDB_4_0</property> <property
   name="connection.password">newpassword</property>
   ```

3. Navigate to *conf/standalone.xml* and search for the datasource corresponding to the application database and update the following tag: `<password>newpassword</password>`.

   #### Example

   ```
   <datasource jndi-name="java:/wolterskluwer/datasources/SYSTEM_DB"
   jta="false"pool-name="SYSTEM_DB"> <connection-
   url>jdbc:oracle:thin:@//localhost:1521/ORACLE</connection-url>
   <driver>oracle.jdbc.OracleDriver</driver> <pool> <min-pool-size>10</min-pool-
   size> <initial-pool-size>10</initial-pool-size> <max-pool-size>32</max-pool-
   size> </pool> <security> <user-name>SYSTEMDB_4_0/user-name>
   <password>newpassword</password> </security> <connection-property
   name="oracle.net.networkCompression">on</connection-property> <connection-
   property name="oracle.net.networkCompressionThreshold">1024</connection-
   property> <validation> <check-valid-connection-sql>select 1from dual</check-
   valid-connection-sql> <background-validation>true</background-validation>
   <background-validation-millis>10000</background-validation-millis>
   </validation></datasource>
   ```

4. To be consistent, navigate to *conf/install.properties* and update the wolterskluwer.application.db.password property.

   ### Example

   wolterskluwer.application.db.password=**newpassword**

5. Start the application by navigating to the *bin* folder and run the *startup-swarm.cmd* script.

8.3.3 Manually updating the system database password

Procedure

1. Ensure that the application is stopped.
2. To be consistent, navigate to *conf/install.properties* and update the wolterskluwer.application.db.password property.

   ### Example

   wolterskluwer.application.db.password=**newpassword**

3. Navigate to the *bin* folder and run the *startup-swarm.cmd* script.

8.4 Importing the OneSumX Regulatory Engine certificate after a JAVA upgrade

In case you update the Java version after the OneSumX Regulatory Engine installation, ensure that you import the OneSumX Regulatory Engine certificate into a new Java *cacerts* file. For details, refer to topic Importing HTTPS certificate into JRE cacerts manually.

9 Troubleshooting

9.1 Running bin\startup-swarm.cmd without first completing the installation

Context

```
12:26:46,167 ERROR [io.undertow.websockets.core.request] (default task-12)
UT025007: Unhandled exception for annotated endpoint
com.wkfsfrc.platform.command.websocket.CommandExecutionWebSocket@3b3a9e85:
java.lang.IllegalStateException: Submitting command failed
...
Caused by: java.lang.RuntimeException: No command type definition found for
'execute-bpe-job' in 'configuration/commands'
```

Getting this error when running processes is due to the content-store component of the OneSumX Regulatory Engine lacking the initial files it requires.

Procedure

1. Run *uninstall-platform.cmd*.
2. Run *install-platform.cmd* to update the content-store.

9.2 Limit the file size of the installation logs

Context

The installation logs are to be found in the */logs* folder. By default there is one *server.log* file for each day when the application is running. In this case it happens that the log becomes too large to read, therefore, there is a possibility to limit the size of the log file to a more convenient value.

Procedure

1. Open the *standalone.xml* file.
2. Search for the `<subsystem xmlns="urn:jboss:domain:logging:6.0">` tag.
3. Replace:

```
<periodic-rotating-file-handler name="FILE" autoflush="true">
                <formatter>
                        <named-formatter name="PATTERN"/>
                </formatter>
                <file relative-to="jboss.server.log.dir" path="logs/
server.log"/>
                <suffix value=".yyyy-MM-dd"/>
                <append value="true"/>
</periodic-rotating-file-handler>
```

With:

```
<size-rotating-file-handler name="FILE" autoflush="true">
                <formatter>
                        <named-formatter name="PATTERN"/>
                </formatter>
                <file relative-to="jboss.server.log.dir" path="logs/
server.log"/>
                <suffix value=".yyyy-MM-dd"/>
                <rotate-size value="1000k"/>
                <max-backup-index value="15"/>
                <append value="true"/>
</size-rotating-file-handler>
```

4. Replace the values for `<rotate-size value="1000k"/>` and `<max-backup-index value="15"/>` with the desired values.

9.3 HTTPS port already occupied

Context

```
15:43:45,144 ERROR [org.jboss.msc.service.fail] (MSC service thread 1-3)
MSC000001: Failed to start service jboss.undertow.listener.default:
org.jboss.msc.service.StartException in service
jboss.undertow.listener.default: Address already in use: bind
/0:0:0:0:0:0:0:0:8080
```

To change the WildFly HTTPS port, you need to pass it the -Djboss.https.port=<new_port> system parameter.

Procedure

1. Edit scripts *.\bin\install-platform.cmd* and *.\bin\startup-swarm.cmd*
2. Add `-Djboss.https.port=<new_port>` after the *"%JAVA_HOME%"\bin\java string*

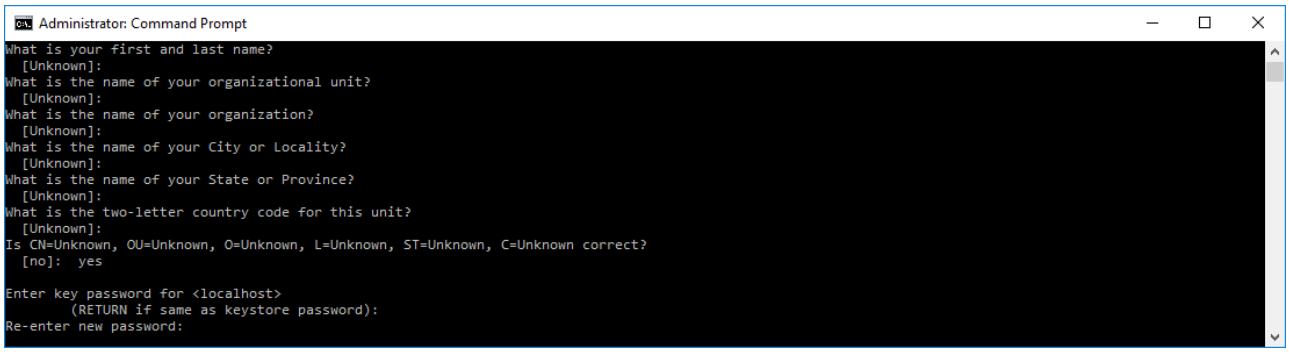
9.4 Workspace creation error linked to a certificate issue

Context

The workspace creation fails with the following error:

```
Caused by: javax.net.ssl.SSLException: Certificate for <localhost> doesn't
match common name of the certificate subject: Unknown
```

The error occurs because, during the certificate generation, incorrect values were provided for “First and Last name”.



Procedure

1. Open the *install.properties* file.
2. If you defined wolterskluwer.application.host = `localhost`, then execute the following command to generate certificates:

   ```
   keytool -genkey -alias localhost -keystore
   C:/WKFS_RE/conf/security/cmd.keystore -storepass changeme -keyalg RSA -
   validity 360 -keysize 2048
   ```

3. For “First and Last name”, provide the following value: localhost.

   Warning:
   It is not recommended to use characters like "$" or any other non-alpha-numeric characters in the properties file.

9.5 Running jobs on the Monitoring page are not refreshed after the processes have finished

Context

In the **Admin Center** application, the **Monitoring** menu enables you to check the status and results of asynchronous tasks triggered in the **Reporting Hub** and **Admin Center.** It may sometimes happen that the **Monitoring** menu does not refresh automatically to show the finished jobs.

Procedure

1. If the application server is stopped, stop any remaining Report Generator instances in Task Manager. When you start the application server again, there should be no Report Generator instances in Task Manager.
2. Refresh the Regulatory Engine browser web page after the application web server has successfully started, if a previous instance was still opened in the browser,

9.6 Input not an X.509 certificate

Context

```
keytool error: java.lang.Exception: Input not an X.509 certificate
```

This error can appear while running `keytool -import`

Procedure

1. Re-generate the password.
2. Repeat the export and import operation (see topic Generate the HTTPS certificate).

9.7 FileNotFoundException: cacerts (Access is denied)

Context

```
[                ] [            main]
wkfsfrc.platform.installer.InstallerMain [ ERROR] (2019-02-04 13:31:19.707)
InstallationException while performing installation. Exiting...
com.wkfsfrc.platform.installer.exception.InstallationException:
com.wkfsfrc.platform.installer.exception.ExecutionException: Import https
certificate to cacert was failed. Please change cacerts password or import
it manually
        ...
Caused by: java.io.FileNotFoundException: C:\Program
Files\Java\jre1.8.0_201\lib\security\cacerts (Access is denied)
```

Procedure

Run cmd as administrator.

9.8 Path to Oracle tnsnames is not valid

Context

```
com.wkfsfrc.platform.installer.exception.InstallationException:
com.wkfsfrc.platform.installer.exception.ExecutionException: Path to Oracle
tnsnames (C:/app/oracle/product/11.2.0/client_1/network/admin) is not
valid. You can set it to empty string if you don't use Oracle database
```

This exception appears when working with SQL Server.

Procedure

Set an empty value in *install.properties*:

```
wolterskluwer.oracle.tnsNames.folder=
```

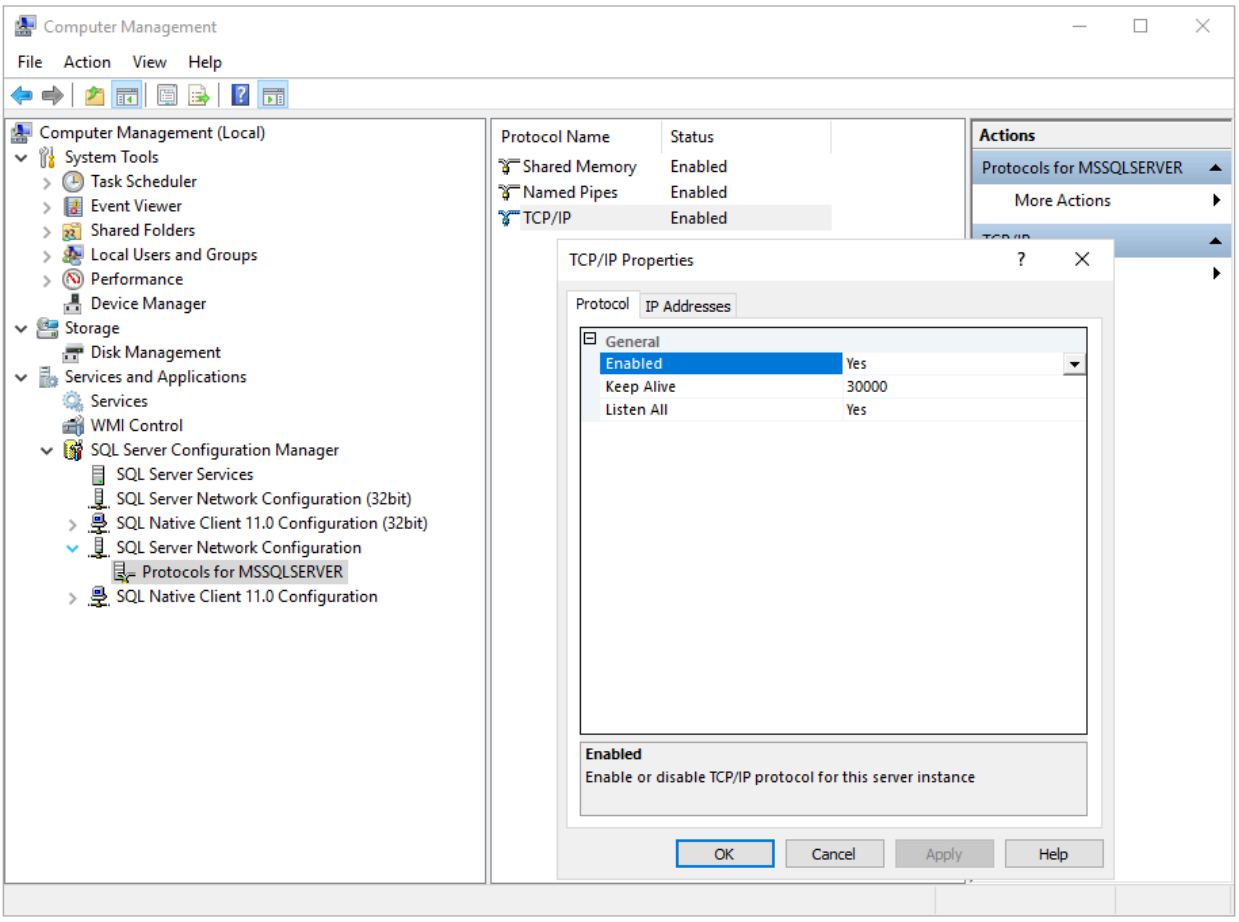
9.9 Prelogin error: host 127.0.0.1 port 1434

Context

```
Prelogin error: host 127.0.0.1 port 1434 Error reading prelogin response:
Software caused connection abort: recv failed ClientConnectionId:…
```

Procedure

1. Run SQL Server Configuration Manager, available via **Computer Management** (run *compmgmt.msc*), and ensure that **TCP/IP** is enabled.



2. In SQL Server Management Studio, Enable port 1434 for the local 127.0.0.1 connection.

9.10 The server principal already exists during reinstallation

Context

If a database has been deleted, it is possible to receive an error during the reinstallation of the application using the SQL Server database:

```
[              ] [          main] com.wkfsfrc.platform.installer.Installer [  INFO] (2019-04-24 15:39:01.148) Done validating installer input properties
[              ] [          main] com.wkfsfrc.platform.installer.Installer [  INFO] (2019-04-24 15:39:01.148) Executing steps...
[              ] [          main] .steps.datasource.DataSourceStepExecutor [  INFO] (2019-04-24 15:39:01.186) Datasource configuration construction is starting
[              ] [          main] source.db.executors.DbScriptExecutorUtil [ ERROR] (2019-04-24 15:39:02.503) Error while script executing [CREATE LOGIN [MY_BANK_SYSTEM_DB] WITH PASSWORD = 'welcome', DEFAULT_DATABASE=[MY_BANK_SYSTEM_DB
], CHECK_POLICY = OFF]
com.microsoft.sqlserver.jdbc.SQLServerException: The server principal 'MY_BANK_SYSTEM_DB' already exists.
        at com.microsoft.sqlserver.jdbc.SQLServerException.makeFromDatabaseError(SQLServerException.java:216)
        at com.microsoft.sqlserver.jdbc.SQLServerStatement.getNextResult(SQLServerStatement.java:1522)
        at com.microsoft.sqlserver.jdbc.SQLServerStatement.doExecuteStatement(SQLServerStatement.java:799)
        at com.microsoft.sqlserver.jdbc.SQLServerStatement$StmtExecCmd.doExecute(SQLServerStatement.java:696)
        at com.microsoft.sqlserver.jdbc.TDSCommand.execute(IOBuffer.java:5690)
        at com.microsoft.sqlserver.jdbc.SQLServerConnection.executeCommand(SQLServerConnection.java:1716)
        at com.microsoft.sqlserver.jdbc.SQLServerStatement.executeCommand(SQLServerStatement.java:180)
        at com.microsoft.sqlserver.jdbc.SQLServerStatement.executeStatement(SQLServerStatement.java:155)
        at com.microsoft.sqlserver.jdbc.SQLServerStatement.executeUpdate(SQLServerStatement.java:649)
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutorUtil.executeStatement(DbScriptExecutorUtil.java:96)
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutorUtil.lambda$executeLineByLine$2(DbScriptExecutorUtil.java:75)
        at java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:184)
        at java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:193)
        at java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:175)
        at java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:193)
        at java.util.Iterator.forEachRemaining(Iterator.java:116)
        at java.util.Spliterators$IteratorSpliterator.forEachRemaining(Spliterators.java:1801)
        at java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:481)
        at java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:471)
        at java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:151)
        at java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:174)
        at java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:234)
        at java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:418)
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutorUtil.executeLineByLine(DbScriptExecutorUtil.java:74)
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutorUtil.executeScript(DbScriptExecutorUtil.java:55)
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutor.createApplicationSchema(DbScriptExecutor.java:54)
        at com.wkfsfrc.platform.installer.steps.datasource.DataSourceStepExecutor.execute(DataSourceStepExecutor.java:77)
        at com.wkfsfrc.platform.installer.Installer.executeSteps(Installer.java:169)
        at com.wkfsfrc.platform.installer.Installer.install(Installer.java:64)
        at com.wkfsfrc.platform.installer.InstallerMain.main(InstallerMain.java:50)
[              ] [          main] datasource.db.executors.DbScriptExecutor [ ERROR] (2019-04-24 15:39:02.519) Couldn't create database schema.
[              ] [          main] wkfsfrc.platform.installer.InstallerMain [ ERROR] (2019-04-24 15:39:02.519) InstallationException while performing installation. Exiting...
com.wkfsfrc.platform.installer.exception.InstallationException: com.wkfsfrc.platform.installer.exception.ExecutionException: com.wkfsfrc.platform.installer.steps.datasource.db.executors.ScriptExecuteException: com.wkfsfrc.platform.instal
ler.steps.datasource.db.executors.DbSqlStatementExecuteException: com.microsoft.sqlserver.jdbc.SQLServerException: The server principal 'MY_BANK_SYSTEM_DB' already exists.
        at com.wkfsfrc.platform.installer.Installer.executeSteps(Installer.java:171)
        at com.wkfsfrc.platform.installer.Installer.install(Installer.java:64)
        at com.wkfsfrc.platform.installer.InstallerMain.main(InstallerMain.java:50)
Caused by: com.wkfsfrc.platform.installer.exception.ExecutionException: com.wkfsfrc.platform.installer.steps.datasource.db.executors.ScriptExecuteException: com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbSqlStatementExecu
teException: com.microsoft.sqlserver.jdbc.SQLServerException: The server principal 'MY_BANK_SYSTEM_DB' already exists.
        at com.wkfsfrc.platform.installer.steps.datasource.DataSourceStepExecutor.execute(DataSourceStepExecutor.java:81)
        at com.wkfsfrc.platform.installer.Installer.executeSteps(Installer.java:169)
        ... 2 more
Caused by: com.wkfsfrc.platform.installer.steps.datasource.db.executors.ScriptExecuteException: com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbSqlStatementExecuteException: com.microsoft.sqlserver.jdbc.SQLServerException:
 The server principal 'MY_BANK_SYSTEM_DB' already exists.
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutorUtil.executeLineByLine(DbScriptExecutorUtil.java:78)
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutorUtil.executeScript(DbScriptExecutorUtil.java:55)
        at com.wkfsfrc.platform.installer.steps.datasource.db.executors.DbScriptExecutor.createApplicationSchema(DbScriptExecutor.java:54)
        at com.wkfsfrc.platform.installer.steps.datasource.DataSourceStepExecutor.execute(DataSourceStepExecutor.java:77)
        ... 3 more
```

Procedure

Delete a user. You can use one of the following methods:

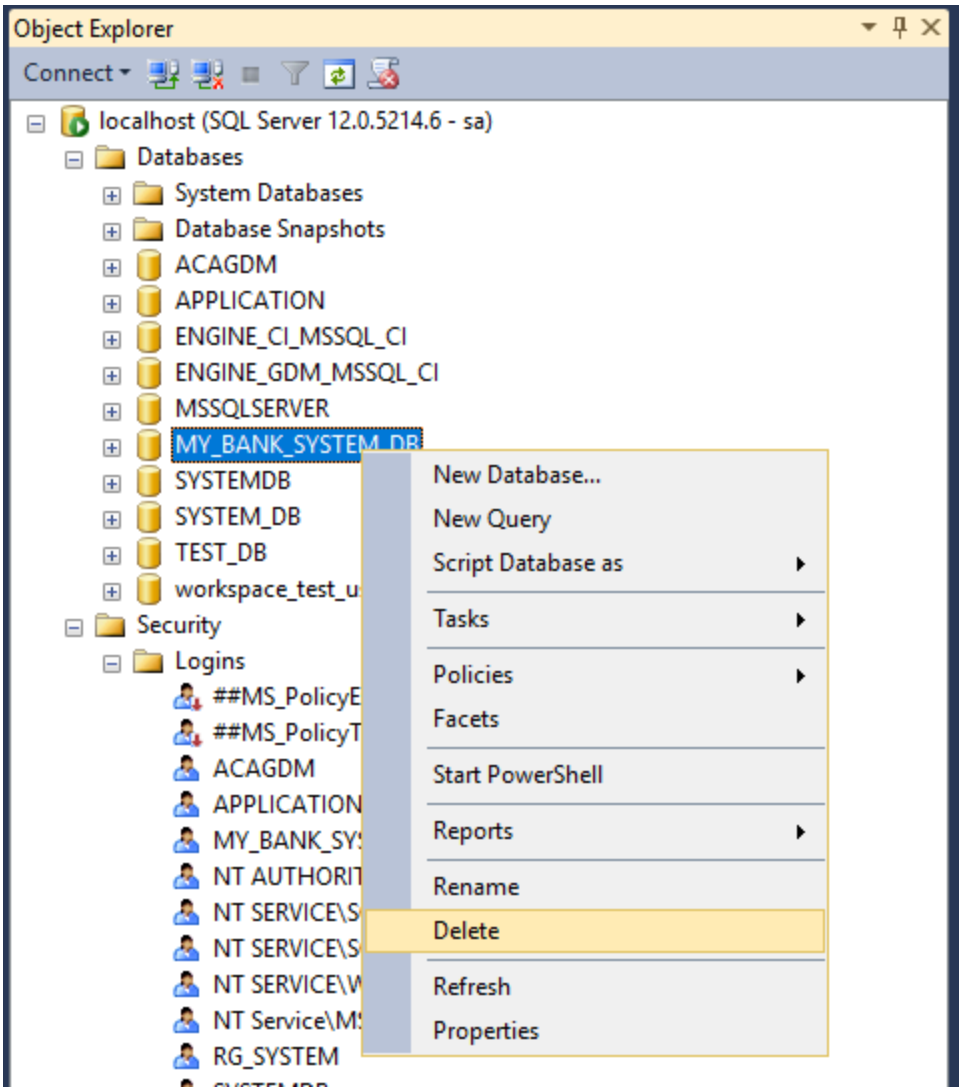
- Using an SQL script:

  a. Run the following SQL script:

  ```
  use master
  go
  drop login MY_BANK_SYSTEM_DB
  go
  ```

  b. Replace `MY_BANK_SYSTEM_DB` with the name of the system database that you plan to delete before the re-installation.

  c. You can also safely delete the database:



- Running *install-platform.cmd*:
  a. Go to *install.properties*.
  b. Rename the system database.
  c. Re-run *install-platform.cmd*.

> ***Warning:***
> Consider that the *install-platform.cmd* script is used for the installation. This means that the standalone config file will be overwritten when running this script. Ensure that *standalone.xml* is backed up before performing the above mentioned action.

9.11 TNS Service Name or Service Name are wrong

Context

When you create a workspace, and provide an incorrect **TNS Service Name** or **Service name**, you can encounter

the following error:

```
Can not find identifier for value
“someValue”:java.lang.IllegalStateException
```

Procedure

1. Go to the *TNS* folder.
2. Open the *tnsnames.ora* file and check that the service name and TNS service name are correct.

9.12 Deployment fails with timeout error

Context

The following message is displayed:

```
Deploy of deployment "rgtng-webapp.war" was rolled back with the following
failure message: "WFLYCTL0344: Operation timed out awaiting service
container stability"
```

Procedure

Increase the WildFly deployment timeout by taking one of the following actions:

- In the *startup-swarm.cmd* script, after the part `set SYS_PROPS=`, add the following property: -Djboss.as.management.blocking.timeout=1000
- In the *conf* folder, in the *config.yml* file, add the following property:

```
swarm:
  deployment:
    timeout:900
  https:
    port: 8080
    only: true
  context:
    path: /reporting-webapp
  admin-webapp:
    context: /admin-webapp
  rgtng-webapp:
    context: /reportgenerator
```

Warning:
Use the correct indentation (2 space characters for deployment and 4 space characters for timeout properties).

The timeout duration is in seconds.

9.13 Platform upgrade stops with a message about "rgtng.file.browser.root"

Context

When a platform upgrade has stopped, you may see one of the following messages:

- "The previous installation does not contain property [rgtng.file.browser.root]. Please, refer to installation guide for the information on manual upgrade".
- "The previous installation has incompatible properties for [rgtng.file.browser.root] and[workspaces.legacy.root]. Please, refer to installation guide for the information on manual upgrade."

```
            ] [          main] com.wkfsfrc.platform.installer.Installer [ ERROR] (2020-02-07 14:53:21.212) There were ERROR messages while validating the input properties.
            ] [          main] com.wkfsfrc.platform.installer.Installer [ ERROR] (2020-02-07 14:53:21.212)         [level=ERROR step=VALIDATE_UPGRADE_INPUT_PROPERTIES message=The previous installation does not contain property [rgtn
g.file.browser.root]. Please, refer to installation guide for the information on manual upgrade.]
            ] [          main] wkfsfrc.platform.installer.InstallerMain [ ERROR] (2020-02-07 14:53:21.212) ValidationException while validating installation input properties. Exiting...
com.wkfsfrc.platform.installer.exception.ValidationException: Errors were encountered while validating input properties. Please correct the errors and rerun the installer
        at com.wkfsfrc.platform.installer.Installer.validationMessagesChecks(Installer.java:153)
        at com.wkfsfrc.platform.installer.Installer.validationProcessOfWorkflow(Installer.java:101)
        at com.wkfsfrc.platform.installer.Installer.install(Installer.java:59)
        at com.wkfsfrc.platform.installer.InstallerMain.main(InstallerMain.java:47)
```

Procedure

Refer to topic Updating the work directory.

9.14 Platform upgrade stops with message "wolterskluwer.[DBMS].default.driver"

Context

The platform upgrade stops with the following message: "Property [wolterskluwer.oracle.default.driver] or [wolterskluwer.sqlserver.default.driver] has not been provided".

```
[                ] [            main] com.wkfsfrc.platform.installer.Installer [ ERROR] (2020-03-29 12:21:17.795)         [level=ERROR step=CONFIGURE_LEGACY message=Property [wolterskluwer.oracle.default.driver] or [wolterskluwer.sqlserver
.default.driver] has not been provided]
[                ] [            main] wkfsfrc.platform.installer.InstallerMain [ ERROR] (2020-03-29 12:21:17.796) ValidationException while validating installation input properties. Exiting...
com.wkfsfrc.platform.installer.exception.ValidationException: Errors were encountered while validating input properties. Please correct the errors and rerun the installer
        at com.wkfsfrc.platform.installer.Installer.validationMessagesChecks(Installer.java:153)
```

This message means that properties [wolterskluwer.oracle.default.driver] and [wolterskluwer.sqlserver.default.driver] were not set.

Procedure

Refer to topic Steps for the ODBC drivers for information on how to set the [wolterskluwer.oracle.default.driver] and [wolterskluwer.sqlserver.default.driver] properties.

9.15 Deployment fails with message "DriverName is not installed"

Context

Deployment is stopped and the following message is displayed: "DriverName is not installed". "DriverName" is the ODBC driver name specified in *install.properties*.

Figure 1. Error Message

```
13:38:25,473 ERROR [com.wolterskluwer.reg.admin.service.impl.StartupJobServiceImpl] (pool-16-thread-1) Startup job failed: com.wolterskluwer.reg.admin.service.impl.workspace.WorkspaceDriversException: Oracle Driver driver is not installe
d.
        at com.wolterskluwer.reg.admin.service.impl.workspace.WorkspaceServiceImpl.checkWorkspaceDatabaseDrivers(WorkspaceServiceImpl.java:399)
        at com.wolterskluwer.reg.admin.service.impl.workspace.WorkspaceServiceImpl$Proxy$_$$_WeldSubclass.checkWorkspaceDatabaseDrivers$$super(Unknown Source)
        at sun.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
        at sun.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
        at sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
```

This error can occur when the ODBC driver specified in the *install.properties* file, for one of these properties, [wolterskluwer.oracle.default.driver] and [wolterskluwer.sqlserver.default.driver], was not installed.

Procedure

Refer to topic Steps for the ODBC drivers for information on how to set the following properties: [wolterskluwer.oracle.default.driver] and [wolterskluwer.sqlserver.default.driver].

9.16 Application doesn't respond on several pages for the SQL Server configuration

Context

In case some pages are not loaded (Workspace details page, Embedded ReportGenerator UI etc.) there is a posibility that the AUTH_USER_PREFERENCE table from the system database is locked.

Procedure

1. To verify if a resource is locked, in the database server, execute the following query:

```
SELECT
        t1.resource_type,
        t1.resource_database_id,
        t1.resource_associated_entity_id,
        t1.request_mode,
        t1.request_session_id,
        t2.blocking_session_id
    FROM sys.dm_tran_locks as t1
    INNER JOIN sys.dm_os_waiting_tasks as t2
        ON t1.lock_owner_address = t2.resource_address;
```

Results

The query will return the ID of the locked resource if any and the specific request and blocking session ID.

2. To find the correspunding object name in a specific database, based on the resource database ID, execute the following query:

```
 SELECT * FROM sys.databases
```

3. In order to kill the session that blocks the resource, execute the following query:

```
Kill session_id;
```

9.17 Authentication process fails if OneSumX Regulatory Engine and the Identity Provider server are not synced

Context

The authentication process fails if OneSumX Regulatory Engine and the Identity Provider server are running on different computers and the time setting of the respective computers differs by more than 60 seconds.

This issue could result in the error message `Response issue time is either too old or with date in the future` in the log of OneSumX Regulatory Engine after a failed authentication.

Procedure

Synchronize the time settings of the computers on which OneSumX Regulatory Engine and the Identity Provider server are running.

9.18 Client requested protocol TLSv1 not enabled or not supported

Context

When the HTTPS with TLS 1.2 communication is enforced, the following error migth appear in *server.log*:

```
Closing SSLConduit after exception on handshake:
javax.net.ssl.SSLHandshakeException: Client requested protocol TLSv1 not enabled
or not supported.
```

Procedure

1. Go to Windows Registry at *[HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\.NETFramework\v4.0.30319]*and add the *SchUseStrongCrypto* registry: *"SchUseStrongCrypto"=dword:00000001*
2. Go to *[HKEY_LOCAL_MACHINE\SOFTWARE\Wow6432Node\Microsoft\.NETFramework\v4.0.30319]* and add the *SchUseStrongCrypto* registry: *"SchUseStrongCrypto"=dword:00000001*
3. Restart OneSumX Regulatory Engine.

9.19 SQL Server user missing execute grant

Context

Starting with version 4.3.0, the user for a workspace connection on an SQL Server database requires an additional permission to run the GDM functions.

When the "EXECUTE" grant is missing, the following functionalities will fail: data set creation for managed workspaces, and connection to unmanaged worskpaces. In this case, the following error appears in *server.log*:

```
com.microsoft.sqlserver.jdbc.SQLServerException: The EXECUTE permission was denied
on the object 'fn_par_isdatabase_partitioned', database 'GDM_NAME', schema 'dbo'
```

For newly created workspaces, the grant is automatically added for the user specified in the connection form. In case of an engine upgrade, the grant must be added to the SQL Server user which was used for the connection.

Procedure

1. Connect to the GDM schema in the SQL Server database using an admin user.
2. Replace **userName** with the actual user for the connection in the following queries and execute them:

```
GRANT EXECUTE ON dbo.fn_par_get_next_file TO userName;
GRANT EXECUTE ON dbo.fn_par_get_partition_field TO userName;
GRANT EXECUTE ON dbo.fn_par_get_partition_function TO userName;
GRANT EXECUTE ON dbo.fn_par_istable_partitioned TO userName;
GRANT EXECUTE ON dbo.fn_par_isdatabase_partitioned TO userName;
GRANT EXECUTE ON dbo.fn_sanitize_filename TO userName;
GRANT EXECUTE ON dbo.fn_get_index_fields_string TO userName;
GRANT EXECUTE ON dbo.isviewexists TO userName;
GRANT EXECUTE ON dbo.istableexists TO userName;
```

```
GRANT EXECUTE ON dbo.gettableid TO userName;
GRANT EXECUTE ON dbo.gettablename TO userName;
GRANT EXECUTE ON dbo.getcolumnname TO userName;
GRANT EXECUTE ON dbo.getcolumnid TO userName;
GRANT EXECUTE ON dbo.getcolumnkeystring TO userName;
GRANT EXECUTE ON dbo.getfkvalidationsql TO userName;
GRANT EXECUTE ON dbo.getpkcolumnslist TO userName;
GRANT EXECUTE ON dbo.getpkdecodecolumnslist TO userName;
GRANT EXECUTE ON dbo.getpkcolumnslistnbr TO userName;
GRANT EXECUTE ON dbo.getpkvalidationsql TO userName;
GRANT EXECUTE ON dbo.fn_get_brs_ini_par TO userName;
GRANT EXECUTE ON dbo.fn_par_get_database_file_location TO userName;
GRANT EXECUTE ON dbo.fn_zeropad TO userName;
GRANT EXECUTE ON dbo.fn_par_get_filegroup_name TO userName;
GRANT EXECUTE ON dbo.fn_par_get_lastkey TO userName;
GRANT EXECUTE ON dbo.fn_par_get_maxbound TO userName;
GRANT EXECUTE ON dbo.fn_par_get_minbound TO userName;
```

9.20 Certificate expiration

Context

By default, the automatically generated files expire after 2 years(AdminConsole.jks, Lineage.jks, RegulatoryPlatform.jks, rgtng_client.jks, trust_store.jks, upload_client,jks) if these certificates expired the bellow steps are needed for them to be recreated.

Procedure

1. Backup old .jks files in from the [INSTALL_DIR]\conf\security\ folder
2. Recreate the 6 certificates

   keytool -genkey -alias adminconsole-localhost -keystore AdminConsole.jks -storepass changeme -keyalg RSA -validity 720 -keysize 2048

   keytool -genkey -alias lineage-localhost -keystore Lineage.jks -storepass changeme -keyalg RSA -validity 720 -keysize 2048

   keytool -genkey -alias wkfs-saml-key -keystore RegulatoryPlatform.jks -storepass changeme -keyalg RSA -validity 720 -keysize 2048

   keytool -genkey -alias rgtng_cert_alias -keystore rgtng_client.jks -storepass changeme -keyalg RSA -validity 720 -keysize 2048

   keytool -genkey -alias upload -keystore trust_store.jks -storepass changeme -keyalg RSA -validity 720 -keysize 2048

   keytool -genkey -alias upload -keystore upload_client.jks -storepass changeme -keyalg RSA -validity 720 -keysize 2048

   Please set the -validity parameter to the desired value. In this example, the validity of the certificates are almost 2 years(720 days)

3. For the 4 main .jks files please check their signing certificate code in the command prompt by

   keytool -list -rfc -keystore AdminConsole.jks -alias adminconsole-localhost -storepass changeme

   keytool -list -rfc -keystore Lineage.jks -alias lineage-localhost -storepass changeme

keytool -list -rfc -keystore RegulatoryPlatform.jks -alias wkfs-saml-key -storepass changeme

keytool -list -rfc -keystore rgtng_client.jks -alias rgtng_cert_alias -storepass changeme

4. After the "----**BEGIN CERTIFICATE**----" line in the output, please copy the certificate and paste it into each ....-sp.xml file among the <ds:X509Certificate> tags.
5. Upload the reconfigured 4 client ....-sp.xml files from the [INSTALL_DIR]\conf\security\ folder to the IdP.
6. Download the new Metadata file.
7. Restart the Application Service.

10 Contact

- For questions or issues regarding the product or solution, please raise a support ticket through our case management tool https://wolterskluwer.my.site.com/FRRCommunity.
- For questions regarding our case management tool or to subscribe additional email addresses to the release notification, please email FRR-Support@wolterskluwer.com.
- To request support from our Professional Services team, for example, implementation assistance and training requirements, please get in touch with your account manager or sales representative from Wolters Kluwer.